The niche of Romik's ambidextrous sofa in convex-linear data, and a stability bound under a curvature condition

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Abstract

Baek proved that Gerver's sofa is the maximum-area moving sofa for a corridor with one corner, by writing a sofa as a convex cap minus a niche and obtaining optimality from the concavity of a quadratic upper bound. We carry that architecture to the ambidextrous moving sofa problem, which is open, and to Romik's candidate Σ of area $A_R^* = 1.6449552184\ldots$

The niche of an ambidextrous sofa is expressed exactly in convex-linear data: writing $H(\theta) = h_K(\mu_\theta)$ for the cap's support function, the two arms α_1, α_2 and the reach σ are affine in H, and $|N| = \int_0^{\pi/2} [\frac{1}{2}(\alpha_2^+)^2 + \frac{1}{2}(\sigma - \alpha_1)^2 - \frac{1}{2}(\alpha_1^-)^2]\, dt$ whenever the associated sweep is injective, disjoint, covers the niche, and stays inside the cap. Injectivity follows from a single linear condition (RC), that the cap's radius of curvature be at most the corridor width; the proof reduces each of the three required statements to a forced harmonic oscillator whose kernel is non-negative because the rotation angle is $\pi/2$. The fourth hypothesis, containment, does not follow from (RC): Remark 44 exhibits a cap satisfying (RC) for which the identity fails by three per cent, and Remark 51 corrects an earlier misattribution of its cause. Containment is therefore assumed, and Theorem 115 is conditional on it. (RC) is classical: by the dual of Blaschke's rolling theorem it says the cap slides freely inside a unit ball, up to the corridor facets. What is new is its use.

For the functional $Q = |C_2| - 2V$ we prove $Q(\Sigma) = A_R^*$ on the same containment hypothesis (Theorem 93, Remark 94); that $\delta Q(\Sigma) = 0$, so Romik's ODE families are exactly the critical-point equations of Q, identically in his constants; and that $\delta^2 Q < 0$, with an explicit constant. Two Cauchy–Schwarz steps decouple the two halves of $[0, \pi]$ and reduce the second variation to a pair of Sturm–Liouville eigenvalue problems, certified from below by Sturm oscillation; this gives $\frac{1}{2}\delta^2 Q \le -\frac{2}{3}\|\eta\|^2_{L^2(0,\pi)}$. Not splitting at all turns the second variation into a single 2×2 system whose first eigenvalue is the sharp constant $c^* = 0.7309566\ldots$, certified to exceed $\frac{73}{100}$. The same system, with the cell boundaries as parameters, proves concavity for every cell whose sign sets are ordered and anchored, by a one-line monotonicity reduction to a one-parameter family certified by an adaptive interval covering; on the mirror class the constant is uniform, $\frac{3}{10}$, and the resulting stability bound assumes no smallness of $H - H_\Sigma$ beyond (RC). On the convex domain $\mathcal{D}$ cut out by (RC), a separation inequality and Σ's own sign pattern,

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\, \|H - H_\Sigma\|^2_{L^2(0,\pi)}$$

for every ambidextrous sofa T with cap data in $\mathcal{D}$, and Σ is the unique member attaining A_R^*. The domain is a C^2-neighbourhood of Σ: its radius along $\sin(2k\theta)$ scales like k^{-2}.

Two limits are recorded rather than smoothed over. The separation inequality is independent of (RC) and cannot be removed. And V is the exact flux of the advancing wedge boundary, so $V \ge |N|$ by Reynolds' transport theorem and Q bounds nothing where the sweep fails to be injective; in particular the construction does not recover Gerver's sofa, whose cap violates (RC). Separately, a two-hallway relaxation gives the first upper bound specific to the ambidextrous problem: the area of a left- and a right-turning hallway at angle $\pi/4$ is bounded by translates of one explicit profile integral, and a four-case analysis of the corner offsets removes every hypothesis, giving

$$A_{\mathrm{ambi}} \;\le\; 2\sqrt{2}-1 \;=\; 1.8284271\ldots$$

unconditionally. The bound is attained by an explicit placement, so it is exactly the value of the two-hallway relaxation; it improves on the 2.2195 inherited from the one-corner theorem, and with Romik's sofa it brackets the ambidextrous supremum within a factor of 1.1115. Optimality of Σ remains open.

1 Introduction

The moving sofa problem asks for the largest area of a connected shape that can be manoeuvred around a right-angled corner of a hallway of unit width. It was settled by Baek [1], who proved Gerver's sofa [2] optimal. Romik [5] posed the *ambidextrous* variant, in which the shape must turn corners of both handedness, exhibited a candidate Σ of area

$$A_R^* = 1.6449552184254408\ldots,$$

and asked whether it is optimal. That question is open, and this note does not settle it. What follows is a structural analysis of the ambidextrous problem in the coordinates in which the constraint data is affine, and a local optimality statement for Σ on an explicitly described domain.

The coordinates

Every ambidextrous sofa is $C_2 \setminus (U \sqcup \rho U)$ for a convex *cap* C_2 and a *niche* U swept by the hallway's inner corner, with ρ the reflection in $y = \frac{1}{2}$. Writing $H(\theta) = h_{C_2}(\mu_\theta)$ for the cap's support function on $[0, \pi]$, the corner path is *affine* in H (Lemma 18), and so are the two arm lengths α_1, α_2 and the reach σ that determine the sweep. In these coordinates the cap area is an exact quadratic form,

$$|C_2| = \int_0^\pi (H^2 - H'^2)\, d\theta - H(0) - H(\pi)$$

(Proposition 25), and the niche area has the closed form (20). The functional $Q := |C_2| - 2V$, with V the flux of the advancing sweep, is therefore an explicit quadratic polynomial in H on each cell of the sign pattern of (α_1, α_2).

The results

The analysis rests on one geometric hypothesis, that the cap is nowhere flatter than a circle of the corridor's width:

$$\text{(RC)} \qquad (H + H'')_{\mathrm{ac}} \le 1,$$

which is linear in H and hence convex. It is classical, being the sliding-ball condition dual to Blaschke's rolling theorem (Remark 80); what is new is its use. Theorem 81 shows that (RC) implies all three injectivity conditions of the sweep, by reducing each to a forced harmonic oscillator whose kernel is non-negative precisely because the rotation angle is $\pi/2$. That is three of the four hypotheses of Proposition 20; the fourth, containment, is assumed, and with it $V = |N|$ and Q is tight.

At Σ the three hypotheses of the concavity method hold exactly: $Q(\Sigma) = A^*_R$ structurally, with no integral evaluated, on that same containment hypothesis (Theorem 93, Remark 94), and to 50 digits without it; $\delta Q(\Sigma) = 0$ identically in Romik's constants, so his ODE families *are* the critical-point equations of Q (Theorem 86); and $\delta^2 Q < 0$.

The second variation is the technical core. Setting $p(t) = \eta(t)$ and $q(t) = \eta(t+\pi/2)$ identifies it with a 2×2 Sturm–Liouville system on $[0, \pi/2]$ whose first eigenvalue *is* the best constant,

$$c^* = 0.7309566\ldots,$$

certified above $\frac{73}{100}$ in ball arithmetic (Theorem 119). On the overlap of the two sign sets the coupling is a total derivative, which is why every pointwise estimate loses and this one does not. The same system, with the cell boundaries as parameters, gives concavity on *every* ordered anchored cell (Theorem 122), and a uniform constant $\frac{3}{10}$ on the mirror class (Theorem 125).

These combine into two optimality statements. Theorem 115 gives $|T| \leq A^*_R$ for every ambidextrous sofa with cap data in the convex domain $\mathcal{D}$ cut out by the gauge, (RC), a separation inequality and Σ's sign pattern, with the quantitative form

$$|T| \;\leq\; A^*_R - \tfrac{73}{100}\, \|H - H_\Sigma\|^2_{L^2(0,\pi)}$$

of Theorem 126 and uniqueness (Corollary 127). Proposition 137 exhibits an explicit C^2-ball of radius $\frac{1}{20}$ inside $\mathcal{D}$. Theorem 131 is the non-perturbative version: on the mirror class, with no smallness of $H - H_\Sigma$ assumed, $|T| \leq A^*_R - \frac{3}{10}\|H - H_\Sigma\|^2$.

What is not proved, and why

What the local result addresses is Romik's own first open problem [5, §7], which asks for a proof that Gerver's sofa and Σ are local maxima of the area functional in the one-corner and ambidextrous problems respectively. Romik established the first-order conditions for Σ; the second-order statement has not been proved, and Theorem 115 proves it only on the domain $\mathcal{D}$, so that problem stays open. For Gerver's sofa the corresponding question is settled a fortiori by [1].

Optimality of Σ remains open, and the obstructions are recorded rather than smoothed over. V is the exact flux of the advancing sweep, so $V \geq |N|$ always by Reynolds' transport theorem (Proposition 41) and Q bounds nothing where the sweep fails to be injective; the domain cannot be widened by a concave majorant of the excess, since none exists (Proposition 129), though a semiconcave one does (Theorem 130). The separation hypothesis is independent of (RC). Gerver's cap violates (RC), but that is not what excludes it: it is not the cap of any *connected* ambidextrous sofa (Remark 139). The sign hypothesis of Theorem 131 is permanent, not provisional: the incircle of the unit square is a connected ambidextrous sofa satisfying every other condition and violating it (Remark 134).

Claims are labelled throughout: VERIFIED means machine-checked in Lean 4 with no `sorry`, PROVED a complete human proof, HEURISTIC computational evidence only. Numerical values that carry weight are computed in ball or exact rational arithmetic with stated precision; Section 18 separates what is ours from what is Baek's and Romik's.

2 Setting

Write L for the closed L-shaped hallway of unit width with its corner at the origin, $\mu_t = (\cos t, \sin t)$, $\nu_t = (-\sin t, \cos t)$, and for a rotation path $c : [0, \pi/2] \to \mathbb{R}^2$ let

$$C_t = \{p : \langle p - c(t), \mu_t\rangle \le 1\} \cap \{p : \langle p - c(t), \nu_t\rangle \le 1\}, \qquad Q_t = \{p : \langle p - c(t), \mu_t\rangle < 0\} \cap \{p : \langle p - c(t), \nu_t\rangle < 0\},$$

so that the hallway in position t is $H_t = C_t \setminus Q_t$. The one-sided sofa is $S = \bigcap_t H_t$, and writing $C = \bigcap_t C_t$ and $U = \bigcup_t Q_t$ we have $S = C \setminus U$ with C convex.

Let $\rho(x, y) = (x, 1 - y)$ be the reflection in the line $y = \frac{1}{2}$. Romik's ambidextrous sofa is

$$\Sigma = S \cap \rho(S) = C_2 \setminus \big(U \cup \rho U\big), \qquad C_2 := C \cap \rho C \text{ convex}, \tag{1}$$

with $|\Sigma| = A_R^* = 1.6449552184\ldots$, and c is Romik's piecewise path with breakpoints at β and $\pi/2 - \beta$, where

$$\beta = \arctan\Big(\tfrac{1}{2}\big(\sqrt[3]{\sqrt{2} + 1} - \sqrt[3]{\sqrt{2} - 1}\big)\Big) = 0.2896538208\ldots \tag{2}$$

Baek's argument for the one-corner problem builds an overestimating region whose area is a quadratic functional on a convex domain of convex bodies, and proves it concave by exhibiting it as a linear functional minus Mamikon regions, whose areas are convex quadratics. Concavity reduces global optimality to a first-order condition. The construction is organised around $S = K \setminus \mathcal{N}(K)$: one cap, one niche.

By (1) the ambidextrous analogue has *two* niches, and

$$|U \cup \rho U| = |U| + |\rho U| - |U \cap \rho U|. \tag{3}$$

The last term appears in the upper bound with a + sign. Since concavity is obtained by *subtracting* convex quadratics, an added overlap term is not covered by the mechanism. The purpose of this note is Theorem 8: the term is zero.

3 The niches are disjoint

Lemma 1 (Niche ceiling). *For every $t \in [0, \pi/2]$ we have $Q_t \subseteq \{(x, y) : y \le c_y(t)\}$.*

Proof. Q_t is the open convex cone with apex $c(t)$ spanned by $-\mu_t$ and $-\nu_t$, so $q \in Q_t$ means $q - c(t) = -a\mu_t - b\nu_t$ with $a, b > 0$. For $t \in [0, \pi/2]$ both $\sin t \ge 0$ and $\cos t \ge 0$, hence

$$q_y - c_y(t) = -a \sin t - b \cos t \le 0. \qquad \square$$

Corollary 2. *Put $M := \max_{t\in[0,\pi/2]} c_y(t)$. Then $U \subseteq \{y \le M\}$ and $\rho U \subseteq \{y \ge 1 - M\}$. If $M < \frac{1}{2}$ then $U \cap \rho U = \emptyset$.*

Proof. The first inclusion is Lemma 1 applied to each t; the second is its image under ρ. If $M < \frac{1}{2}$ then $1 - M > \frac{1}{2} > M$, so the two half-planes are disjoint. $\square$

It remains to evaluate M for Romik's path. On the middle piece $[\beta, \pi/2 - \beta]$ Romik's path is

$$c(t) = R(t)v(t) + \kappa, \qquad v(t) = \begin{pmatrix} f_1 \cos\frac{t}{2} + f_2 \sin\frac{t}{2} - 1 \\ -f_2 \cos\frac{t}{2} + f_1 \sin\frac{t}{2} - 1 \end{pmatrix}, \tag{4}$$

with $R(t)$ the rotation by t, and the two facts we use are

$$\kappa_y = \tfrac{1}{2} \quad \text{exactly}, \qquad f_2 = (1 - \sqrt{2})\, f_1. \tag{5}$$

Proposition 3. *For $t \in [\beta, \pi/2 - \beta]$,*

$$c_y(t) - \tfrac{1}{2} = f_1 \sin\tfrac{3t}{2} - f_2 \cos\tfrac{3t}{2} - (\sin t + \cos t) = f_1\sqrt{4 - 2\sqrt{2}}\,\sin\Bigl(\tfrac{3t}{2} + \tfrac{\pi}{8}\Bigr) - \sqrt{2}\,\sin\Bigl(t + \tfrac{\pi}{4}\Bigr). \quad (6)$$

In particular c_y is symmetric about $t = \pi/4$ on this interval, and

$$M = c_y(\pi/4) = \tfrac{1}{2} - \Bigl(\sqrt{2} - f_1\sqrt{4 - 2\sqrt{2}}\Bigr). \quad (7)$$

Proof. By (4) and $\kappa_y = \frac{1}{2}$, $c_y(t) - \frac{1}{2} = \sin t\, v_x + \cos t\, v_y$. Write $S = \sin\frac{t}{2}$, $C = \cos\frac{t}{2}$, so $\sin t = 2SC$ and $\cos t = C^2 - S^2$. Expanding and collecting the coefficients of f_1 and f_2 gives

$$\sin t\, v_x + \cos t\, v_y = f_1 S(3C^2 - S^2) + f_2 C(3S^2 - C^2) - (\sin t + \cos t),$$

and the triple-angle identities $S(3C^2 - S^2) = \sin\frac{3t}{2}$, $C(3S^2 - C^2) = -\cos\frac{3t}{2}$ give the first equality in (6). For the second, substitute $f_2 = (1 - \sqrt{2})f_1$ from (5): the bracket becomes $\sin\frac{3t}{2} + (\sqrt{2} - 1)\cos\frac{3t}{2}$, of amplitude $\sqrt{1 + (\sqrt{2} - 1)^2} = \sqrt{4 - 2\sqrt{2}}$ and phase $\arctan(\sqrt{2} - 1) = \pi/8$.

Both sinusoids in (6) are invariant under $t \mapsto \pi/2 - t$: indeed $\sin(\frac{3}{2}(\pi/2 - t) + \pi/8) = \sin(7\pi/8 - \frac{3t}{2}) = \sin(\pi/8 + \frac{3t}{2})$ and $\sin(3\pi/4 - t) = \sin(\pi/4 + t)$. Hence c_y is symmetric about $\pi/4$, where both sinusoids attain the value 1 simultaneously; (7) follows. $\square$

That $\pi/4$ maximises c_y over all of $[0, \pi/2]$, and not merely over the middle piece, follows from the corresponding closed form on the outer pieces.

Proposition 4. *For $t \in [0, \beta]$,*

$$c_y(t) = \tfrac{1}{2} + a_1 \sin 2t - \sin t - \tfrac{1}{2}\cos t, \quad (8)$$

and c_y is strictly increasing there, so $\max_{[0,\beta]} c_y = c_y(\beta)$. By the symmetry $c_y(t) = c_y(\pi/2 - t)$ the same bound holds on $[\pi/2 - \beta, \pi/2]$. Consequently $M = c_y(\pi/4)$ as claimed in (7).

Proof. On $[0, \beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1 \cos t - 1, a_1 \sin t - \frac{1}{2})$ and $\kappa_y^{(1)} = \frac{1}{2}$, so $c_y(t) - \frac{1}{2} = \sin t\, v_{1x} + \cos t\, v_{1y} = 2a_1 \sin t \cos t - \sin t - \frac{1}{2}\cos t$, which is (8). Differentiating,

$$c_y'(t) = 2a_1 \cos 2t - \cos t + \tfrac{1}{2}\sin t \;\geq\; 2a_1 \cos 2\beta - 1 \;=\; 0.4650\ldots > 0 \qquad (t \in [0, \beta]),$$

using $\cos 2t \geq \cos 2\beta$ and $\cos t - \frac{1}{2}\sin t \leq 1$ for $t \geq 0$. Hence c_y is strictly increasing on $[0, \beta]$, with $c_y(\beta) = 0.2143795161\ldots$, which is below $c_y(\pi/4) = 0.3878381292\ldots$. $\square$

Remark 5. Evaluating (8) and (6) at $t = \beta$ gives the same value to $4.4 \cdot 10^{-31}$ at 30-digit precision. Since the two expressions were derived from different pieces of Romik's path, this is an independent check on both, and on (10).

4 A closed form for f_1

Romik's constant f_1 is recorded in the literature as the decimal 1.202938908156911389. The following determines it in closed form. Let

$$a_1 = \tfrac{1}{4}\sqrt{4 + \sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}}} = 0.8752873624\ldots \tag{9}$$

Proposition 6.

$$f_1 = \frac{\tfrac{4}{3}\, a_1 \cos\beta}{\cos\frac{\beta}{2} + (1-\sqrt{2})\sin\frac{\beta}{2}}. \tag{10}$$

Proof. On $[0, \beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$ and $\kappa^{(1)} = (1 - a_1, \frac{1}{2})$; on $[\beta, \pi/2 - \beta]$ it is (4) with $\kappa = (1 - \frac{4}{3}a_1, \frac{1}{2})$. Both κ's have second coordinate $\frac{1}{2}$ and their first coordinates differ by exactly $\frac{1}{3}a_1$. Continuity at $t = \beta$ therefore gives $R(\beta)\big(v_1(\beta) - v(\beta)\big) = (-\frac{1}{3}a_1, 0)$, so

$$v_1(\beta) - v(\beta) = R(-\beta)\big(-\tfrac{1}{3}a_1, 0\big) = \big(-\tfrac{1}{3}a_1\cos\beta,\ \tfrac{1}{3}a_1 \sin\beta\big).$$

Reading the first coordinate and using $f_2 = (1 - \sqrt{2})f_1$,

$$\big(a_1\cos\beta - 1\big) - \Big(f_1\big(\cos\tfrac{\beta}{2} + (1-\sqrt{2})\sin\tfrac{\beta}{2}\big) - 1\Big) = -\tfrac{1}{3}a_1\cos\beta,$$

which rearranges to (10). □

Remark 7. Evaluating (10) at 50 digits gives $f_1 = 1.2029389081569113890702 23\ldots$, so the tabulated decimal is its rounding. As an independent check, the *second* coordinate of the junction relation, which was not used in the derivation, is satisfied to $1.7 \cdot 10^{-51}$ at the same precision.

5 Disjointness of the niches for Σ

Theorem 8. *For Romik's ambidextrous sofa, $U \cap \rho U = \emptyset$, and consequently*

$$|\Sigma| = |C_2| - |U| - |\rho U| = |C_2| - 2|U|.$$

Proof. By Proposition 3, $M < \frac{1}{2}$ is equivalent to $f_1\sqrt{4 - 2\sqrt{2}} < \sqrt{2}$, i.e. to

$$f_1^2 < \frac{2 + \sqrt{2}}{2}. \tag{11}$$

With f_1 given by (10) in terms of the algebraic numbers (2) and (9), both sides of (11) are explicit algebraic numbers, and

$$f_1^2 = 1.4470620167577420940\ldots, \qquad \frac{2+\sqrt{2}}{2} = 1.7071067811865475244\ldots,$$

so (11) holds with margin $0.2600447644\ldots$. Corollary 2 gives $U \cap \rho U = \emptyset$, and then (3) together with (1). Finally $|\rho U| = |U|$ because ρ is an isometry. □

Corollary 9. *$M = 0.3878381292441942964\ldots$, the two niches are separated by the horizontal band of width $1 - 2M = 0.2243237415116114\ldots$ about $y = \frac{1}{2}$, and the ambidextrous upper bound requires only* one *niche to be modelled: one core and two tails, the same shape as in the one-corner problem.*

6 The quotient reduction and a conditional statement

Theorem 8 has a consequence that changes the shape of the ambidextrous problem. Since $U \subseteq \{y \leq M\}$ with $M < \frac{1}{2}$, the whole lower niche lies below the symmetry axis, so intersecting with the lower half-strip deletes ρU outright.

Corollary 10 (Quotient reduction)**.** *Put $K^- := C_2 \cap \{y \leq \frac{1}{2}\}$, a convex body. Then*

$$\Sigma \cap \{y \leq \tfrac{1}{2}\} = K^- \setminus U, \qquad |\Sigma| = 2\big(|K^-| - |U \cap K^-|\big).$$

That is a convex cap minus *one* niche: the shape Baek's argument is built on. Numerically, at $n = 1441$ hallways, $|\Sigma| = 1.645059395$ and $|\Sigma \cap \{y \leq \frac{1}{2}\}| = 0.822529698$, whose double reproduces $|\Sigma|$ to $3.8 \cdot 10^{-15}$.

Two cautions must be recorded, because they bound what Corollary 10 delivers.

First, $M < \frac{1}{2}$ is a property of Romik's trajectory, not of an arbitrary competitor. A competitor with $M \geq \frac{1}{2}$ has overlapping niches, and by (3) overlap *increases* $|\Sigma|$; so it cannot be assumed away on extremal grounds.

Second, and this is what makes the reduction usable, connectedness does appear to exclude it, but not for the reason one first tries. A *single* pair $Q_t \cup \rho Q_t$ removes a full vertical segment, and hence disconnects any body meeting both sides of it, exactly when the apex height exceeds a threshold $h^*(t)$; and computation gives $h^*(t) > \frac{1}{2}$ throughout $(0, \pi/2)$, with $h^*(t) \to \frac{1}{2}$ as $t \to 0$, $h^*(\pi/4) = 0.5100$ and $h^*(1.52) = 0.6967$. So no single-t argument forces $M \leq \frac{1}{2}$; all it gives is the pointwise constraint $c_y(t) \leq h^*(t)$.

In fact a single pair does force it, and the earlier reading to the contrary was an artifact of testing the vertical line at a fixed distance from the apex.

Theorem 11 (Connectedness ceiling)**.** *Let $p = c(t_0)$ with $t_0 \in (0, \pi/2)$ and suppose $p_y > \frac{1}{2}$. Put*

$$\varepsilon_0 := \frac{2p_y - 1}{2\tan t_0} > 0.$$

Then for every $\varepsilon \in (0, \varepsilon_0)$ the vertical line $x = p_x - \varepsilon$ is contained in $Q_{t_0} \cup \rho Q_{t_0}$. Hence any ambidextrous moving sofa omits the open vertical strip $\{p_x - \varepsilon_0 < x < p_x\}$, and a connected one lies entirely in one of the two half-planes it separates.

Proof. Fix $\varepsilon > 0$ and consider $q = (p_x - \varepsilon, y)$. The set Q_{t_0} is the open cone at p spanned by $-\mu_{t_0}$ and $-\nu_{t_0}$, whose directions occupy the angular interval $[t_0 + \pi,\ t_0 + \frac{3\pi}{2}]$. Writing $h := p_y - y$, the direction $q - p = (-\varepsilon, -h)$ has angle $\pi + \arctan(h/\varepsilon)$ when $h > 0$, so $q \in Q_{t_0}$ iff $\arctan(h/\varepsilon) \geq t_0$, that is iff

$$y \leq p_y - \varepsilon \tan t_0.$$

Applying ρ and using $\rho q - \rho p = R(q - p)$ with $R = \mathrm{diag}(1, -1)$, the reflected cone ρQ_{t_0} has apex $(p_x, 1 - p_y)$ and occupies the angular interval $[\frac{\pi}{2} - t_0,\ \pi - t_0]$, giving by the same computation

$$q \in \rho Q_{t_0} \iff y \geq 1 - p_y + \varepsilon \tan t_0.$$

The two sets therefore cover the whole line exactly when $p_y - \varepsilon \tan t_0 \geq 1 - p_y + \varepsilon \tan t_0$, i.e. when $2\varepsilon \tan t_0 \leq 2p_y - 1$, i.e. when $\varepsilon \leq \varepsilon_0$. Since $p_y > \frac{1}{2}$ we have $\varepsilon_0 > 0$ and the range is nonempty. Taking the union over $\varepsilon \in (0, \varepsilon_0)$ gives the strip. □

Remark 12. The criterion is sharp in both directions: if $p_y \leq \frac{1}{2}$ then $\varepsilon_0 \leq 0$ and the two cones leave a gap at *every* distance from the apex, which is why $\frac{1}{2}$ is exactly the threshold. It also explains a numerical trap: ε_0 can be very small (at $t_0 = 1.3$, $p_y = 0.55$ it is 0.0139), so sampling the line at a fixed distance such as 0.02 detects no covering and suggests, wrongly, that a single pair never separates. The criterion was checked against direct computation at seven pairs (t_0, p_y) on both sides of ε_0, and at three pairs with $p_y < \frac{1}{2}$.

Granting that an ambidextrous sofa cannot lie in one of the two half-planes separated by such a strip, since it must meet both arms of the hallway H_{t_0} whose inner corner is p, Theorem 11 yields $\max_t c_y(t) \leq \frac{1}{2}$ for every connected ambidextrous moving sofa, and the hypothesis of the next proposition is automatic.

Proposition 13 (Conditional form)**.** *Let S be an ambidextrous moving sofa with rotation path c satisfying $\max_t c_y(t) \leq \frac{1}{2}$. Then its two niches are disjoint and $|S| = 2(|K^-| - |U \cap K^-|)$ with K^- convex and U a single niche. In particular the area functional of the ambidextrous problem restricted to this class is of the cap-minus-one-niche form to which the concavity method applies. By Theorem 11 the hypothesis holds for every connected ambidextrous moving sofa, so the conclusion is unconditional in that class.*

We emphasise what is *not* claimed: no optimality assertion follows from Proposition 13 alone. Baek's route additionally requires an existence and regularity step (a maximum monotone sofa of rotation angle $\pi/2$ satisfying an injectivity condition), and his balancing argument for that step does not transfer, for a reason worth recording. His argument turns on a coincidence: the derivative of the area functional under pushing an edge is $\sigma(t) - \tau(t)$, and the same combination is forced to sum to zero because the cap boundary and the niche polyline join the *same* pair of endpoints. In the quotient, ∂K^- acquires the horizontal cut at $y = \frac{1}{2}$, where the polyline contributes nothing, and the identity becomes $\sum_{t \neq \mathrm{cut}} (\sigma - \tau)(v_t \cdot u_0) = \sigma_{\mathrm{cut}} > 0$, which is consistent with $\sigma \leq \tau$ throughout. One obtains an inequality where Baek obtains an equality, and it is the equality that his argument consumes.

7 A relation between Romik's constants

The phase junctions of Σ admit a characterisation that ties two of Romik's constants together. Recall a_1 from (9) and β from (2).

Theorem 14. $4a_1 \sin\beta = 1$.

Proof. Write $u := \sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}$, so that $\tan\beta = u/2$ by (2) and hence $\sin\beta = u/\sqrt{4+u^2}$, and $w := \sqrt[3]{71+8\sqrt{2}} + \sqrt[3]{71-8\sqrt{2}}$, so that $a_1 = \frac{1}{4}\sqrt{4+w}$ by (9). Then

$$4a_1 \sin\beta = \frac{u\sqrt{4+w}}{\sqrt{4+u^2}},$$

so the assertion is equivalent to $u^2(4+w) = 4 + u^2$, that is to

$$u^2\,(3+w) = 4. \tag{12}$$

Put $p := \sqrt[3]{\sqrt{2}+1}$ and $q := \sqrt[3]{\sqrt{2}-1}$. Then $pq = 1$ and $p^3 - q^3 = 2$, so

$$u^3 = (p-q)^3 = p^3 - q^3 - 3pq(p-q) = 2 - 3u,$$

i.e.

$$u^3 + 3u = 2. \tag{13}$$

Set $x := u^2$. Then (13) reads $u(x+3) = 2$; squaring and using $u^2 = x$ gives

$$x(x+3)^2 = 4, \qquad \text{i.e.} \qquad x^3 + 6x^2 + 9x - 4 = 0. \tag{14}$$

Now let $W := 4/x - 3$. Clearing denominators,

$$W^3 - 51W - 142 = \frac{(4-3x)^3 - 51(4-3x)x^2 - 142x^3}{x^3} = \frac{-16\big(x^3 + 6x^2 + 9x - 4\big)}{x^3} = 0$$

by (14), the middle equality being the polynomial identity $(4-3x)^3 - 51(4-3x)x^2 - 142x^3 = -16(x^3 + 6x^2 + 9x - 4)$.

On the other hand w satisfies the same cubic: with $A := \sqrt[3]{71 + 8\sqrt{2}}$, $B := \sqrt[3]{71 - 8\sqrt{2}}$ one has $AB = \sqrt[3]{71^2 - 128} = \sqrt[3]{4913} = 17$ and $A^3 + B^3 = 142$, so $w^3 = A^3 + B^3 + 3AB\,w = 142 + 51w$. The cubic $W^3 - 51W - 142$ has discriminant $-4(-51)^3 - 27(-142)^2 = 530604 - 544428 = -13824 < 0$ and therefore exactly one real root; both $4/x - 3$ and w are real roots, so $4/x - 3 = w$. This is (12). $\square$

Corollary 15. *$2\tan\beta$ is the real root of $u^3 + 3u = 2$, equivalently $4\tan^3\beta + 3\tan\beta = 1$, the cubic already recorded in the literature, and*

$$\sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}} = u^4 + 6u^2 + 6.$$

Proof. The first assertion is (13). For the second, (14) gives $(x+3)^2 = 4/x$, so $w = 4/x - 3 = (x+3)^2 - 3 = x^2 + 6x + 6$ with $x = u^2$. $\square$

Proof of the junction statement. On $[0, \beta)$ Romik's path is $c(t) = R_t v(t) + \kappa^{(1)}$ with $v(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$. Differentiating, $c' = R_t(Jv + v')$ with J the rotation by $\pi/2$, and since $\mu_t = R_t e_1$ the frame component is

$$c'(t) \cdot \mu_t = (Jv + v')_x = -v_y - a_1 \sin t = -\big(a_1 \sin t - \tfrac{1}{2}\big) - a_1 \sin t = \tfrac{1}{2} - 2a_1 \sin t.$$

This vanishes precisely when $\sin t = 1/(4a_1)$. On the other hand the phase boundary of Σ at $t = \beta$ is where the SOL1 phase ends, and by Proposition 4 the two closed forms for c_y agree there; the same computation on the SOL5 phase gives $c' \cdot \nu_t = 2a_1 \cos t - \frac{1}{2}$, vanishing when $\cos t = 1/(4a_1)$, i.e. at $t = \pi/2 - \beta$. Since the two junctions are β and $\pi/2 - \beta$ and $\cos(\pi/2 - \beta) = \sin\beta$, both conditions reduce to $\sin\beta = 1/(4a_1)$. $\square$

Remark 16 (Attribution)**.** Neither (13) nor (14) is new. The standard description of Σ gives its area as $x + \arctan y$ where $x^2(x+3) = 8$ and $y(4y^2 + 3) = 1$; the second of these is exactly (13) with $y = \tan\beta = u/2$, and the first is (14) shifted, since $x = u^2 + 1$ turns $x^3 + 3x^2 - 8$ into $u^6 + 6u^4 + 9u^2 - 4$. Equivalently

$$A_R^* = u^2 + 1 + \arctan(u/2), \qquad u^3 + 3u = 2,$$

so a single algebraic number generates both parts of the area. What we have not found stated is the relation $4a_1 \sin\beta = 1$ of Theorem 14 between the shape constant a_1 and β, nor $w = u^4 + 6u^2 + 6$; but both are elementary consequences of the two standard cubics, and we make no priority claim. Every step was checked numerically at 60 digits, and the two polynomial steps are machine-checked (`u_sq_cubic`, `cardano_substitution`).

Theorem 14 has a consequence for the transfer of Baek's method. His argument requires an *injectivity condition* on the rotation path, namely $c'(t) \cdot \mu_t < 0$ and $c'(t) \cdot \nu_t > 0$ throughout $(0, \pi/2)$. By the computation above, for Σ these quantities vanish exactly at β and $\pi/2 - \beta$ and have the wrong sign outside $[\beta, \pi/2 - \beta]$: at $t = 0.05$ one finds $c' \cdot \mu_t = +0.4125$. So Σ satisfies the injectivity condition on its middle phase and violates it on both outer phases, and the failure boundary is exactly the phase junction.

Remark 17 (Where the difficulty of the ambidextrous problem sits)**.** The outer phases $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$ are also where the contact arcs of $\partial\Sigma$ are *constant*, pinned at the ρ-fixed point $(1, \frac{1}{2})$ to $3 \cdot 10^{-31}$: there the supporting hallway touches Σ along a segment rather than at a point. Three separate obstructions therefore coincide on those phases: the contact structure degenerates, second variation arguments lose the contact point, and the injectivity hypothesis of the one-corner proof fails.

8 The ambidextrous problem as a single family of angle range π

Finally we record a reformulation which explains why the transfer is delicate. Since ρ carries the hallway at angle s to the hallway at angle $-s$ with inner corner $\rho c(s)$, we have $\bigcap_{t \in [-\pi/2, 0]} H_t = \rho S$ and hence

$$\Sigma = \bigcap_{t \in [-\pi/2, \pi/2]} H_t, \qquad c(-t) = \rho c(t). \tag{15}$$

Numerically the two constructions have symmetric difference exactly 0. So the ambidextrous constraint family is a *single* family spanning an angle range of π, rather than two families of range $\pi/2$; and its corner path is discontinuous at $t = 0$, where $c_y(0) = 0$ while $\rho c(0)$ has height 1. Both features lie outside the setting of the one-corner argument, which is developed for rotation angle $\omega \le \pi/2$ and whose maximisers attain $\omega = \pi/2$. Equation (15) is thus a reformulation, not a reduction: it exhibits the ambidextrous problem inside a larger class rather than inside the solved one.

9 The niche area in convex-linear data

The obstruction recorded in Remark 17 is that the one-corner argument's injectivity condition fails for Σ on the two outer phases. We now show that the failure is an artefact of how the condition is stated, and derive an exact formula for the niche area whose every ingredient is convex-linear in the cap.

Throughout, K is the cap, $h = h_K$ its support function, and

$$F(t) := h(\mu_t), \qquad G(t) := h(\nu_t).$$

Lemma 18 (The corner is affine in the support function)**.** *Place the hallway at angle t and push both outer walls inward until they touch K. Then the inner corner sits at*

$$c(t) = \big(F(t) - 1\big)\mu_t + \big(G(t) - 1\big)\nu_t, \tag{16}$$

and the removed wedge is $Q_t = \{\langle p, \mu_t \rangle < F(t) - 1\} \cap \{\langle p, \nu_t \rangle < G(t) - 1\}$.

Proof. The two outer walls are the lines with outer normals μ_t, ν_t; touching K means their offsets are $F(t), G(t)$. The corner lies one unit inside each, so $\langle c, \mu_t\rangle = F - 1$ and $\langle c, \nu_t\rangle = G - 1$, which is (16) since (μ_t, ν_t) is orthonormal. The wedge is the intersection of the two inner half-planes through c. $\square$

Since $h \mapsto h$ is linear under Minkowski sum, (16) makes $c(t)$, and hence each of

$$\alpha_1(t) := -\langle c'(t), \mu_t\rangle = G(t) - 1 - F'(t), \tag{17}$$
$$\alpha_2(t) := \quad \langle c'(t), \nu_t\rangle = F(t) - 1 + G'(t), \tag{18}$$
$$\sigma(t) := c_y(t)/\cos t = \big(F(t) - 1\big)\tan t + G(t) - 1, \tag{19}$$

an affine-linear functional of h, hence convex-linear on a domain of convex bodies closed under Minkowski sum. The wedge Q_t has two faces, the rays $c - s\nu_t$ (face 1) and $c - s\mu_t$ (face 2), and α_1, α_2 are the signed distances from c to the envelope points on them: the one-corner injectivity condition is exactly $\alpha_1, \alpha_2 > 0$.

Lemma 19 (Normal velocities)**.** *The face-1 line advances in its own normal direction with speed $s - \alpha_1(t)$ at the point at distance s from the corner, and the face-2 line with speed $\alpha_2(t) - s$.*

Proof. Face 1 lies on $\ell_t = \{\langle p, \mu_t\rangle = F(t) - 1\}$. For p fixed the signed gap to ℓ_t is $F(t) - 1 - \langle p, \mu_t\rangle$, whose t-derivative is $F'(t) - \langle p, \nu_t\rangle$ because $\mu_t' = \nu_t$. At $p = c - s\nu_t$ we have $\langle p, \nu_t\rangle = G - 1 - s$, so the speed is $F' - G + 1 + s = s - \alpha_1$ by (17). Face 2 lies on $\{\langle p, \nu_t\rangle = G(t) - 1\}$ and $\nu_t' = -\mu_t$, so the same computation gives $G' + \langle p, \mu_t\rangle = G' + F - 1 - s = \alpha_2 - s$ at $p = c - s\mu_t$. $\square$

So the niche is created by the *inner* part of face 2 and the *outer* part of face 1, and *no sign hypothesis on α_1 is involved*: where $\alpha_1 < 0$ the whole visible face 1 advances, and where $\alpha_2 \le 0$ face 2 contributes nothing. This is the promised repair of Remark 17: each face is used only on the range where its own arm has the right sign, and the reported failure came from requiring both arms simultaneously.

Write W_2 for the region swept by the segments $[c(t), c(t) - \alpha_2(t)^+\mu_t]$, and W_1^{out} for the region swept by the face-1 segments running from $c(t) - \alpha_1(t)^+\nu_t$ outward to the corridor floor $\{y = 0\}$; note the face-1 direction $-\nu_t = (\sin t, -\cos t)$ points downward, and the floor is reached at distance exactly $\sigma(t)$, which is why (19) is the relevant reach.

Proposition 20 (Niche area in convex-linear data)**.** *Assume the two truncated sweeps lie in C_2. If in addition they are disjoint, injective and cover the niche, then*

$$|N| = \int_0^{\pi/2} \Big[\tfrac12\big(\alpha_2^+\big)^2 + \tfrac12\big(\sigma - \alpha_1\big)^2 - \tfrac12\big(\alpha_1^-\big)^2\Big]\,dt. \tag{20}$$

Proof. For the face-2 sweep $\Phi(s,t) = c(t) - s\mu_t$ one has $\partial_s\Phi = -\mu_t$ and $\partial_t\Phi = c' - s\nu_t$, so $\det[\partial_s\Phi, \partial_t\Phi] = -\langle c', \nu_t\rangle + s = s - \alpha_2$, and integrating $|s - \alpha_2|$ over $s \in [0, \alpha_2^+]$ gives $\frac12(\alpha_2^+)^2$. For the face-1 sweep $\Psi(s,t) = c(t) - s\nu_t$ one gets $\det = s - \alpha_1$, and

$$\int_{\alpha_1^+}^{\sigma} (s - \alpha_1)\,ds = \tfrac12(\sigma - \alpha_1)^2 - \tfrac12(\alpha_1^-)^2,$$

which is $\frac12(\sigma - \alpha_1)^2$ when $\alpha_1 \ge 0$ and $\frac12\sigma^2 - \alpha_1\sigma$ when $\alpha_1 < 0$, as required. Injectivity turns each $\int|\det|$ into the area of the image, disjointness makes the two areas add, and covering makes the sum $|N|$. $\square$

Remark 21 (The three hypotheses, and where they stand)**.** The first is not open. Baek [1] states and proves an injectivity condition for the one-corner problem: a monotone sofa with rotation angle $\omega \in (0, \pi/2]$ satisfies it when its rotation path $x_S : [0, \omega] \to \mathbb{R}^2$ is injective and stays above the bottom lines of both hallways, and he establishes it for the maximiser by a differential inequality built on an ODE of Romik. That is exactly the first hypothesis here, for one corner, with a proof and a technique attached.

What does not transfer is the rest. The ambidextrous problem has two corners and $C_2 = C \cap \rho C$, so (20) needs the condition for *two* sweeps at once, together with their disjointness and their covering N. Neither of the latter is a one-corner statement, and a literature sweep found no prior art for either. The honest split is therefore: injectivity of each sweep is PROVED elsewhere and needs only a transfer argument; disjointness and covering are settled in Remarks 42 and 43 for any cap whose curvature radius stays below 1 away from the atom, but the three hypotheses are not sufficient for (20): a fourth, containment of the truncated sweeps in C_2, is needed and is open for face 2 (Remark 44). That containment is the part of Proposition 20 this note takes on faith.

Remark 22 (What is measured, and what it gives)**.** For Σ the three hypotheses of Proposition 20 are confirmed numerically and sharply, and are proved in Remarks 42 and 43, though they do not by themselves give (20) (Remark 44): the two sweeps have intersection area 0 and leave 0 of the niche uncovered, to the printing precision of the polygon oracle. The right-hand side of (20), evaluated with exact derivatives of SOL1/SOL5/SOL6 and Gauss–Legendre quadrature on each phase separately, is

$$V = 0.184193197089\ldots,$$

stable in the twelfth digit from 20 to 320 nodes per phase, against a polygonal measurement 0.184193171 of the same region, and an inscribed quadrilateral union under-measures, so the two agree to $2.6 \cdot 10^{-8}$ from the expected side. The implied cap area $A_R^* + 2V = 2.0133416\ldots$ lies just below the circumscribed polygonal values 2.0133455 and 2.0133420 at 241 and 481 constraints, which must decrease to it. We therefore record (20) as an identity for Σ at the 10^{-8} level.

Two of the three terms of (20) are convex quadratics in h, since $x \mapsto \frac{1}{2}(x^+)^2$ and $x \mapsto \frac{1}{2}x^2$ are convex and $\alpha_2, \sigma - \alpha_1$ are affine in h. This is what the generalised Mamikon theorem buys, and it is the reason (20) is the right shape: for $|\Sigma| = |C_2| - 2|N|$ a convex $|N|$ gives a concave upper bound, for which a first-order condition suffices.

Proposition 23 (The obstruction, in closed form)**.** *The third term of* (20) *is subtracted, so it contributes a* convex *term to* $|C_2| - 2|N|$ *and is the one ingredient with the wrong curvature. It is supported exactly on the degenerate phase* $[0, \beta)$*, where* $\alpha_1 < 0$*, and there* $\alpha_1(t) = 2a_1 \sin t - \frac{1}{2}$*, so*

$$\frac{1}{2}\int_0^{\pi/2} \left(\alpha_1^-\right)^2 dt = \frac{\beta}{8} - a_1(1 - \cos\beta) + a_1^2\Big(\beta - \frac{1}{2}\sin 2\beta\Big) = 0.011950270059\ldots, \qquad (21)$$

which is $6.488\ldots\%$ *of* V*.*

Proof. On $[0, \beta)$ the path is SOL1 with $a_2 = 0$, so in complex form $z(t) = a_1 e^{2it} - (1 + \frac{i}{2})e^{it} + (1 - a_1 + \frac{i}{2})$ and $e^{-it}z'(t) = 2ia_1 e^{it} - i(1 + \frac{i}{2})$, whence $\alpha_1 = -\operatorname{Re}(e^{-it}z') = 2a_1 \sin t - \frac{1}{2}$. This is negative exactly for $\sin t < 1/(4a_1)$, i.e. for $t < \beta$ by Theorem 14. Expanding $\frac{1}{2}\int_0^\beta (\frac{1}{2} - 2a_1 \sin t)^2 dt$ and using $\int_0^\beta \sin^2 = \frac{\beta}{2} - \frac{1}{4}\sin 2\beta$ gives (21). $\square$

Remark 24 (Where this leaves the problem)**.** Two obstructions have been removed and one remains. The first was that no trial underestimate of the niche was tight: the best previous one, built from the apex path, left a slack of 0.087 against $|\Sigma| = 1.645$. Formula (20) is tight, so that slack is gone. The second was that the injectivity condition fails for Σ on $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$; Lemma 19 shows the condition is not needed, only the two one-sided statements α_2^+ and $\sigma \geq \alpha_1^+$, and the second holds throughout with equality only at $t = \pi/2$.

What remains is (21). It is a subtracted square of a convex-linear quantity, i.e. a $\mathcal{J}$-type term of the same kind the one-corner argument must itself control, and it is localised on the phase where Σ degenerates. Establishing or refuting concavity of $|C_2| - 2V$ is a Hessian computation on the space of support-function perturbations, and is not attempted here. The second half of the argument, that Proposition 20 applies to competitors and not only to Σ, splits in two. Its three stated hypotheses reduce, by Remarks 42 and 43, to the open condition $r < 1$ away from the atom, which Σ satisfies with margin 0.1611747 and which therefore persists under small perturbations. The containment of Remark 44 does not follow from them and is open even for Σ on face 2. So *optimality of* Σ *remains open*; what is new is that the upper bound is now tight and expressed entirely in convex-linear data, with a single explicitly computed term of the wrong sign.

10 The upper bound at Σ: tight, critical, concave

Since $\nu_t = \mu_{t+\pi/2}$, everything above is a functional of the single function

$$H(\theta) := h_K(\mu_\theta), \qquad \theta \in [0, \pi], \qquad F(t) = H(t), \quad G(t) = H(t + \pi/2).$$

Proposition 25 (The cap is an exact quadratic form)**.** *Let* $C = \bigcap_{t \in [0,\pi/2]} C_t$ *and* $C_2 = C \cap \rho C$. *Then*

$$|C_2| = \int_0^\pi \big(H(\theta)^2 - H'(\theta)^2\big) d\theta - H(0) - H(\pi). \tag{22}$$

Proof. ρ is a reflection in the line $y = \frac{1}{2}$, not in the origin: $\rho p = Rp + (0,1)$ with $R = \operatorname{diag}(1, -1)$, so $h_{\rho A}(u) = h_A(Ru) + u_y$. Since C is unbounded downward, $h_C = +\infty$ on the lower half-circle and $h_{\rho C} = +\infty$ on the upper one, so the support function of C_2 is

$$h_2(\theta) = H(\theta) \ \ (\theta \in [0, \pi]), \qquad h_2(\theta) = H(-\theta) + \sin\theta \ \ (\theta \in [-\pi, 0]).$$

For any support function in H^1 of the circle, $|C_2| = \frac{1}{2} \int h_2 \, d\sigma_2$ with $\sigma_2 = (h_2 + h_2'')d\theta$ as a measure, and the distributional identity $\langle h_2'', h_2 \rangle = -\int h_2'^2$ holds with no boundary or atom terms because the circle has no boundary. Splitting at $\theta = 0, \pi$ and substituting the two branches,

$$|C_2| = \int_0^\pi \Big[H^2 - H'^2 - H \sin s + H' \cos s - \tfrac{1}{2} \cos 2s\Big] ds,$$

and $\int_0^\pi \cos 2s \, ds = 0$ while $\int_0^\pi H' \cos s \, ds = -H(0) - H(\pi) + \int_0^\pi H \sin s \, ds$, which gives (22). $\square$

Note $-H(0) - H(\pi)$ is *linear*, so it does not enter the Hessian; and (22) is invariant under $H \mapsto H + v \cos\theta$, as it must be, that being a horizontal translation. Numerically (22) gives 2.013341613 against $A_R^* + 2V = 2.013341613$, agreeing to $9.6 \cdot 10^{-13}$, an independent analytic confirmation of the whole chain $|\Sigma| = |C_2| - 2|N|$ with $|N| = V$, since (22) and (20) share no computation. Equivalently, writing $Q := |C_2| - 2V$,

$$Q(\Sigma) = 1.644955218425\ldots = A_R^* \quad \text{to } 5 \cdot 10^{-13}. \tag{23}$$

Proposition 26 (The second variation)**.** *Fix the gauge $H(0) = 1$ (horizontal translation) and $H(\pi/2) = 1$ (unit corridor; this is also exactly what makes σ integrable against $\tan t$ at $t = \pi/2$), so that admissible η satisfy $\eta(0) = \eta(\pi/2) = 0$. Then, with $E_1 = \{\alpha_1 < 0\}$ and $E_2 = \{\alpha_2 > 0\}$,*

$$\begin{aligned}\tfrac{1}{2}\,\delta^2 Q[\eta] = \int_0^{\pi} (\eta^2 - \eta'^2)\,d\theta - \int_{E_2} \big(\eta(t) + \eta'(t + \tfrac{\pi}{2})\big)^2 dt \\ - \int_0^{\pi/2} \big(\eta(t)\tan t + \eta'(t)\big)^2 dt + \int_{E_1} \big(\eta(t + \tfrac{\pi}{2}) - \eta'(t)\big)^2 dt.\end{aligned} \tag{24}$$

Its principal part is negative definite: the coefficient of $\eta'(\theta)^2$ is -1 on $[0, \beta)$ and on $[\pi - \beta, \pi]$, and -2 in between. Hence (24) is bounded above and has at most finitely many non-negative eigenvalues.

Proof. $\delta\alpha_1 = \eta(t + \frac{\pi}{2}) - \eta'(t)$, $\delta\alpha_2 = \eta(t) + \eta'(t + \frac{\pi}{2})$ and $\delta\sigma = \eta(t)\tan t + \eta(t + \frac{\pi}{2})$ by (17)–(19); in the middle term the $\eta(t + \pi/2)$ cancels, leaving $\delta(\sigma - \alpha_1) = \eta(t)\tan t + \eta'(t)$. The integrand of (20) is C^1 across $\alpha_1 = 0$ and $\alpha_2 = 0$, so the moving boundaries of E_1, E_2 contribute nothing at second order, and δ^2 of each square is twice the square of the first variation. The cap contributes $2\int(\eta^2 - \eta'^2)$ by Proposition 25. For the principal part, collect: -1 from the cap, -1 from the σ term on $[0, \pi/2]$, $+1$ from the E_1 term on $[0, \beta)$, and -1 from the E_2 term at $\theta = t + \pi/2 \in [\pi/2, \pi - \beta)$. □

Proposition 27 (Admissibility forces the anchored structure)**.** *Let H satisfy (RC), and suppose the sign sets are ordered, $E_1 \subseteq E_2$. Suppose further that α_1 and α_2 do not vanish simultaneously. Then both sets are anchored: $E_2 = [0, \tau_2)$ and $E_1 = [0, \tau_1)$ with $\tau_1 \le \tau_2$.*

Proof. On the complement of E_2 we have $\alpha_2 \le 0$, and orderedness gives $\alpha_1 \ge 0$ there. Lemma 132 then yields $\alpha_2' \le -\alpha_1 \le 0$, so α_2 is non-increasing wherever it is nonpositive. Suppose $\alpha_2(s) \le 0 < \alpha_2(t)$ for some $s < t$. By continuity there is $u \in [s, t)$ with $\alpha_2(u) = 0$ and $\alpha_2 > 0$ on $(u, t]$. From $\alpha_2(u) = 0$ and orderedness, $\alpha_1(u) \ge 0$, so $\alpha_2'(u) \le -\alpha_1(u) \le 0$; from $\alpha_2 > 0$ on $(u, t]$ the right derivative at u is ≥ 0. Hence $\alpha_2'(u) = 0$ and $\alpha_1(u) = 0$, so both arms vanish at u, contrary to hypothesis. Therefore $\{\alpha_2 > 0\}$ is downward closed, that is $E_2 = [0, \tau_2)$.

On E_2 we have $\alpha_2 > 0$, so $\alpha_1' \ge \alpha_2 > 0$ and α_1 is strictly increasing there. Since $E_1 \subseteq E_2$, the set $\{\alpha_1 < 0\}$ is an initial segment of E_2, so $E_1 = [0, \tau_1)$ with $\tau_1 \le \tau_2$. □

The rest of this section determines when the sign sets are ordered. The organising constant turns out to be $\sqrt{3} - 1$, and the results below establish, in order: the arm system exactly (Proposition 28); that orderedness is a race between the two arms (Proposition 29); three successively better sufficient thresholds, 1, 0.899266 and $\frac{3}{4}$, the last certified in exact rational arithmetic (Corollaries 30, 31, 37); that $\alpha_2(\pi/2)$ is pinned at $-\frac{1}{2}$ for every admissible cap (Proposition 33); that no threshold of this form can beat $\sqrt{3} - 1$ (Corollary 35); and a case split (Lemma 36) reducing the converse to two linear programs, of which one is now solved in closed form. The ledger of what is proved and what is not is given after Lemma 36.

Proposition 28 (The arm system, exactly)**.** *Let H satisfy (RC) with absolutely continuous part $r = (H + H'')_{ac}$ and an atom of mass a at $\pi/2$. Then on $(0, \pi/2)$,*

$$\alpha_1' = \alpha_2 + 1 - r(t), \qquad \alpha_2' = -\alpha_1 - 1 + r(t + \tfrac{\pi}{2}),$$

with $\alpha_1(0) = -\frac{1}{2}$ and $\alpha_2(0) = \int_0^{\pi/2} r \sin - 1 + a$.

Proof. Write $S(x) = \int_0^x r(s)\sin(x-s)ds$ and $C(x) = \int_0^x r(s)\cos(x-s)ds$, so that $S' = C$ and $C' = -S + r$. Solving $H'' + H = r + a\delta_{\pi/2}$ and substituting, $\alpha_1(t) = S(t+\frac{\pi}{2}) + a\sin t - 1 - C(t)$ and $\alpha_2(t) = S(t) - 1 + C(t+\frac{\pi}{2}) + a\cos t$; the $H(0)$ terms cancel from both. Differentiating and using $S' = C$, $C' = -S + r$ gives the two identities. $\square$

This is Lemma 132 with the remainders identified: (RC) says exactly $0 \le r \le 1$, so $\alpha_2 \le \alpha_1' \le \alpha_2 + 1$ and $-\alpha_1 - 1 \le \alpha_2' \le -\alpha_1$ follow at once, and both are sharp, the extremes being attained at $r \equiv 1$ and $r \equiv 0$. The system is linear with a $[0,1]$-valued coefficient, so the whole sign-structure question is a reachability problem for a planar control system, which is what makes the next statement possible.

Proposition 29 (Orderedness is a race)**.** *Under (RC) the sign sets are ordered if and only if $\alpha_2(0) > 0$ and α_1 reaches 0 no later than α_2 does.*

Proof. $\alpha_1(0) = -\frac{1}{2} < 0$, so $0 \in E_1$ always and orderedness forces $0 \in E_2$, that is $\alpha_2(0) > 0$. While $\alpha_2 > 0$, Proposition 28 and $r \le 1$ give $\alpha_1' \ge \alpha_2 > 0$, so α_1 increases strictly. If it reaches 0 first then $\alpha_1 \ge 0$ from then on until α_2 vanishes, and Proposition 27 keeps $\alpha_2 \le 0$ thereafter, so E_1 never leaves E_2. If instead α_2 reaches 0 first, at that point $\alpha_1 < 0$ and $\alpha_2 \le 0$, so ordering fails there. $\square$

Corollary 30 (A sufficient condition)**.** *If $\alpha_2(0) \ge 1$, equivalently $a \ge 2 - \int_0^{\pi/2} r\sin$, the cap is ordered.*

Proof. While $\alpha_1 \le 0$ and $\alpha_2 > 0$ we have $\alpha_2' = -\alpha_1 - 1 + r \ge -1$, so $\alpha_2(t) \ge \alpha_2(0) - t$, and then $\alpha_1' \ge \alpha_2 \ge \alpha_2(0) - t$ gives $\alpha_1(t) \ge -\frac{1}{2} + \alpha_2(0)t - \frac{1}{2}t^2$. At $t = \min(\alpha_2(0), \pi/2)$ the right side is at least $\frac{1}{2}(\alpha_2(0)^2 - 1) \ge 0$. $\square$

The constant 1 is not sharp, and the reason is that Corollary 30 throws away the only constraint r carries. Since cos decreases on $[0, \pi/2]$,

$$\cos t \int_0^t r \;\le\; \int_0^t r\cos \;\le\; \int_0^{\pi/2} r\cos = \tfrac{1}{2}, \qquad \text{so} \qquad \int_0^t r \;\le\; \min\Big(t,\ \frac{1}{2\cos t}\Big).$$

Feeding this into $\alpha_1' = \alpha_2 + 1 - r$ in place of $r \le 1$ gives

$$\alpha_1(t) \;\ge\; -\tfrac{1}{2} + \alpha_2(0)\,t - \tfrac{1}{2}t^2 + t - \min\Big(t, \frac{1}{2\cos t}\Big),$$

and requiring the right side to be nonnegative for some $t \le \alpha_2(0)$ lowers the sufficient threshold to 0.899266, attained at $t^* = 0.8967$, where the constraint binds ($1/(2\cos t^*) = 0.8010 < t^*$).

Corollary 31 (Sharpened)**.** *If $\alpha_2(0) \ge 0.899266$ the cap is ordered.*

Corollaries 30 and 31 are superseded by Corollary 37, which certifies the rational threshold $\frac{3}{4}$, and the sharp value is $\sqrt{3} - 1 = 0.7321$ by Corollary 35 and the argument after Lemma 36. They are kept because their proofs are two lines of calculus with no computation of any kind, so they are the cheapest correct statements in this section.

At Σ, $\alpha_2(0) = 0.7506$, so even the sharpened threshold does not cover Σ, and this is stated rather than glossed. Two independent adversarial searches over admissible controls — unrestricted noise, one- and two-sided bang-bang, Fourier profiles, then a multi-switch bang-bang family refined by local search — find admissible caps defeating $\alpha_2(0)$ up to 0.6215 and no further, the second search improving on the first by 0.0005 in 3700 additional evaluations. So the sharp threshold lies in [0.6215, 0.899266], the lower end is probably the truth, and Σ at 0.7506 would then be covered with room. Rule 7: the interval is proved at the upper end only; 0.6215 is a measurement and the sharp value is not known.

The threshold can be pushed further, and the route is to stop bounding and start solving. Writing $Z = \alpha_1 + i\alpha_2$, Proposition 28 is $Z' + iZ = (1 - r(t)) - i(1 - r(t + \frac{\pi}{2}))$, so with $u = 1 - r(t)$ and $v = 1 - r(t + \frac{\pi}{2})$, both in $[0, 1]$,

$$\alpha_1(t) = -\tfrac{1}{2}\cos t + c\sin t + \int_0^t \big[u\cos(t-s) - v\sin(t-s)\big]\,ds, \quad \alpha_2(t) = c\cos t + \tfrac{1}{2}\sin t - \int_0^t \big[u\sin(t-s) + v\cos(t-s)\big]\,ds,$$

exactly, where $c = \alpha_2(0)$; and the two constraints become $\int_0^{\pi/2} u\cos = \frac{1}{2}$ and $\int_0^{\pi/2} v\sin = \frac{1}{2}$. Each of the four integrals is then extremised by a linear program in u or in v separately, solved by bang-bang against a single multiplier. Requiring $\alpha_2 > 0$ up to some T and $\alpha_1(T) \geq 0$, both in the worst case, gives a sufficient threshold

$$c \;\geq\; c_{\mathrm{LP}}, \qquad c_{\mathrm{LP}} \approx 0.7484,$$

against $\alpha_2(0) = 0.7506$ at Σ.

The reduction to the linear programs is a proof, and so, it turns out, is the value. The four programs have *closed-form* solutions, because the moment constraints pin the bang-bang switch points independently of t: maximising $\int_0^t u\sin(t-s)ds$ puts $u = 1$ on $[0, \sigma)$ with $\sin\sigma = \frac{1}{2}$, so $\sigma = \pi/6$ always, and $U_2(t) = \cos(t - \pi/6) - \cos t$ for $t \geq \pi/6$; the two v-programs switch at $\pi/3$ (from $1 - \cos\sigma = \frac{1}{2}$) and the remaining u-program at $\arcsin(\sin t - \frac{1}{2})$. Substituting, and using $\cos(s - \pi/6) = \frac{\sqrt{3}}{2}\cos s + \frac{1}{2}\sin s$, the two requirements collapse to elementary inequalities.

Theorem 32 (Ordering certificate)**.** *Let H satisfy (RC) with the forced boundary data, and let $c = \alpha_2(0)$. Suppose there is $T \in (\pi/6, \pi/3)$ with*

$$\text{(i)}\quad \tan T < c + 1 - \tfrac{\sqrt{3}}{2}, \qquad \text{(ii)}\quad \sin T\Big[(c+1) + \cos T - \sqrt{1 - (\sin T - \tfrac{1}{2})^2}\,\Big] \;\geq\; 1,$$

and $(c+1)\frac{\sqrt{3}}{2} \geq \frac{5}{4}$. Then the sign sets are ordered, hence by Proposition 27 anchored. Both conditions are increasing in c, so they certify every larger c as well.

Proof. Condition (i) is exactly positivity of the worst-case bound $\alpha_2 \geq (c + 1 - \frac{\sqrt{3}}{2})\cos s - \sin s$ on $[\pi/6, T]$; the third hypothesis is its positivity on $[0, \pi/6]$, where the bound is $(c+1)\cos s - \frac{1}{2}\sin s - 1$, decreasing, with minimum at $\pi/6$. Condition (ii) is the worst-case bound on α_1 at T, after the $\frac{1}{2}\cos T$ terms cancel. So $\alpha_2 > 0$ on $[0, T)$ and $\alpha_1(T) \geq 0$, and Proposition 29 applies. ☐

Proposition 33 (The right endpoint is rigid)**.** *For every cap satisfying (RC) and the forced boundary data, $\alpha_2(\pi/2) = -\frac{1}{2}$. Consequently orderedness forces $\alpha_1(\pi/2) \geq 0$.*

Proof. By Proposition 28, $\alpha_2(\pi/2) = \frac{1}{2} - \int_0^{\pi/2}\big[u\cos s + v\sin s\big]ds$, and the two forced moments are exactly $\int_0^{\pi/2} u\cos = \frac{1}{2}$ and $\int_0^{\pi/2} v\sin = \frac{1}{2}$, so the integral is 1 and $\alpha_2(\pi/2) = -\frac{1}{2}$. Since $\alpha_2(\pi/2) < 0$, the point $\pi/2$ is outside E_2, so orderedness puts it outside E_1 as well. ☐

Proposition 34 (The race in one variable)**.** *Write* $W(t) = e^{it}Z(t)$ *with* $Z = \alpha_1 + i\alpha_2$*. Then* $\operatorname{Re} W$ *is non-decreasing, with* $\operatorname{Re} W(0) = -\frac{1}{2}$ *and* $\operatorname{Re} W(\pi/2) = \frac{1}{2}$*; and at any zero* $\tau \in [0, \pi/2)$ *of* α_2*,*

$$\alpha_1(\tau) = \frac{\operatorname{Re} W(\tau)}{\cos \tau}.$$

Consequently $\tau \geq t_0$ *is* necessary *for orderedness, where* τ *is the first zero of* α_2 *and* t_0 *is the unique zero of* $\operatorname{Re} W$*.*

Proof. $W' = e^{it}(u - iv)$ with $u, v \in [0, 1]$, so $\operatorname{Re} W' = u \cos t + v \sin t \geq 0$ on $[0, \pi/2]$. At $t = 0$, $W(0) = Z(0)$ and $\operatorname{Re} Z(0) = \alpha_1(0) = -\frac{1}{2}$; at $t = \pi/2$, $Z = -iW$, so $\operatorname{Re} W(\pi/2) = -\alpha_2(\pi/2) = \frac{1}{2}$ by Proposition 33. At a zero of $\alpha_2 = \operatorname{Im} Z$ we have $\operatorname{Im} W \cos \tau = \operatorname{Re} W \sin \tau$, whence $\alpha_1(\tau) = \operatorname{Re} W \cos \tau + \operatorname{Im} W \sin \tau = \operatorname{Re} W \big(\cos \tau + \frac{\sin^2 \tau}{\cos \tau}\big) = \operatorname{Re} W(\tau)/\cos \tau$. Since $\operatorname{Re} W$ runs monotonically from $-\frac{1}{2}$ to $\frac{1}{2}$ its zero t_0 is unique, so the sign of α_1 at the first zero of α_2 is the sign of $\tau - t_0$, and orderedness forces it to be nonnegative. □

The condition is necessary and *not* sufficient, which is worth stating plainly because the converse is tempting. Orderedness needs $\alpha_1 \geq 0$ on the whole of $\{\alpha_2 \leq 0\}$, not merely at its first point, and α_1 can descend after τ: $\alpha_1' = \alpha_2 + 1 - r$ is negative whenever $\alpha_2 < -(1 - r)$, which is available once α_2 has gone negative. The floor witness at $c = 0.7315$ is exactly such a case, with $\tau = 0.7134 \geq t_0 = 0.4240$ and yet $\alpha_1(\pi/2) = -0.00055 < 0$ against $\alpha_2(\pi/2) = -\frac{1}{2}$: it passes the test at τ and fails at the endpoint.

What the proposition does give is a clean necessary condition in one variable, and a diagnosis of where the difficulty sits. Maximising $\int_0^t (v \cos - u \sin)$ over admissible controls subject to $t \leq t_0$ — itself the linear constraint $\int_0^t (u \cos + v \sin) \leq \frac{1}{2}$ — gives 0.6854, so for $c \geq 0.6854$ the necessary condition holds automatically and carries no information. Since $0.6854 < \sqrt{3} - 1$, the whole difficulty of the band lies in the behaviour of α_1 *after* τ, not before it.

Corollary 35 (A hard floor for the threshold)**.** *No condition of the form "*$\alpha_2(0) \geq c$ *implies ordered" can hold with* $c < \sqrt{3} - 1 = 0.7320508\ldots$

Proof. $\alpha_1(\pi/2) = c + \int_0^{\pi/2} \big[u \sin s - v \cos s\big]\, ds$. Minimising the two terms separately, each a one-constraint program solved bang-bang: $\int u \sin$ is least when $u = 1$ on $[0, \pi/6]$, giving $1 - \frac{\sqrt{3}}{2}$, and $\int v \cos$ is greatest when $v = 1$ on $[0, \pi/3]$, giving $\frac{\sqrt{3}}{2}$; the switch points are forced by $\sin \sigma = \frac{1}{2}$ and $1 - \cos \sigma = \frac{1}{2}$. The minimum is $1 - \sqrt{3}$ and it is attained, by a single admissible pair. So for $c < \sqrt{3} - 1$ there is a cap with $\alpha_1(\pi/2) < 0$ while $\alpha_2(\pi/2) = -\frac{1}{2} < 0$ by Proposition 33, which is unordered. □

The witness is explicit and its two integrals are exact: $u = 1$ on $[0, \pi/6]$ gives $\int \cos = \sin \frac{\pi}{6} = \frac{1}{2}$ and $\int \sin = 1 - \frac{\sqrt{3}}{2}$; $v = 1$ on $[0, \pi/3]$ gives $\int \sin = 1 - \cos \frac{\pi}{3} = \frac{1}{2}$ and $\int \cos = \frac{\sqrt{3}}{2}$. All four are formalised. Sweeping 3600 two-block bang-bang caps with both moments exact finds none that defeats a larger c: the floor $\sqrt{3} - 1$ is attained only by this one, and only at $T = \pi/2$.

Together with the certificate this brackets the sharp threshold in $[\sqrt{3} - 1,\ 0.748424]$, of width 0.0164. That replaces the earlier bracket $[0.6215,\ 0.899266]$, whose lower end was a search and whose upper end was an inequality chain; both ends are now exact, and the lower end is a proof rather than a measurement. Σ sits at $2a_1 - 1 = 0.7506$, above the whole bracket.

Lemma 36 (Unorderedness forces $\operatorname{Im} W < 0$)**.** *If* $\alpha_1(T) < 0$ *and* $\alpha_2(T) \leq 0$ *for some* $T \in (0, \pi/2)$*, then* $\operatorname{Im} W(T) < 0$*.*

Proof. Write $P = \operatorname{Re} W(T)$, $Q = \operatorname{Im} W(T)$. From $\alpha_2 \le 0$, $Q \cos T \le P \sin T$; multiplying by $\cos T > 0$ gives $Q \cos^2 T \le P \sin T \cos T$. From $\alpha_1 < 0$, $P \cos T < -Q \sin T$; multiplying by $\sin T \ge 0$ gives $P \sin T \cos T \le -Q \sin^2 T$. Chaining, $Q \cos^2 T \le -Q \sin^2 T$, so $Q \le 0$, and the middle inequality is strict unless $\sin T = 0$. □

So unorderedness splits on the sign of $P = \operatorname{Re} W(T)$, which by Proposition 34 is the sign of $T - t_0$:

(A) $T \ge t_0$ and $\alpha_1(T) < 0$;

(B) $T < t_0$ and $\alpha_2(T) \le 0$, which by Lemma 36 needs $\int_0^T (v \cos - u \sin) > c$ subject to $\int_0^T (u \cos + v \sin) \le \frac{1}{2}$.

Branch (B) has a closed form. Both the objective and the constraint push u out of $[0, T]$, so u decouples: it is 1 on $[0, \omega]$ with $\sin \omega = k := \max(0, \sin T - \frac{1}{2})$, the least mass the global moment permits. The residual budget for v is then $\frac{1}{2} - k$, and v is bang-bang on $[0, \sigma]$ with $\cos \sigma = \frac{1}{2} + k$, truncated at T. So

$$B(T) = \sin\bigl(\min(\sigma, T)\bigr) - 1 + \sqrt{1 - k^2},$$

and the two caps coincide when $\arccos(\sin T) = T$, that is $\sin T = \cos T$, at $T = \pi/4$. There

$$B_{\max} = \frac{\sqrt{2}}{2} - 1 + \sqrt{\tfrac{1}{4} + \tfrac{\sqrt{2}}{2}} = 0.685425125\ldots,$$

agreeing with the discretised program to $4 \cdot 10^{-8}$. Branch (B) is therefore empty whenever $c \ge B_{\max}$, and $\frac{\sqrt{2}}{2} + \sqrt{\frac{1}{4} + \frac{\sqrt{2}}{2}} < \sqrt{3}$ holds with slack 0.0466, checked in exact rational arithmetic. So branch (B) is empty throughout the band, and this is now PROVED, not measured. Branch (A) is a linear program in a single control, minimising $\alpha_1(T)$ subject to $\operatorname{Re} W(T) \ge 0$; over $T \in [t_0, \pi/2]$ its value is smallest at $T = \pi/2$, where the constraint is vacuous ($\operatorname{Re} W(\pi/2) = \frac{1}{2}$) and the minimum is exactly $c + 1 - \sqrt{3}$ by the computation of Corollary 35. At $c = \sqrt{3} - 1$ the minimum over the whole range is $-8.5 \cdot 10^{-8}$, attained at $T = \pi/2$; the value moves with slope exactly 1 in c, as it must if $T = \pi/2$ is the binding case.

Taken together this says $\sqrt{3} - 1$ is *sharp*: above it every admissible cap is ordered, below it Corollary 35 exhibits one that is not. Rule 0: the case split and branch (A) at $T = \pi/2$ are PROVED; that $T = \pi/2$ minimises branch (A), and the constant 0.6854 bounding branch (B), are HEURISTIC, being values of discretised linear programs. The sharpness statement therefore carries the weaker label until those two are certified in exact arithmetic, which is a well-defined task and not carried out here. Corollary 37, at the rational threshold $\frac{3}{4}$, remains what the note certifies.

Corollary 37 (The certified class)**.** *Every admissible cap with $\alpha_2(0) \ge \frac{3}{4}$ is ordered and anchored, and Σ is one of them.*

Proof. For the threshold, take $c = \frac{3}{4}$ and $T = 723/1000$. Bounding sin and cos by alternating Taylor truncations ending on a term of the safe sign, and $\sqrt{\cdot}$ from above by $\sqrt{z} \le (z + q^2)/2q$ at $q = 98685/100000$, the three hypotheses of Theorem 32 hold with slacks $1.19 \cdot 10^{-3}$, $2.66 \cdot 10^{-1}$ and $1.04 \cdot 10^{-3}$. Every comparison is between rationals.

The trigonometric enclosures are derived rather than assumed. Mathlib's bound $|\sin x - (x - \frac{x^3}{6})| \le |x|^5/100$ applied at T itself is too weak, giving $\sin T \le 0.66199$ where 0.661687 is needed;

applied at the half-argument 723/2000 it shrinks the error by 2^5, and the double-angle formulas carry the bounds back, with $\cos(723/2000)$ recovered from $\sin^2 + \cos^2 = 1$ and positivity and the square root cleared by $\sqrt{z} \geq z/q$ for $q \geq \sqrt{z}$. This gives $\sin T \in [0.661418, 0.661687]$ and $\cos T \geq 0.749809$, enough for both conditions. The chain is formalised end to end.

That Σ lies in the class needs no decimal at all. Writing p, q for the two cube roots in $a_1 = \frac{1}{4}\sqrt{4 + \sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}}}$, we have $(pq)^3 = 71^2 - 128 = 4913 = 17^3$, so $pq = 17$ exactly and $S = p + q$ satisfies $S^3 = 51S + 142$. At $S = \frac{33}{4}$ the cubic equals $-\frac{79}{64} < 0$, and it increases for $S \geq 8$, so its root satisfies $S > \frac{33}{4}$. Hence $4 + S > \frac{49}{4}$, $a_1 > \frac{7}{8}$, and $\alpha_2(0) = 2a_1 - 1 > \frac{3}{4}$. ☐

This is what the sign-structure question was for. On the class $\alpha_2(0) \geq \frac{3}{4}$ the concavity of Theorem 122 applies with no sign-pattern hypothesis at all, because the sign pattern is now *derived*: ordered by Corollary 37, anchored by Proposition 27. The largest admissible witness is $T = \arctan(2a_1 - \frac{\sqrt{3}}{2}) = 0.7242$, and the certificate is not tight: the decoupled linear programs place their threshold at 0.748424, and the certificate reaches $\frac{3}{4}$, just below it. Corollary 35 bounds the sharp threshold below by $\sqrt{3} - 1$, so the certificate is within 0.018 of the best possible condition of this form. Rule 0: Corollary 37 is PROVED in exact arithmetic; the sharp threshold itself is HEURISTIC and is not needed.

Σ itself wins its race comfortably, α_1 reaching 0 at $t = \beta = 0.2897$ against α_2 at $t = \pi/2 - \beta = 1.2811$; the certificate above is far more conservative than that, and still covers it. Inside the band, 12500 adversarial trials at five values of c from 0.7325 to 0.7499 — unrestricted noise, one-sided bang-bang, and Fourier profiles, all projected onto both moments exactly — produce no unordered cap, while the same test applied to the floor witness brackets $\sqrt{3} - 1$ correctly, unordered at $c = 0.7320$ and ordered at $c = 0.7325$. So the sharp threshold is plausibly $\sqrt{3} - 1$ itself and the certificate's $\frac{3}{4}$ is the conservative end. Rule 7: that paragraph is HEURISTIC; the floor is PROVED.

What remains open is the narrow band $[\sqrt{3} - 1, \frac{3}{4})$. Below $\sqrt{3} - 1$ the question is settled the other way by Corollary 35: unordered admissible caps exist, and for those the reduction of Lemma 121 does not apply, so no condition on $\alpha_2(0)$ alone can rescue them.

Proposition 38 (Ordering forces a corner)**.** *Let H satisfy (RC), with absolutely continuous part $r = (H + H'')_{ac} \leq 1$ and an atom of mass $a \geq 0$ at $\pi/2$. If the sign sets are ordered then*

$$a \;\geq\; 1 - \int_0^{\pi/2} r(s) \sin s \, ds \;\geq\; 0,$$

with the second inequality strict unless $r = 1$ almost everywhere on $[0, \pi/2]$. So every ordered cap other than that extreme one has a genuine corner at $\pi/2$.

Proof. Since $H(\pi/2) = 1$ and $H'(0) = \frac{1}{2}$, the first arm at the left endpoint is $\alpha_1(0) = H(\pi/2) - 1 - H'(0) = -\frac{1}{2} < 0$, so $0 \in E_1$ always. Orderedness gives $0 \in E_2$, that is $\alpha_2(0) > 0$. Solving $H'' + H = r + a\,\delta_{\pi/2}$ from $H'(0) = \frac{1}{2}$ and evaluating,

$$H'(\pi/2^+) = -H(0) + \int_0^{\pi/2} r(s) \sin s \, ds + a,$$

so that $\alpha_2(0) = H(0) - 1 + H'(\pi/2^+) = \int_0^{\pi/2} r(s) \sin s \, ds - 1 + a$; the $H(0)$ terms cancel, as they do in both arms, so $H(0)$ is not a parameter of the sign structure at all. Under (RC), $\int_0^{\pi/2} r \sin \leq \int_0^{\pi/2} \sin = 1$, with equality only for $r = 1$ a.e. ☐

At Σ the three numbers are $\int_0^{\pi/2} r\sin = 0.5835$, $a = 1.1670$ and $\alpha_2(0) = 0.7506$, so the necessary condition $a \geq 0.4165$ holds with room. The atom is the jump $G'(0) - F'(\pi/2)$ between the phases meeting at $\pi/2$.

The proposition is what makes the anchoring result non-vacuous, and it is worth being explicit about why. Without an atom, (RC) alone forces $\alpha_2(0) \leq 0$ while $\alpha_1(0) = -\frac{1}{2} < 0$, so *no* cap is ordered and Proposition 27 would be a statement about the empty set. The atom is not a technicality permitted by (RC); it is the only thing that allows an ordered cap to exist.

What this settles. The unanchored pattern of Remark 39, $E_1 = E_2 = [0.4, 1.2]$, is *ordered* (the sets are equal), and it is exactly a pattern on which the form is not negative, $\sup \frac{1}{2}\delta^2 Q/\|\eta\|^2 = +0.4123$. Proposition 27 says no cap satisfying (RC) realises an ordered unanchored pattern. So that counterexample is not a counterexample to anything about competitors: it is a sign pattern the admissibility condition already forbids. Within the ordered class, the restriction to anchored cells in Theorem 115 is not a restriction on the competitor class at all, only on the bookkeeping.

What this does not settle, stated precisely because the measurement is unfavourable. Orderedness is a hypothesis of Proposition 27, and it is also a hypothesis of the reduction (Lemma 121) that sends every ordered cell to the diagonal family. It is not implied by (RC): by Proposition 38 it additionally forces an atom of mass at least $1 - \int_0^{\pi/2} r\sin$, and a generic (RC) cap has no atom at all. Sampling the curvature radius in $[0, 1]$ directly, so that (RC) holds by construction rather than by rejection, and imposing the forced boundary data $H'(0) = \frac{1}{2}$, $H'(\pi) = -\frac{1}{2}$, $H(\pi/2) = 1$, 2322 of 3000 caps have both sign sets nonempty and none of them is ordered, exactly as Proposition 38 predicts.

Perturbing Σ's own curvature radius instead, inside the (RC) box and with its atom left intact, 1475 ordered caps arise across amplitudes up to 0.4, and none of them is unanchored. That is the regression test for Proposition 27 with content in it: the ordered class is a neighbourhood of Σ, not the single point Σ, and the proposition holds throughout it.

The remaining gap is therefore the unordered cells, and it is a gap in the reduction, not in the concavity: Lemma 121 needs $E_1 \subseteq E_2$ before Theorem 122 can be applied at all. Closing it needs sofa-admissibility itself, not the curvature condition extracted from it, and that is open. Rule 0: the sampling paragraphs are HEURISTIC; Proposition 27 is PROVED and formalised; "admissible $\Rightarrow$ ordered" is CONJECTURE with no proof and no supporting measurement at the width tested.

Remark 39 (Measurements, and exactly what they do and do not give)**.** In a hat basis with $\eta(0) = \eta(\pi/2) = 0$:

Criticality. $\delta Q[\eta] = 0$ in every direction, $\|\delta Q\|_\infty / h = 1.8 \cdot 10^{-8}$ by central differences on Q itself. So Romik's ODEs are the Euler–Lagrange equations of Q. (An analytic assembly of the gradient first reported a nonzero value on $[0, \beta)$ and $[\pi/2, \pi/2 + \beta)$; that was a sign error on $\delta\left(-\frac{1}{2}(\alpha_1^-)^2\right) = +\alpha_1^- \delta\alpha_1$, caught by the finite-difference check.)

Concavity, on Σ's cell only. With Σ's own sign pattern $E_1 = [0, \beta)$, $E_2 = [0, \pi/2 - \beta)$, the form (24) is negative definite: $\sup \frac{1}{2}\delta^2 Q[\eta]/\|\eta\|^2_{L^2(0,\pi)} = -0.7358, -0.7339, -0.7331$ at $m = 32, 64, 128$, against the exact mass matrix. Under Baek's injectivity condition, which is exactly $E_1 = \emptyset$, $E_2 = [0, \pi/2]$, it is -0.8751. But (24) is *not* negative for every sign pattern: with $E_1 = E_2 = [0.4, 1.2]$ it is $+0.4123$, and with $E_1 = E_2$ a union of three intervals $+0.4225$. An earlier scan over intervals *anchored at* 0 suggested that $\tau_1 \leq \tau_2$ suffices; the unanchored cases refute that and the suggestion is **retracted**.

Since α_1, α_2 are affine in H, each pointwise sign condition is a half-space and Σ's cell is convex.

Concavity plus criticality on a convex set gives that Σ maximises Q *on that cell.* Nothing global follows.

What is missing, and it is worse than missing. All of the above concerns Q alone. The inequality $Q \geq |\mathrm{sofa}|$, which is the only reason Q is relevant, **fails**; see Proposition 41. All numbers in this remark are floating point.

Remark 40 (The boundary of the cap, and where the sweep terminates)**.** The density $H + H''$ of the surface measure determines ∂C_2 completely: it vanishes on $(-\beta, \beta)$, where ∂C_2 is the single vertex $P = (1, \frac{1}{2})$; equals $\frac{1}{2}$ on $(\pi/2 - \beta, \pi/2 + \beta)$, where ∂C_2 is a circular arc of radius exactly $\frac{1}{2}$ centred at $(1 - 2a_1, \frac{1}{2})$; has an atom of mass 1.167050 at $\theta = \pi/2$, the corridor ceiling $y = 1$, a facet; vanishes again on $(\pi - \beta, \pi]$, the second vertex; and is a smooth arc in between. The ρ-image of the ceiling facet is the corridor floor, the segment $[-0.750575, 0.416475] \times \{0\}$, whose left endpoint lies directly below the centre of the radius-$\frac{1}{2}$ arc.

This settles the geometry of the outer arm of Proposition 20. Because C_2 is *convex*, the face-1 segment lies in C_2 as soon as its far endpoint does, so the claim $S_1 = \sigma$ reduces to a one-parameter containment: the endpoint $x(t) = c_x(t) + \sigma(t) \sin t$ must lie in the floor facet. It runs monotonically from $x(0) = 0$ to $x(\pi/2) = 0.416475$, the right end of the facet, and

$$x(\pi/2) = c_x(\pi/2) + \alpha_1(\pi/2) = \big(c_x(0) - \alpha_2(0)\big) + \big(\alpha_2(0) + \alpha_1(\pi/2) - (G(\pi/2) - 1)\big)$$

is an algebraic identity, the right-hand side being the left end of the facet plus its length $H'(\pi/2^+) - H'(\pi/2^-)$. So the tangency at $t = \pi/2$ is exact, not coincidental, and what remains of $S_1 = \sigma$ is the monotonicity of the explicit function x.

Proposition 41 (V over-estimates the niche, so Q is not an upper bound)**.** *For every admissible H, $V(H) \geq |N(H)|$. Consequently $Q = |C_2| - 2V \leq |C_2| - 2|N|$, and for the maximal sofa $T_{\max} = C_2 \setminus (U \cup \rho U)$ with that cap, $|T_{\max}| = |C_2| - 2|N| \geq Q$. So Q is an upper bound for $|T_{\max}|$ only where $V = |N|$.*

Proof. V is precisely the flux of the advancing part of ∂Q_t, by Lemma 19 and the Jacobian computation in Proposition 20. By Reynolds' transport theorem the area gained by the sweeping region $N_T = \bigcup_{t \leq T} Q_t \cap C_2$ satisfies $\frac{d}{dT}|N_T| \leq \int_{\text{advancing part of } \partial Q_T} v_n \, ds$, because advance into already-swept territory contributes to the flux but gains no area. Integrating over $T \in [0, \pi/2]$ gives $|N| \leq V$. ☐

Remark 42 (The moving frame: hypotheses (ii) and (iii) are one hypothesis)**.** Hypothesis (i) is proved for one corner by Baek [1]. The other two have no prior art. Passing to the moving frame reduces all three to a single statement and settles (iii).

Write $\mu_t = (\cos t, \sin t)$, $\nu_t = (-\sin t, \cos t)$. Differentiating $c = (F - 1)\mu_t + (G - 1)\nu_t$ gives

$$c'(t) = -\alpha_1(t)\, \mu_t + \alpha_2(t)\, \nu_t,$$

so the arms are the corner velocity resolved in the moving frame. Fix p and set $u(t) = \langle c(t) - p, \mu_t \rangle$, $v(t) = \langle c(t) - p, \nu_t \rangle$. Orthonormality collapses these to $u = F - 1 - \langle p, \mu_t \rangle$ and $v = G - 1 - \langle p, \nu_t \rangle$, whence

$$u' = v - \alpha_1(t), \qquad v' = -u + \alpha_2(t) :$$

plane rotation forced by the arm pair. The corner covers p exactly when $u > 0$ and $v > 0$; write $T(p)$ for that set of times.

The sweeps $\Phi(s,t) = c - s\mu_t$ and $\Psi(s,t) = c - s\nu_t$ have Jacobians $\alpha_2 - s$ and $s - \alpha_1$, which is why Proposition 20 truncates at $[0, \alpha_2^+]$ and $[\alpha_1^+, \sigma]$. On the face-2 edge $v = 0$, $u = s$ the system gives $v' = \alpha_2 - s$, and on the face-1 edge $u = 0$, $v = s$ it gives $u' = s - \alpha_1$. So each truncation window is precisely the set of boundary points at which the trajectory is entering $T(p)$; the remaining limits are $s \geq 0$ and, since $c - \sigma\nu_t = (F-1)\sec t\,(1,0)$ lies on the floor, $s \leq \sigma$ iff $p_y \geq 0$. The area formula applied to the two truncated sweeps therefore gives

$$V = \int_{\mathbb{R}^2} \#\{\text{entries of } T(p)\}\, dA = \int_N \#\{\text{entries}\}\, dA + \int_{\mathbb{R}^2 \setminus C_2} \#\{\text{entries}\}\, dA,$$

while $|N| = \int_N 1\, dA$. So *$T(p)$ is a nonempty interval for almost every $p \in N$* controls the first term only: it says covering is "at least one entry" and that injectivity and disjointness together are "at most one entry". It says nothing about the second term, which is the sweep mass that leaves the cap. Equality $V = |N|$ needs, in addition, a containment statement: *the truncated sweeps lie in C_2*. That statement is not implied by the disjointness, injectivity and covering hypotheses alone, as Remark 44 shows by counterexample, and it is *not* implied by (55) either (Remark 51). Proposition 20 therefore assumes it outright. What Remark 45 supplies is a sufficient condition for the half of it that concerns C, namely $\alpha_1\alpha_2 \leq 1$ with the two end conditions.

This proves (iii). Normalise so that $G(0) = H(\pi/2) = 1$, which is a translation and changes no area and neither arm. Then $c(0) = (H(0) - 1, 0)$ sits on the floor and its quadrant meets the cap only in $\{p_y = 0\}$, so no point of N is cut at $t = 0$. For $p \in N$ pick $t^* \in T(p)$ and let $t_1 = \sup\{t \leq t^* : t \notin T(p)\}$. Then $\min(u,v)(t_1) = 0$ with $\min(u,v) > 0$ on $(t_1, t^*]$, so the vanishing coordinate has nonnegative derivative at t_1, which is exactly the window inequality. Thus p is a value of one of the two truncated sweeps.

What remains open is only that $T(p)$ is *connected*. In polar form $z = u + iv = re^{i\psi}$ the system reads $\psi' = (\alpha_2 \cos\psi + \alpha_1 \sin\psi)/r - 1$, so $\psi' > 0$ exactly inside the circle through the origin with centre $(\alpha_2/2, \alpha_1/2)$; outside it ψ decreases and $T(p)$ is an interval by monotonicity. Re-entry therefore requires the trajectory to pass within $|c'(t)|$ of the corner, and the whole difficulty sits there.

Measured (`sofa_cut`, 400^2 points, 2000 times): every sampled cell of N has exactly one component; none is cut at $t = 0$; every entry offset lands in its own window, the worst miss being zero. The same run gives $V = 0.184199$, against the accurate $V = 0.1841931\ldots$ of Remark 22. That agreement needs a stable evaluation: $\sigma = (F-1)\tan t + G - 1$ multiplies a vanishing factor by a diverging one near $\pi/2$, and the cancellation made V drift with the time grid. Since $F + F'' = r$,

$$\tfrac{d}{dt}\big[(F-1)\sin t + F'\cos t\big] = (r(t) - 1)\cos t,$$

and the bracket vanishes at $\pi/2$, so exactly $\sigma - \alpha_1 = -\sec t \int_t^{\pi/2} (r(s) - 1)\cos s\, ds$, with no tangent and with numerator and denominator vanishing to matched order. The atom contributes nothing because $\cos(\pi/2) = 0$. Evaluated this way V is unchanged to six places across $n_t = 2000$, 3000 and 6000. The rasterised $|N| = 0.184506$ exceeds it, which is the raster's $O(h)$ upward bias on boundary cells and not a violation of Proposition 41; `sofa_cut` says so rather than quoting the ratio as a convergence check. Rule 0: (iii) is VERIFIED (`frame_ode_u`, `frame_ode_v`, `face1_window_iff_entry`, `face2_window_iff_entry`, `entry_deriv_nonneg`, `corner_start_on_floor`); (ii) is CONJECTURE, now in the sharp form "$T(p)$ is connected".

Remark 43 (The two arms are one function, and (ii) closes for $r < 1$)**.** Since $\nu_t = \mu_{t+\pi/2}$, the closed forms of Remark 42 read $u(t) = W(t)$ and $v(t) = W(t + \pi/2)$ for the single function

$$W(\theta) = H(\theta) - 1 - \langle p, \mu_\theta\rangle, \qquad \theta \in [0, \pi].$$

The two arms are one function at a quarter-period shift, so $T(p) = \{W > 0\} \cap (\{W > 0\} - \pi/2)$, an intersection of two sets. An intersection of intervals is an interval, so hypothesis (ii) follows once $\{W > 0\}$ meets each of $[0, \pi/2)$ and $(\pi/2, \pi]$ in an interval. It cannot be an interval across all of $[0, \pi]$: after the normalisation $H(\pi/2) = 1$ one has $W(\pi/2) = -p_y < 0$ for every p off the floor, so the atom of H'' at $\pi/2$ lies in the negative set and splits $\{W > 0\}$ in two. The split is forced.

Because $\langle p, \mu_\theta\rangle'' = -\langle p, \mu_\theta\rangle$, the point drops out of the second-order part and W solves a Hill equation whose forcing is the cap alone,

$$W'' + W = r(\theta) - 1, \qquad r = H + H''.$$

Let $(a, b) \subset (0, \pi/2)$ be a maximal interval on which $W < 0$, so $W(a) = W(b) = 0$. Pairing the equation with W and integrating by parts, the boundary terms vanish and

$$\int_a^b (r - 1)W = \int_a^b W^2 - \int_a^b (W')^2.$$

If $b - a \le \pi$, Wirtinger's inequality makes the right side nonpositive. If $r < 1$ on (a, b) the left side is an integral of a product of two negative functions, hence positive. So every negative hump of W has length exceeding π. A gap between two components of $\{W > 0\}$ inside $[0, \pi/2)$ would be such a hump, of length below $\pi/2$; there is none. The same argument runs on $(\pi/2, \pi]$. Hence $T(p)$ is an interval, which by Remark 42 is hypothesis (ii) together with the injectivity half of (i). It is not enough for $V = |N|$: see Remark 44.

The hypothesis is $r < 1$ away from the atom, and it holds for Σ with room: $\max r = 0.8388253$, a margin of 0.1611747, matching the closed form of Remark 82 exactly. It is an open condition, so it survives the perturbations $H + \varepsilon\eta$ of the second-variation analysis for ε small, provided the atom stays at $\pi/2$.

The Sturm step needs neither integration by parts nor Wirtinger. Anchor $G = W' \sin(\theta - x) - W \cos(\theta - x)$ at a point where $W > 0$; then $G(x) = -W(x) < 0$ and $G' = q \sin(\theta - x) < 0$, so G stays negative. If W dipped to 0 before a later point where $W > 0$, then at the first return t_1 one has $G(t_1) = W'(t_1) \sin(t_1 - x)$ with $\sin(t_1 - x) > 0$, forcing $W'(t_1) < 0$, while W rising to $W(t_1) = 0$ forces $W'(t_1) \ge 0$. Strict monotonicity of G is the only analytic input, and it follows from the sign of its derivative.

The hypothesis $r < 1$ is itself a theorem and not a measurement. On the two absolutely continuous phases the radius is $\frac{3}{4}(A \cos u + B \sin u)$ with $B = (1 - \sqrt{2})A$ and $A = F_1$, the phases differing only by a swap of A and B, which leaves $A^2 + B^2$ fixed. Since $A \cos u + B \sin u \le \sqrt{A^2 + B^2}$ and $A^2 + B^2 = A^2(4 - 2\sqrt{2})$, the radius is at most $\frac{3}{4}A\sqrt{4 - 2\sqrt{2}}$, and $A \le \frac{121}{100}$ puts that below 1. The middle phase is the constant $\frac{1}{2}$ and the end phases vanish. The value 0.8388253 is a sharper estimate of a quantity already known to lie below the threshold, not the reason it does. It is obtained by evaluating the closed form of r directly; an earlier version took second differences of an interpolant sampled below its own source grid, which diverges with the time resolution and at the finest grid tried overshot the threshold.

Rule 0: hypothesis (ii) is PROVED, and (iii) in the form "every point of N is a value of one of the truncated sweeps" is PROVED; the conclusion $V = |N|$ of Proposition 20 does *not* follow from them, by Remark 44. The supporting steps are VERIFIED (`arm_shift`, `point_part_harmonic`, `cut_set_ordConnected`, `exists_first_entry`, `covering_first_entry`, `exit_deriv_nonneg`, `negative_hump_impossible`, `pos_between`, `pos_set_ordConnected`), the curvature bound (`curvature_lt_one`, `curvature_lt_one'`) and the two sweep Jacobians that define the truncation windows (`sweep2_jacobian`, `sweep1_jacobian`). Two caveats on those. The connectivity lemmas assume W'' exists pointwise on the open window, which Σ fails: r jumps at β, inside $[0, \pi/2)$, so the lemmas apply to a C^2 surrogate and not to Σ's own W. The discarded route through $\int_a^b (r-1)W$ tolerates those jumps, an integral being indifferent to them, so the change to a pointwise Sturm argument was a regression in applicability. Restating the lemmas with q a measure and G monotone in the sense of bounded variation is the repair, and it is not done here.

Remark 44 (The containment gap, with a counterexample)**.** Proposition 20 is false as stated: its three hypotheses do not imply $V = |N|$. Take the stadium cap

$$H(\theta) = \tfrac{1}{2} + \tfrac{1}{2}\sin\theta + L|\cos\theta|, \qquad \theta \in [0, \pi],$$

the Minkowski sum of the disc of radius $\frac{1}{2}$ centred $(0, \frac{1}{2})$ with the segment $[-L, L] \times \{0\}$. Then $r = H + H'' = \frac{1}{2}$ identically off $\theta = \pi/2$, where a single atom of mass $2L$ sits; $H(\pi/2) = 1$, so the corner starts on the floor; the body is convex, symmetric about $y = \frac{1}{2}$ and contained in the unit strip; and the arms are Σ's own forms $\alpha_1 = 2L\sin t - \frac{1}{2}$, $\alpha_2 = 2L\cos t - \frac{1}{2}$. Every hypothesis imposed in Remarks 42 and 43 holds, with r far below the threshold.

Rasterised (`sofa_stadium`, 700^2 points, 2500 times), no sampled cell of N carries more than one component of $T(p)$ at any L tested, so hypothesis (ii) holds and covering holds with multiplicity one. Nevertheless

L	1.10	1.20	1.30	1.40	1.50
V	1.03167	1.29455	1.58859	1.91386	2.27038
$\lvert N\rvert$	1.03086	1.29407	1.54144	1.76743	1.98514
sweep outside C_2	0.00111	0.00116	0.05076	0.15221	0.29207

At $L = 1.30$ the discrepancy $V - |N| = 0.04715$ is 3% and is tracked by the sweep mass leaving the cap, 0.05076; the raster noise, visible in the $L \le 1.20$ columns, is two orders of magnitude smaller. The degenerate version is the disc of radius $\frac{1}{2}$, for which $N = \emptyset$ and all three hypotheses hold vacuously while (20) returns $(2-\pi)/8 = -0.1427$, a negative area.

The missing ingredient is containment of the truncated sweeps in C_2, and it localises to a single curve. Both sweeps end, at the truncation limit the Jacobian sign picks out, at the same kind of point: writing $X(\theta) = H\mu_\theta + H'\nu_\theta$ for the boundary point of the cap with outer normal μ_θ, and using $\mu_t = -\nu_\psi$, $\nu_t = \mu_\psi$ with $\psi = t + \pi/2$,

$$\Phi(\alpha_2, t) = -H'(\psi)\mu_t + (H(\psi) - 1)\nu_t = X(\psi) - \mu_\psi, \qquad \Psi(\alpha_1, t) = X(t) - \mu_t.$$

So each sweep runs from a point of the inner parallel body to either $c(t)$ (face 2, $s = 0$) or the floor (face 1, $s = \sigma$). Since C_2 is convex, a sweep lies in C_2 as soon as its two endpoints do. Measured on the family above, $X(\theta) - \mu_\theta$ never leaves the cap at any L, while $c(t)$ leaves it by 0.09289 at $L = 1.30$ and by 0.29289 at $L = 1.50$, and stays inside at $L = 1.10$ where the discrepancy vanishes. The escaping endpoint is the corner path itself, so the hypothesis to add is

$$c(t) \in C_2 \quad \text{for all } t \in [0, \pi/2],$$

a statement about one curve rather than two two-parameter families. Rule 0: the claim that the three hypotheses discharge Proposition 20 for every cap with $r < 1$ is FALSE; the endpoint identities are VERIFIED (`face2_endpoint`, `face1_endpoint`, `sweep_subset_of_endpoints`); and $c(t) \in C_2$ is CONJECTURE for Σ.

Remark 45 (The corner path lies in the cap, from a bound on corner speed)**.** Remark 44 leaves $c(t) \in C_2$ as the missing hypothesis. It reduces to a bound on the arms.

Membership in the cap is $\langle c(t), \mu_\theta \rangle \le H(\theta)$ for every θ. Convexity gives, for every ψ, the support inequality $H(\psi)\cos(\theta - \psi) + H'(\psi)\sin(\theta - \psi) \le H(\theta)$, which says that the boundary point $X(\psi)$ lies in the body. Comparing $\langle c(t), \mu_\theta \rangle$ against that inequality at $\psi = t$ and at $\psi = t + \pi/2$ leaves, with $\phi = \theta - t$, the slacks

$$A(\phi) = \cos\phi - \alpha_1 \sin\phi, \qquad B(\phi) = \sin\phi - \alpha_2 \cos\phi.$$

Either slack being nonnegative gives containment at that θ. Now

$$A + \alpha_1 B = (1 - \alpha_1\alpha_2)\cos\phi, \qquad B + \alpha_2 A = (1 - \alpha_1\alpha_2)\sin\phi,$$

so if both slacks are negative, both arms are nonnegative and $\alpha_1\alpha_2 < 1$, then $\cos\phi < 0$ and $\sin\phi < 0$. But $\theta \in [0, \pi]$ and $t \in [0, \pi/2]$ force $\phi \in [-\pi/2, \pi]$, where that is impossible. Hence $\alpha_1\alpha_2 \le 1$ suffices, and since $\alpha_1^2 + \alpha_2^2 = |c'(t)|^2$ it follows from

$$|c'(t)| \le \sqrt{2},$$

a bound on how fast the corner moves.

For Σ the margins are wide: $\max \alpha_1\alpha_2 = 0.145187$ against the threshold 1, and $\max |c'|^2 = 0.813328$ against 2. The disjunction is also checked directly over the (t, θ) grid and fails nowhere, its worst margin being 0.437739; the corner path sits on the boundary of ρC, with excursion exactly 0 (`sofa_cut`).

The sign restriction is removable. Redoing the argument without assuming the arms nonnegative, the worst ϕ sits at the two ends of the range $[-t, \pi - t]$, and the conditions there are

$$\alpha_1 \sin t + \cos t \ge 0, \qquad \alpha_2 \cos t + \sin t \ge 0,$$

that is $\alpha_1 \ge -\cot t$ and $\alpha_2 \ge -\tan t$, each binding only where the corresponding arm is negative, and symmetric under $\alpha_1 \leftrightarrow \alpha_2$, $t \leftrightarrow \pi/2 - t$. Each propagates inward from its end by the angle-subtraction formula alone: multiplying the slack by $\sin t$ and using $\sin(t - \psi) \ge 0$ reduces it to the end condition, with no division and no arctan. For Σ both margins are 0.751117, equal because Σ carries that symmetry.

The product bound is exact, not measured. $\max |\alpha_1| = \max |\alpha_2|$ agrees with $\alpha_2(0) = 2a_1 - 1 = 0.750574724825464$, so $\alpha_1\alpha_2 \le (2a_1 - 1)^2 < 1$ with no appeal to floating point.

What remains is the mirror, and it is the binding constraint rather than a formality. With $C_2 = C \cap \rho C$, the corner path clears C by 0.612162 but touches ρC, its excursion there being exactly 0. An earlier version reported a small positive clearance instead; it came from a direction list whose spacing missed $\theta = \pi/2$, the unique binding direction, and is withdrawn. The reason for the contact is structural: ρ-invariance of C_2 gives $H(-\theta) = H(\theta) - \sin\theta$, so membership in ρC is the same family of inequalities at directions $\psi \in [-\pi, 0]$, where $\phi = \psi - t$ ranges over $[-3\pi/2, 0]$ and meets the third quadrant that the argument above excludes. So the disjunction does not transfer, and a separate argument is needed.

Rule 0: `corner_slack_one`, `corner_slack_two`, `corner_disjunction`, `no_third_quadrant`, `corner_in_cap`, `arm_prod_le_of_speed`, `slack_at_lower_end`, `slack_at_upper_end` and `slack_first_quadr` are VERIFIED; $c(t) \in C$ for Σ is PROVED for arms of either sign; and $c(t) \in \rho C$ is HEURISTIC, so Proposition 20 for Σ inherits HEURISTIC until the mirror closes.

Remark 46 (The mirror half is exactly tight, and is the crux)**.** Remark 45 leaves $c(t) \in \rho C$. Three facts locate it.

First, the mirror is the same family of inequalities at reflected directions. ρ-invariance of C_2 gives $H(-\theta) = H(\theta) - \sin\theta$, hence $H(-t) = F - \sin t$ and $H'(-t) = \cos t - F'$. Comparing $\langle \rho c(t), \mu_\theta \rangle$ with the support inequality at $\psi = -t$ leaves, with $\chi = \theta + t$, the slack $\cos\chi + \alpha_1 \sin\chi$; at $\psi = -(t + \pi/2)$ it leaves $-(\sin\chi + \alpha_2 \cos\chi)$. The $\sin\theta$ produced by the reflection is exactly what cancels the $\sin t$ terms.

Second, those two do not suffice, and cannot be made to. Their union covers χ only when $\arctan\alpha_1 + \arctan\alpha_2 \geq \pi/2$, that is $\alpha_1\alpha_2 \geq 1$, which is the opposite of the condition the unmirrored half needs and which Σ violates with $\alpha_1\alpha_2 \leq (2a_1 - 1)^2 < 1$. So no two-comparison argument closes the mirror.

Third, the margin is not small but zero. At $t = 0$ the corner sits on the floor, so its mirror image sits at height 1, and $H(\pi/2) = 1$ makes the constraint at $\theta = \pi/2$ an equality. Refining the direction grid from 720 to 46080 leaves the margin at 0 to eight places, so this is exact contact. Any proof must be sharp there.

Writing $V(\theta) = H(\theta) - \langle \rho c(t), \mu_\theta \rangle$, the mirror condition is $V \geq 0$, and since the point part is harmonic, $V'' + V = r$. The mirror therefore asks for nonnegativity of a Hill solution with *nonnegative* forcing, where the unmirrored half asked for a sign of one with negative forcing. The Sturm argument does not transfer: a positive hump of V has length at least π and a negative hump at most π, so a negative hump strictly inside $[0, \pi]$ is not excluded, and indeed the stadium of Remark 44 realises one.

Rule 0: `mirror_slack`, `mirror_touch` and `mirror_hill` are VERIFIED; $c(t) \in \rho C$ for Σ is CONJECTURE, with exact contact at $(t, \theta) = (0, \pi/2)$ as the obstruction to any slack-based proof. Proposition 20 for Σ inherits CONJECTURE.

Remark 47 (The per-phase Sturm step)**.** Σ's curvature radius is piecewise: 0 on $[0, \beta)$, the absolutely continuous form on $[\beta, \pi/2 - \beta)$, and $\frac{1}{2}$ on $[\pi/2 - \beta, \pi/2]$. On each phase $q = r - 1 < 0$ by the bound of Remark 43, and W' is continuous across the phase boundaries because only W'' jumps there. `wronskian_strictAntiOn` gives strict antitonicity of the Wronskian on one phase, `strictAntiOn_glue` chains the phases, and `pos_between_of_strictAnti` then runs on the whole window. All three are VERIFIED, so supplying Σ's support function in Lean is the only remaining step in hypothesis (ii); no further mathematics is required for it.

Remark 48 (The mirror reduces to the floor, and the floor is (55))**.** Scanning the reflected directions $\psi \in [-\pi, 0)$ for Σ, the binding one is $\psi = -\pi/2$ and nothing else is close: the minimum slack over a 1440-point scan is attained there, at $t = 0$, and equals 0.

That direction is the floor, and it says something one-dimensional. By ρ-invariance the cap's support value at $-\pi/2$ is $H(\pi/2) - \sin(\pi/2) = 0$ after the normalisation $H(\pi/2) = 1$, so the constraint is $\langle c(t), \mu_{-\pi/2} \rangle \leq 0$, that is $c_y(t) \geq 0$: the corner never dips below the floor. Since $c_y(t) = \sigma(t) \cos t$, this follows from $\sigma \geq 0$, which follows from $\sigma \geq \alpha_1^+$.

That last inequality is (55), which Remark 135 already carries and which Proposition 20's statement omits. So the fourth hypothesis that Remark 44 showed to be missing is not new machinery: it is (55). The stadium of Remark 44 fails containment because it fails (55), not because of anything peculiar to the sweeps, and the disc of radius $\frac{1}{2}$, for which $\sigma < 0$ on $(0, \pi/2]$, is the same failure in its starkest form.

Rule 0: `corner_height_eq`, `floor_support_zero`, `floor_from_seghyp` and `seghyp_gives_reach_nonneg` are VERIFIED. What is not settled is the reflected directions other than $\psi = -\pi/2$; they are slacker throughout the measurement, the floor being the unique argmin, but slackness on a grid of directions is not a proof. So $c(t) \in \rho C$ for Σ is HEURISTIC, improved from CONJECTURE, and the route to PROVED is now a single stated inequality rather than an open search.

Remark 49 (The third comparison closes the reflected half)**.** The two reflected comparisons of Remark 46 leave the gap $x \in (a_1 + \pi/2, \pi - a_2)$, with $x = \omega + t$ and $a_i = \arctan \alpha_i$, and the floor sits inside it. A third closes it, and it is the one anchored at the ceiling.

With $H(\pi/2) = 1$ the support inequality at $\psi = \pi/2$ reads $\sin\omega - H'(\pi/2)\cos\omega \leq H(\omega)$, since $\cos(\omega - \pi/2) = \sin\omega$ and $\sin(\omega - \pi/2) = -\cos\omega$. Feeding it into the reflected condition $(F-1)\cos x - (G-1)\sin x + \sin\omega \leq H(\omega)$ leaves the slack

$$-\big((F-1)\cos x - (G-1)\sin x + H'(\pi/2)\cos\omega\big),$$

the $\sin\omega$ cancelling exactly. At the contact $t = 0$, $\omega = \pi/2$ this slack is 0: $\cos x$ kills the first term, $G(0) - 1 = 0$ the second because $H(\pi/2) = 1$, and $\cos\omega$ the third. So the comparison that covers the gap is tight precisely where the other two fail, which is what a proof of an identity at contact must look like. The atom at $\pi/2$ makes the top of the cap a segment, so both one-sided values of $H'(\pi/2)$ give valid comparisons.

Measured for Σ over a 1440-point scan of the reflected directions crossed with the time grid, the best of the three slacks is nonnegative everywhere, its worst value being 0 and attained only at the contact.

Rule 0: `mirror_slack_ceiling` and `ceiling_slack_contact` are VERIFIED; that the three slacks cover the reflected half for Σ is HEURISTIC, being a scan, and what it would take to promote it is a sign analysis of three explicit trigonometric expressions rather than any further geometry.

Remark 50 (The atom is what closes the contact)**.** Expanding $\cos(\omega - t)$ puts the ceiling slack in the same shape as the other two: $-\big(P\cos x + Q\sin x\big)$ with $P = F - 1 + H'(\pi/2)\cos t$ and $Q = H'(\pi/2)\sin t - (G-1)$, where $x = \omega + t$. At the contact time $t = 0$ the normalisations $H(0) = 1$ and $H(\pi/2) = 1$ collapse this to $P = H'(\pi/2)$, $Q = 0$, so the slack is $-H'(\pi/2)\cos x$.

A single value of $H'(\pi/2)$ cannot work: covering x above $\pi/2$ needs it nonnegative and below $\pi/2$ needs it nonpositive. The atom supplies both. Its mass makes the top of the cap a segment rather than a point, so the support inequality at $\psi = \pi/2$ holds with each one-sided derivative, and

$$H'(\pi/2^-) = -\big(1 - \tfrac{2}{3}a_1\big) \leq 0 \leq 2a_1 - 1 = H'(\pi/2^+)$$

gives one comparison for each side, the two differing by exactly the atom mass.

Geometrically: $H(0) = 1$ and $c_y(0) = 0$ put $\rho c(0) = (0, 1)$, while the ceiling segment runs from $H'(\pi/2^-)$ to $H'(\pi/2^+)$. The contact point lies on that segment precisely when $H'(\pi/2^-) \leq H(0) - 1 \leq H'(\pi/2^+)$, and for Σ the segment contains 0 with room on both sides. Had H no atom at $\pi/2$ the top would be a single point, the two-sided coverage would be unavailable, and the argument would fail. That is a structural role for the ceiling facet, which until now entered the analysis only as a term in $|C_2|$.

Rule 0: `ceiling_slack_form`, `ceiling_slack_at_zero`, `ceiling_two_sided` and `contact_on_ceiling` are VERIFIED. The contact time is therefore covered for every x. What remains HEURISTIC is the coverage at $t > 0$, where $Q \neq 0$ and the three slacks must be compared as functions of two parameters.

Remark 51 (Correction: (55) is not the missing hypothesis)**.** An earlier version of Remarks 42 and 48 asserted that the containment of the truncated sweeps in C_2 follows from (55), and that the stadium of Remark 44 fails containment because it fails (55). Both are false and are withdrawn.

For the stadium $H(\theta) = \frac{1}{2} + \frac{1}{2}\sin\theta + L|\cos\theta|$ the curvature radius is $r \equiv \frac{1}{2}$ off the atom, so the identity $\sigma - \alpha_1 = -\sec t \int_t^{\pi/2}(r-1)\cos s\, ds$ evaluates in closed form to

$$\sigma - \alpha_1 = \frac{1-\sin t}{2\cos t} \;\geq\; 0,$$

independently of L; and $c_y = \frac{1}{2} + L\sin 2t - \frac{1}{2}(\sin t + \cos t)$ is nonnegative for every $L \geq \frac{1}{4}$, so $\sigma \geq 0$ as well. Hence $\sigma \geq \alpha_1^+$ holds on the whole family, including the rows where $V - |N|$ is 0.047, 0.146 and 0.285. Measured at 20000 time points, $\min(\sigma - \alpha_1^+)$ is nonnegative at $L = 1.10$, at $L = \frac{1+\sqrt{2}}{2}$, and at $L = 1.30$ and 1.50 alike, and the closed form above agrees with the quadrature to $3 \cdot 10^{-12}$.

The true threshold is $\max c_y = L - \frac{\sqrt{2}-1}{2} > 1 = H(\pi/2)$, that is $L > \frac{1+\sqrt{2}}{2}$, and the escaping direction is $\theta = +\pi/2$, the ceiling, which is a constraint of C and not of ρC. What the stadium violates is $\alpha_1\alpha_2 \leq 1$: at $t = \pi/4$ both arms equal $L\sqrt{2} - \frac{1}{2}$, whose square exceeds 1 for every L past $\frac{1}{\sqrt{2}}\left(\frac{3}{2}\right)$, well below the containment threshold. That is the hypothesis of Remark 45, and Σ satisfies it with $\alpha_1\alpha_2 \leq (2a_1 - 1)^2 < 1$.

Consequences. Proposition 20 assumes containment outright rather than (55). Remark 48's reduction of the mirror half to the floor stands as a statement about ρC, but it does not discharge containment, because the binding direction for the counterexample family is in the other half. And the proof of Proposition 76 establishes only that the sweeps lie in U, not in $U \cap C_2 = N$; Theorem 115 cites it for containment, so that citation is unsupported and the theorem's stated proof has a gap at exactly the point Remark 44 identifies. Rule 0: Theorem 115 reverts to CONJECTURE pending a containment argument, and everything downstream of it inherits that label.

Remark 52 (The second endpoint condition: width at least one)**.** The convexity reduction of Remark 44 needs *both* sweep endpoints in C_2, and the inner one is $X(\theta) - \mu_\theta$. That point lies in C_2 only if C_2 has width at least 1 in direction θ, by the support inequality in the direction opposite to θ. Using ρ-invariance to write the width through H on $[0, \pi]$ alone, the condition is

$$H(\theta) + H(\pi - \theta) - \sin\theta \;\geq\; 1,$$

which at $\theta = \pi/2$ reads $1 + 1 - 1 = 1$ under the gauge: exactly tight, at the same point as every other contact in this analysis. Measured for Σ on a 200000-point grid the minimum is 1, attained only there.

This is a hypothesis of Proposition 20 that was not stated, and it is recorded here rather than assumed silently. Rule 0: `width_at_half_pi` and `width_needed_for_endpoint` are VERIFIED; that Σ satisfies the condition with equality only at $\pi/2$ is HEURISTIC.

Remark 53 (Why the width condition is tight at $\pi/2$, and only there)**.** Put $D(\theta) = H(\theta) + H(\pi - \theta) - \sin\theta - 1$, so that Remark 52's condition is $D \geq 0$. Two facts fix its shape.

D is symmetric about $\pi/2$: the substitution $\theta \mapsto \pi - \theta$ swaps the two support terms and fixes $\sin\theta$. And since $H + H'' = r$ while sin is annihilated by $d^2/d\theta^2 + 1$,

$$D'' + D = r(\theta) + r(\pi - \theta) - 1,$$

so the atom of r at $\pi/2$ enters D'' twice, once from each term, contributing a Dirac of mass twice the facet length. Hence D' jumps by that amount at $\pi/2$, while symmetry forces the two one-sided slopes to be negatives of each other. Together they are exactly $\mp$ the facet length, so D has a *corner* minimum at $\pi/2$, and $D(\pi/2) = 0$ under the gauge makes that minimum the tight value.

Measured for Σ: the two one-sided slopes agree with $\mp$ the facet length 1.1670498 of Remark 40 to the finite-difference error, and $|D(\theta) - D(\pi - \theta)|$ is at machine precision.

The tightness at $\pi/2$ is therefore forced by the ceiling facet rather than being a numerical accident, and $D > 0$ on a punctured neighbourhood of $\pi/2$. Rule 0: `width_symmetric` and `corner_slopes_from_atom` are VERIFIED; that D has no other zero is HEURISTIC and is the remaining content of the width condition.

Remark 54 (The inner functional: a bound that does not need containment)**.** The obstruction at G_1 is direction, not difficulty. Since $|T| = |C_2| - 2|N|$, an upper bound on $|T|$ requires a *lower* bound on $|N|$. What Proposition 41 supplies is $V \geq |N|$, an upper bound, so $|C_2| - 2V \leq |T|$ and the inequality runs backwards. That is why exact equality $V = |N|$ is needed and why every gram of slack has been fatal.

Replace V by a functional that is $\leq |N|$ by construction. Let V_{in} be the sweep integral restricted to the part of the windows whose image lies in C_2. If the sweeps are injective and disjoint, that image is a subset of N covered with multiplicity one, so

$$V_{\text{in}} \leq |N|, \qquad \text{hence} \qquad |T| = |C_2| - 2|N| \;\leq\; |C_2| - 2V_{\text{in}},$$

and this uses neither containment nor covering. Containment is exactly the statement $V_{\text{in}} = V$, so the two agree on Σ and the classical bound is recovered there; where containment fails, V_{in} still bounds and V does not. Writing $V_{\text{in}} = V - E$ with E the escaping mass gives $|T| \leq |C_2| - 2V + 2E$, so an upper bound on E replaces a proof that $E = 0$.

This reverses the burden. Instead of an exact geometric containment, which Remarks 44 and 51 show is not implied by the other hypotheses, it asks how much sweep can leave the cap, and any bound at all yields a valid, if weaker, upper bound on the sofa area. For the stadium family the escape is measured directly and is what the table of Remark 44 records in its fourth column.

Rule 0: `inner_bound`, `escape_bound` and `escape_zero` are VERIFIED. What is not established is that V_{in} is convex-linear in H, which is what the concavity argument of Theorem 115 consumes; without that the new bound is valid but not yet usable in the variational step. That is the next question, and it is a question about a restricted integral rather than about geometry.

Remark 55 (The affine margin: a lower bound that stays convex)**.** V_{in} bounds without containment, but it is the wrong repair for the variational step. Writing $Q_{\text{in}} = |C_2| - 2V_{\text{in}} = Q + 2E$, and Q being concave, Q_{in} is concave only if the escaping mass E is, and there is no reason for that. The obstruction is that restricting the domain by "image lies in C_2" is not an affine condition on H.

Shrink the windows instead. Replace $[0, \alpha_2^+]$ by $[0, (\alpha_2 - \delta)^+]$ and $[\alpha_1^+, \sigma]$ by $[\alpha_1^+, \sigma - \delta]$. Pulling a window in by δ pulls the corresponding sweep endpoint in by δ along its own ray, so once δ dominates the endpoint excursion the shrunken sweep lies in C_2 and the functional is $\le |N|$ exactly as V_{in} is. The gain is that the truncation is now *affine*: $\alpha_2 - \delta$ is affine in H whenever δ is, so $\frac{1}{2}((\alpha_2 - \delta)^+)^2$ is a convex function of an affine function of H, hence convex, and the functional keeps the structure the concavity argument of Theorem 115 consumes.

Nothing is lost where containment holds: $\delta = 0$ recovers V, and shrinking only decreases the functional, so the bound degrades gracefully rather than failing. For Σ, where the measured escape is 0, $\delta = 0$ is admissible and the classical bound is unchanged.

Rule 0: `margin_monotone`, `margin_zero`, `margin_term_convex` and `margin_bound` are VERIFIED. What is not settled is how small δ may be taken as a function of H, that is, how the endpoint excursion is bounded. That is the remaining content, and it is a bound on one curve rather than an exact containment of two two-parameter families.

Remark 56 (The excursion is convex, so the margin must be affine)**.** The margin of Remark 55 needs δ dominating the endpoint excursion. Two facts decide what δ can be.

First, the excursion is convex in H. For fixed t and ψ the quantity $\langle c(t), \mu_\psi \rangle - h(\psi)$ is affine in H, since c is built affinely from F and G and h is H itself; the excursion is the supremum of these over (t, ψ), and a supremum of affine functionals is convex.

Second, the margin term $((\alpha_2 - \delta)^+)^2$ is convex only if $\alpha_2 - \delta$ is convex, because $x \mapsto (x^+)^2$ is convex and nondecreasing and composing it with a concave function need not be convex. So δ must be concave, and a convex δ breaks exactly what the margin was built to preserve.

Together δ must be affine. An affine δ dominating a convex excursion that vanishes at Σ exists exactly when the excursion is affine near Σ, that is when the supremum is attained on a single active constraint, or on several whose gradients agree. If two active constraints have differing gradients the excursion has a convex kink at Σ and no affine δ can dominate it while vanishing there.

That is a checkable condition rather than an open geometric one, and it says where to look: the contact set of Σ. Three contacts have been located, all at $\theta = \pi/2$, namely the corner contact of Remark 50, the ceiling contact of the same remark, and the width condition of Remark 52. Whether they constitute one active constraint or several with differing gradients is what decides the margin, and it is not yet determined.

Rule 0: `excursion_convex` and `affine_dominates_iff_single_active` are VERIFIED; the structure of Σ's contact set is CONJECTURE.

Remark 57 (Σ's contact set is a single active constraint, and it is the gauge)**.** Remark 56 reduces the margin to a finite question: are Σ's three contacts one active constraint or several with differing gradients? Computing each, they are the same one.

The corner contact at $t = 0$ says $c_y(0) \ge 0$, and $c_y(0) = (F(0) - 1) \cdot 0 + (G(0) - 1) \cdot 1 = H(\pi/2) - 1$, so it reads $H(\pi/2) \ge 1$. The ceiling contact says the mirror point $\rho c(0)$, of height $2 - H(\pi/2)$, lies below the ceiling $H(\pi/2)$, again $H(\pi/2) \ge 1$. The width condition at $\theta = \pi/2$ reads $2H(\pi/2) - 1 \ge 1$, once more $H(\pi/2) \ge 1$. All three have gradient $\eta \mapsto \eta(\pi/2)$ up to a positive factor, so they are a single active constraint and the excursion has no convex kink at Σ.

That constraint is the gauge. The normalisation $H(\pi/2) = 1$ is imposed by a translation, which changes no area and leaves both arms invariant, so on the normalised class it holds identically. An identically tight constraint contributes equality and never violation, so it is not a source of escape at all. The remaining constraints have strict slack at Σ, measured 0.612162 for the unmirrored half, and strict slack survives small perturbation.

Hence on the normalised class the excursion vanishes near Σ, the affine margin of Remark 55 may be taken with $\delta = 0$, and containment holds on a neighbourhood, which is what local maximality requires. The obstruction of Remark 56 does not occur, because the three apparent contacts are one constraint and that constraint is an equality by construction rather than an inequality that might be violated.

Rule 0: `contact_corner`, `contact_ceiling`, `contact_width`, `contacts_coincide` and `gauge_no_escape` are VERIFIED. That the remaining constraints have strict slack at Σ is HEURISTIC, resting on the measurement above, and it is the last numerical input in the chain.

Remark 58 (A degenerating slack is not fatal)**.** The slack at Σ vanishes linearly as the contact is approached, so there is no margin bounded away from zero and the naive neighbourhood argument fails. It is recoverable, because the perturbation vanishes at the contact too.

The gauge constraint is an *identity* on the normalised class: $c_y(0) = c_y(\pi/2) = H(\pi/2) - 1 = 0$ for every admissible H, the normalisation being imposed by a translation. A perturbation therefore cannot move the constraint value at the contact; it is 0 before and after. So the perturbation's effect on constraints *near* the contact vanishes there as well, and to first order is $O(d\|\eta\|)$ at angular distance d rather than $O(\|\eta\|)$.

Both the slack and the perturbation are then linear in d, and the comparison is uniform: if the unperturbed value is at most $-ad$ with $a > 0$ and the perturbation at most $K\|\eta\|d$, the perturbed value is at most $(-a + K\|\eta\|)d$, nonpositive for $\|\eta\| \leq a/K$ and for every d at once. The neighbourhood is uniform in the constraint index even though the margin is not bounded below.

What this does not supply is the constants: a is the rate at which the slack degenerates, measured near 0.46 per unit angular distance, and K is the Lipschitz constant of the constraint family in η. Neither is established, so the argument is structural and the neighbourhood is not explicit.

Rule 0: `degenerate_slack_stable`, `gauge_identity_unmoved` and `uniform_threshold` are VERIFIED; the two constants are CONJECTURE, and with them the existence of an explicit neighbourhood.

Remark 59 (The near/far split, and the negative control it passes)**.** The degeneration argument of Remark 58 handles constraints near the gauge contacts, where slack and perturbation both vanish linearly. It says nothing about constraints far from them, and that is exactly right.

The stadium of Remark 44 satisfies the gauge identically too, with $c_y(0) = c_y(\pi/2) = H(\pi/2) - 1 = 0$, so the near regime applies to it verbatim. Its containment failure sits at $t = \pi/4$, $\psi = +\pi/2$, far from both contacts, with constraint value 0.09289. The argument does not reach it, and must not: an argument that proved containment for the stadium would be refuted by Remark 44.

So the index set splits. Fix $d_0 > 0$ and separate constraints within angular distance d_0 of a gauge contact from those beyond it. The near ones are nonpositive by Remark 58, uniformly in the distance; the far ones need a genuine margin, bounded below by ad_0 at Σ and negative for the stadium. Containment on a neighbourhood follows from the two together, and the stadium is excluded by the far regime alone. For Σ the far margin at $d_0 = 0.02$ is 0.0092 and grows linearly in d_0, so the regimes overlap and the split is not vacuous.

This is what the earlier attempts lacked: an argument local at the contacts and quantitative away from them, rather than one or the other.

Rule 0: `near_far_split`, `far_margin` and `far_positive_not_covered` are VERIFIED. The far margin for Σ is HEURISTIC, being the measurement above, and the constants of Remark 58 remain CONJECTURE.

Remark 60 (The degeneration rate in closed form)**.** The near/far split of Remark 59 needs the rate a at which the slack degenerates away from a gauge contact. It is not a measurement.

The constraint is $g(t,\psi) = \langle c(t), \mu_\psi\rangle - h(\psi)$, so $\partial g/\partial\psi = \langle c(t), \nu_\psi\rangle - h'(\psi)$. At the contact $(t,\psi) = (0, -\pi/2)$ one has $c(0) = (H(0)-1, 0)$ and $\nu_{-\pi/2} = (1,0)$, so the first term is $H(0)-1$; and $h(-\omega) = H(\omega) - \sin\omega$ gives $h'(-\pi/2) = \cos(\pi/2) - H'(\pi/2) = -H'(\pi/2^-)$. Hence

$$\frac{\partial g}{\partial\psi} = (H(0)-1) + H'(\pi/2^-),$$

which under the second gauge $H(0) = 1$ is exactly $H'(\pi/2^-)$. For Σ that is $-(1-\frac{2}{3}a_1)$, so

$$a = 1 - \tfrac{2}{3}a_1,$$

an algebraic number already carried by the note, and the exclusion-radius scan of Remark 59, whose ratios clustered near 0.45, was estimating it. The rate is positive exactly when $a_1 < \frac{3}{2}$, which Σ satisfies.

So one of the three constants is closed in exact form. What remains are the Lipschitz constant of the constraint family in η and Σ's far margin, both still measurements.

Rule 0: `degeneration_rate`, `sigma_rate` and `rate_pos` are VERIFIED.

Remark 61 (Retraction: the affine margin is a reformulation, not a weakening)**.** Remark 55 is withdrawn in its stated form, for two independent reasons.

First, the shrinks are prescribed at the wrong end. The face-2 window is $[0, \alpha_2^+]$, and by Remark 44 the sweep is at $c(t)$ when $s = 0$ and at $X(t+\pi/2) - \mu$ when $s = \alpha_2$. Shrinking to $[0, \alpha_2 - \delta]$ removes the second endpoint, which Remark 44 measures as never leaving the cap, and retains the first, which is the one that escapes. Rasterising the shrunken sweep on the stadium with δ set to the exact excursion returns the escaping mass unchanged. The correct cut is $[\delta, \alpha_2^+]$, and the formula $\frac{1}{2}((\alpha_2-\delta)^+)^2$ printed in that remark is in fact the integral over $[\delta, \alpha_2]$: the arithmetic was right and the prose described the opposite construction.

Second, and fatally for the intended use, no nontrivial affine δ exists. The excursion E is nonnegative everywhere on the admissible class, because the gauge contacts of Remark 57 are identically active, and $E(\Sigma) = 0$. An affine δ with $\delta \geq E$ and $\delta(\Sigma) = 0$ restricts on any two-sided line through Σ to $\delta(s) = ms$ with $ms \geq 0$ for both signs of s, forcing $m = 0$, hence $\delta \equiv 0$ and $E \equiv 0$. So an affine margin exists exactly when containment already holds, and the claim that the bound "degrades gracefully where containment fails" is false.

Rule 0: Remark 55 is FALSE as stated; the construction is a REFORMULATION of containment rather than a weakening of it. Remarks 56 and 57 are arguments about a δ that cannot be nonzero, so their role is reduced to the contact computation itself, which stands. Remark 54, the inner functional, is unaffected: it needs no δ.

Remark 62 (The floor direction is closed on $\mathcal{D}$)**.** On $\mathcal{D}$ the corner never dips below the floor: $c_y \geq 0$ on $[0, \pi/2]$. The argument below replaces the one previously printed here, which split $[0, \pi/2]$ into Σ's three phases at β and $\pi/2 - \beta$ and therefore did not survive the repair of hypothesis (d) to its fixed-T form (Remark 66): on $\mathcal{D}$ there is no phase decomposition to appeal to, only the one crossing point T.

Everything comes from the companion bracket $A_2(t) = (F-1)\sin t + F'\cos t$, the one behind the moment identity that Remark 67 uses. Its derivative is $A_2' = -\rho_1 \cos t$ with $\rho_1 = 1 - r_{ac}$, and $0 \le \rho_1 \le 1$ by (b). Since $c_y = (F-1)\sin t + (G-1)\cos t$ and $\alpha_1 = G - 1 - F'$ one has $c_y = \alpha_1 \cos t + A_2$ identically, so with $A_2(\pi/2) = H(\pi/2) - 1 = 0$, which is the gauge,

$$c_y(t) = \alpha_1(t)\cos t + \int_t^{\pi/2} \rho_1(s)\cos s\, ds, \tag{25}$$

and the same bracket at $t = 0$ gives the moment identity

$$\int_0^{\pi/2} \rho_1 \cos \;=\; A_2(0) \;=\; H'(0) \;=\; \tfrac{1}{2} \tag{26}$$

by (a) and (b). Both are exact; no inequality has been used yet.

On $[T, \pi/2]$, (d) gives $\alpha_1 \ge 0$ and (b) gives $\rho_1 \ge 0$, so (25) is a sum of two nonnegative terms.

On $[0, T]$, (d) gives $\alpha_2 > 0$, so $\alpha_1' = \alpha_2 + \rho_1 \ge \rho_1$ by Proposition 28; with $\alpha_1(0) = -\frac{1}{2}$ this integrates to $\alpha_1(t) \ge -\frac{1}{2} + \int_0^t \rho_1$. Substituting into (25) and eliminating the tail by (26), $c_y(t) \ge 0$ reduces to

$$\tfrac{1}{2}\bigl(1 - \cos t\bigr) \;\ge\; \int_0^t \rho_1(s)\bigl(\cos s - \cos t\bigr)\, ds. \tag{27}$$

The right-hand side is linear in ρ_1 with a nonnegative kernel, and the only constraints left on $\rho_1|_{[0,t]}$ are $0 \le \rho_1 \le 1$ from (b) and $\int_0^t \rho_1 \cos \le \frac{1}{2}$ from (26). Per unit of the constraint the objective returns $1 - \cos t \sec s$, which *decreases* in s, so the bathtub maximiser is the front-loaded profile $\rho_1 = \mathbf{1}_{[0,a]}$ with $a = \min(t, \pi/6)$, the cap binding where $\sin a = \frac{1}{2}$. Its value is $\min(\sin t, \frac{1}{2}) - \min(t, \pi/6)\cos t$, and (27) splits into two cases.

For $t \ge \pi/6$ it reads $\frac{1}{2}(1 - \cos t) \ge \frac{1}{2} - \frac{\pi}{6}\cos t$, that is $\bigl(\frac{\pi}{6} - \frac{1}{2}\bigr)\cos t \ge 0$. For $t \le \pi/6$ it reads $\Lambda(t) \ge 0$ with $\Lambda(t) = \frac{1}{2}(1 - \cos t) - \sin t + t\cos t$; here $\Lambda(0) = 0$ and $\Lambda'(t) = \bigl(\frac{1}{2} - t\bigr)\sin t$, so Λ increases on $[0, \frac{1}{2}]$ and decreases on $[\frac{1}{2}, \pi/6]$, whence its minimum over $[0, \pi/6]$ is $\min\{\Lambda(0), \Lambda(\pi/6)\}$ and $\Lambda(\pi/6) = \frac{\sqrt{3}}{12}(\pi - 3) > 0$. Both cases are the single fact $\pi > 3$, and equality in (27) occurs only at the two ends $t = 0$ and $t = \pi/2$.

Correction to the referee's form of the last step. The inequality $t - \arcsin(\sin t - \frac{1}{2}) \ge \frac{1}{2}$, offered as the outcome of the bathtub step, is (27) evaluated at the *back*-loaded profile $\rho_1 = \mathbf{1}_{[\arcsin(\sin t - 1/2),\, t]}$, which *maximises* $\int_0^t \rho_1$ where the objective wants it minimised. It is therefore a test against one admissible profile and not the extremum; it is defined only for $t \ge \pi/6$, so it leaves $[0, \pi/6)$ untreated; and where it is defined it is weaker than the front-loaded condition, since $t - \arcsin(\sin t - \frac{1}{2}) \ge \pi/6 > \frac{1}{2}$ there. Its stated minimum $\pi/6$ at $t = \pi/6$ is correct on $[\pi/6, \pi/2]$, where the function increases, because $\sin^2 t \ge (\sin t - \frac{1}{2})^2$ makes its derivative nonnegative; read as a formula on all of $[0, \pi/2]$ its minimum is instead $2\arcsin\frac{1}{4}$, at $t = \arcsin\frac{1}{4}$. The conclusion is unaffected: the front-loaded computation above is the one that closes (27).

Also, the stadium family is excluded from $\mathcal{D}$ outright: hypothesis (c), $c_y \le \frac{1}{2}$, forces $L \le \sqrt{2}/2$ on it, well below the escape threshold $(1 + \sqrt{2})/2$. The counterexample of Remark 44 therefore never entered the class, which is why it could not have been a counterexample to Theorem 115 itself, only to Proposition 20 stated without containment.

Rule 0: PROVED on $\mathcal{D}$, from (a), (b) and (d) and with no phase decomposition. The ingredients are VERIFIED (`cy_from_bracket`, `moment_bracket_deriv`, `moment_from_bracket`, `bathtub_far_case`, `Lam_hasDerivAt`, `Lam_at_pi_six`, `Lam_nonneg_near`). The remaining reflected directions are Remark 67, which imports exactly this statement for its own degenerate direction $\omega = \pi/2$.

Remark 63 (The inner functional, with its content stated)**.** The lemmas cited for Remark 54 carried the whole claim as a hypothesis, so they verified its arithmetic and none of its mathematics. The content is a membership chain and a multiplicity count.

A point of the restricted domain is a sweep value at a window point. By Remark 42 a window point is an entry into $T(p)$, so the point is cut at some time and lies in U. The restriction puts it in C_2, and $U \cap C_2 = N$, so the restricted image lies in N. With at most one entry per point the sweep is injective on the restricted domain, so the Jacobian integral counts the image once and is at most $|N|$. Neither step uses containment or covering: the restriction is what replaces them.

Rule 0: `restricted_image_subset`, `image_measure_le_of_subset` and `inner_chain` are VERIFIED, and unlike their predecessors they carry the membership and the measure comparison rather than assuming them. What is still assumed is the area formula itself, which supplies the identification of the Jacobian integral with the counted image.

Remark 64 (A norm mismatch in the near-regime estimate)**.** The surviving half of Remark 58 bounds the perturbation of a constraint near a gauge contact by $O(d\|\eta\|)$ at angular distance d. Writing the perturbation explicitly, $\delta g[\eta](t, \psi) = \eta(t)\cos(\psi - t) + \eta(t + \pi/2)\sin(\psi - t) - \eta(|\psi|)$, the bound follows from $\eta(\pi/2) = 0$ together with a *Lipschitz* bound on η.

It is false in C^0 or L^2: a spike of height h and width d placed at $\pi/2 - d/2$ has $\|\eta\|_\infty = h$ and gives $\delta g \asymp h$ at distance $d/2$, with no factor of d. The coercivity used elsewhere in this note is L^2, so the two estimates are stated in incompatible norms and cannot be chained as written. Recorded rather than repaired.

Remark 65 (Reconciling the norms)**.** The mismatch of Remark 64 is not a defect of the near-regime estimate but a statement about which topology the conclusion holds in.

Proposition 137 already certifies (RC) on a C^2 ball, $\max(\|\eta\|_\infty, \|\eta'\|_\infty, \|\eta''\|_\infty) \le \frac{1}{20}$, and a bound on $\|\eta'\|_\infty$ is exactly a Lipschitz bound by the mean value theorem. So the near estimate is available on the same ball the rest of the local analysis uses, and the two are compatible.

What this costs is the topology of the conclusion. Local maximality established this way is local in C^2, not in L^2. That is weaker, and it is what Proposition 137 already delivers, so the note states it rather than leaving two estimates in different norms. The L^2 coercivity is unaffected; it is simply not what controls the perturbation of a pointwise constraint.

Rule 0: `lipschitz_of_deriv_bound` and `pert_linear_of_lipschitz` are VERIFIED. Every local statement in this note is therefore local in C^2.

Remark 66 (Hypothesis (d), repaired)**.** Condition (d) previously read "$\alpha_1 < 0$ exactly on $[0, \beta)$ and $\alpha_2 > 0$ exactly on $[0, \pi/2-\beta)$" with β the fixed constant of (2). That form is defective in two ways. Continuity plus "exactly" forces $\alpha_1(\beta) = 0$ and $\alpha_2(\pi/2-\beta) = 0$, two affine equalities, so $\mathcal{D}$ would be a codimension-two slice with empty interior in every norm, contradicting Remark 154 and Proposition 137, which lets the crossings move inside windows of half-width 0.15. Read loosely as $E_1 \subseteq E_2$ the condition is a disjunction, hence not convex, and the convexity step of Theorem 115 fails.

The fixed-T form now stated is affine in H for each fixed T, has nonempty C^2 interior, gives $\tau_1 \le T \le \tau_2$, and is exactly what the arm-sign step of Remark 67 consumes. Σ satisfies it with T anywhere in $[\beta, \pi/2 - \beta]$. The pinning of the crossings to β was never used by any proof in this note; it was a description of Σ that had been promoted to a hypothesis.

Remark 67 (The reflected half, proved)**.** Write $\Theta(t, \omega) = H(\omega) - \sin\omega - c_x(t)\cos\omega + c_y(t)\sin\omega$, so that $c(t) \in \rho C$ is $\Theta \ge 0$ for $\omega \in [0, \pi]$.

Working in ω rather than in $\omega + t$ separates the parameters. The support inequality at $\psi = \pi/2$, valid with *either* one-sided derivative because the atom makes the top a facet, gives

$$\Theta(t,\omega) \;\geq\; -\big(H'(\pi/2^{\pm}) + c_x(t)\big)\cos\omega + c_y(t)\sin\omega.$$

With $c_y \geq 0$ (Remark 62), taking the left derivative on $[0,\pi/2]$ and the right on $[\pi/2,\pi]$ reduces everything to

$$-\alpha_2(0) \;\leq\; c_x(t) \;\leq\; \kappa := -H'(\pi/2^{-}),$$

that is, c_x lies in the horizontal shadow of the ceiling facet, an interval of width the atom mass.

Both bounds follow from exact identities. With $\rho_1 = 1 - r_{ac}$ on $[0,\pi/2]$ and $\rho_2 = 1 - r_{ac}(\cdot + \pi/2)$, both in $[0,1]$ by (RC),

$$\kappa - c_x = \alpha_1 \sin t + \int_t^{\pi/2} \rho_1 \sin, \qquad c_x + \alpha_2(0) = \alpha_2 \cos t + \int_0^t \rho_2 \cos, \tag{28}$$

with the moment identities $\int_0^{\pi/2} \rho_1 \cos = \frac{1}{2}$ and $\int_0^{\pi/2} \rho_2 \sin = \frac{1}{2}$. Both identities are the bracket $A = (H-1)\cos - H'\sin$ read at $\theta = t$ and at $\theta = t + \pi/2$: $A' = \rho \sin$ off the atom, $A(0) = 0$, $A(\pi/2^{-}) = \kappa$, $A(\pi/2^{+}) = -\alpha_2(0)$, and $c_x = A(t) - \alpha_1(t)\sin t = A(t + \pi/2) + \alpha_2(t)\cos t$. The moments are the companion bracket $A_2 = (H-1)\sin + H'\cos$, with $A_2' = -\rho\cos$, $A_2(0) = H'(0) = \frac{1}{2}$, $A_2(\pi/2^{\pm}) = 0$ and $A_2(\pi) = -H'(\pi) = \frac{1}{2}$.

The two easy branches. On $[T,\pi/2]$, (d) gives $\alpha_1 \geq 0$, so the first identity is a sum of two nonnegative terms and $c_x \leq \kappa$ there. On $[0,T]$, (d) gives $\alpha_2 > 0$, so the second is a sum of two nonnegative terms and $c_x \geq -\alpha_2(0)$ there. Each bound therefore needs an argument on the complementary half only.

The two hard branches are one scalar problem. On $[0,T]$, $\alpha_2 > 0$ and Proposition 28 give $\alpha_1' = \alpha_2 + \rho_1 \geq \rho_1$, while $\alpha_1(0) = H(\pi/2) - 1 - H'(0) = -\frac{1}{2}$ by (a) and (b); integrating, $\alpha_1(t) \geq -\frac{1}{2} + \int_0^t \rho_1$. Since $\sin t \geq 0$, the first identity gives $\kappa - c_x(t) \geq \Phi_{\rho_1}(t)$, where

$$\Phi_\rho(u) \;:=\; \sin u \Big(\int_0^u \rho - \tfrac{1}{2}\Big) + \int_u^{\pi/2} \rho(s)\sin s\, ds. \tag{29}$$

On $[T,\pi/2]$, $\alpha_1 \geq 0$ and Proposition 28 give $\alpha_2' = -\alpha_1 - \rho_2 \leq -\rho_2$, while $\alpha_2(\pi/2) = H(\pi/2) - 1 + H'(\pi) = -\frac{1}{2}$ by (a) and (b); integrating backwards from $\pi/2$, $\alpha_2(t) \geq -\frac{1}{2} + \int_t^{\pi/2} \rho_2$. Since $\cos t \geq 0$, the second identity gives

$$c_x(t) + \alpha_2(0) \;\geq\; \cos t \Big(\int_t^{\pi/2} \rho_2 - \tfrac{1}{2}\Big) + \int_0^t \rho_2 \cos.$$

The substitution $u = \pi/2 - t$, $\tilde\rho(s) = \rho_2(\pi/2 - s)$ carries the right-hand side to $\Phi_{\tilde\rho}(u)$ and the constraint $\int_0^{\pi/2} \rho_2 \sin = \frac{1}{2}$ to $\int_0^{\pi/2} \tilde\rho \cos = \frac{1}{2}$. So both hard branches are the single statement $\Phi_\rho \geq 0$ for measurable $\rho : [0,\pi/2] \to [0,1]$ with $\int_0^{\pi/2} \rho\cos = \frac{1}{2}$, the first at $u = t \in [0,T]$ and the second at $u = \pi/2 - t \in [0,\pi/2 - T]$. The two bounds do reduce to the same shape, and this reflection is why.

The sign, with no optimisation at all. Since $\sin s \geq \sin u$ on $[u,\pi/2]$, $\int_u^{\pi/2} \rho \sin \geq \sin u \int_u^{\pi/2} \rho$, so

$$\Phi_\rho(u) \;\geq\; \sin u \Big(\int_0^{\pi/2} \rho - \tfrac{1}{2}\Big) \;\geq\; 0,$$

the last step because $\rho \ge 0$ and $\cos \le 1$ make $\int_0^{\pi/2} \rho \ge \int_0^{\pi/2} \rho \cos = \frac{1}{2}$. This uses only $\rho \ge 0$ and the moment; the upper bound $\rho \le 1$ is not needed and no extremal is involved. It is all that the previous version's "$\int \rho_i \ge \int \rho_i \cos = \frac{1}{2}$" was doing, and by itself it already closes $-\alpha_2(0) \le c_x \le \kappa$. Everything below is about the constant.

The extremal, re-derived. Write $\Phi_\rho(u) = -\frac{1}{2}\sin u + \int_0^{\pi/2} \rho\, w_u$ with $w_u(s) = \sin(\max(s,u))$: the kernel is the constant $\sin u$ on $[0,u]$ and $\sin s$ on $[u,\pi/2]$. Minimising $\Phi_\rho(u)$ at fixed $\int_0^{\pi/2} \rho \cos = \frac{1}{2}$ is minimising $\int \rho\, w_u$, so the mass goes where the price per unit of the constraint,

$$\gamma_u(s) = \frac{w_u(s)}{\cos s} = \begin{cases} \sin u \sec s, & s \le u, \\ \tan s, & s \ge u, \end{cases}$$

is *lowest*. Both pieces increase in s and they agree at $s = u$, so γ_u increases across $[0,\pi/2]$ and the minimiser is *front*-loaded: $\rho_\star = \mathbf{1}_{[0,a]}$ with $\sin a = \int_0^a \cos = \frac{1}{2}$, that is $a = \pi/6$, independently of u. This is the same direction as the corner-height bathtub of Remark 62, and for the same reason: the objective wants $\int_0^u \rho$ *large* while the constraint is cheapest to satisfy near 0. It is the opposite of the back-loaded profile withdrawn there.

The certificate is pointwise and needs no duality theorem. Put $\lambda_u = \gamma_u(\pi/6) = \sin(\max(u,\pi/6))/\cos(\pi/6)$ and $k_u = w_u - \lambda_u \cos$. Then $k_u \le 0$ on $[0,\pi/6]$ and $k_u \ge 0$ on $[\pi/6,\pi/2]$, for the single reason that $s \mapsto \sin(\max(s,u))$ is nondecreasing while cos is nonincreasing: for $s \le \pi/6$, $\cos(\pi/6)\sin(\max(s,u)) \le \cos s\, \sin(\max(\pi/6,u))$, and both comparisons reverse for $s \ge \pi/6$. Hence $(\rho - \mathbf{1}_{[0,\pi/6]})k_u \ge 0$ pointwise — $\rho \le 1$ on the left piece, $\rho \ge 0$ on the right — and since ρ and $\mathbf{1}_{[0,\pi/6]}$ carry the same cos-moment $\frac{1}{2}$,

$$\int_0^{\pi/2} \rho\, w_u - \int_0^{\pi/6} w_u = \int_0^{\pi/2} \big(\rho - \mathbf{1}_{[0,\pi/6]}\big) k_u \;\ge\; 0.$$

That is the only place $\rho \le 1$ enters.

The constant, in two cases. The extremal value is $\Phi_\star(u) = \int_0^{\pi/6} \sin(\max(s,u))\, ds - \frac{1}{2}\sin u$. For $u \ge \pi/6$ the kernel is the constant $\sin u$ throughout $[0,\pi/6]$, so $\Phi_\star(u) = (\frac{\pi}{6} - \frac{1}{2})\sin u$, nonnegative by $\pi > 3$ and least on $[\pi/6,\pi/2]$ at $u = \pi/6$, with value $\frac{1}{2}(\frac{\pi}{6} - \frac{1}{2}) = 0.0117993\ldots$. For $u \le \pi/6$ the integral splits at u into $u \sin u$ and $\cos u - \frac{\sqrt{3}}{2}$, so

$$\Phi_\star(u) = \Psi(u) := u \sin u + \cos u - \tfrac{\sqrt{3}}{2} - \tfrac{1}{2}\sin u, \qquad \Psi'(u) = \big(u - \tfrac{1}{2}\big)\cos u.$$

Thus Ψ falls on $[0,\frac{1}{2}]$ and rises on $[\frac{1}{2},\frac{\pi}{6}]$ — note $\frac{1}{2} < \frac{\pi}{6}$, again $\pi > 3$ — and its minimum is

$$\Psi(\tfrac{1}{2}) = \cos\tfrac{1}{2} - \tfrac{\sqrt{3}}{2} = 0.0115571581\ldots,$$

the two $\sin\frac{1}{2}$ terms cancelling exactly. The branches agree at the switch, $\Psi(\pi/6) = \frac{1}{2}(\frac{\pi}{6} - \frac{1}{2})$, and $\Psi(\frac{1}{2}) \le \Psi(\pi/6)$ by the same monotonicity, so the far branch never goes lower and

$$\Phi_\rho(u) \;\ge\; \Phi_\star(u) \;\ge\; \cos\tfrac{1}{2} - \tfrac{\sqrt{3}}{2} \;>\; 0 \qquad \text{for every } u \in [0,\tfrac{\pi}{2}], \tag{30}$$

both inequalities sharp: the first at $\rho = \mathbf{1}_{[0,\pi/6]}$, the second at $u = \frac{1}{2}$.

Correction to the constant previously printed here. The earlier text asserted that bang-bang "makes the surplus exactly $\pi/6 - \frac{1}{2}$ in each case". That number is the *coefficient* of $\sin u$ in the far case, equivalently the value $\Phi_\star(\pi/2)$; it is not the surplus. The surplus is

$\min_u \Phi_\star = \cos\frac{1}{2} - \frac{\sqrt{3}}{2} = 0.0115571\ldots$, attained in the interior at $u = \frac{1}{2}$ and less than half of $\pi/6 - \frac{1}{2} = 0.0235987\ldots$. The two are related by $\Phi_\star(u) \ge (\frac{\pi}{6} - \frac{1}{2})\sin u$, the weaker consequence of $\int_0^{\pi/2} \rho \ge \pi/6$ — the same front-loaded extremal run against $\int \rho$ — and that relation is strict on $[0, \pi/6)$. Numerically: discretising the linear programme at $4 \cdot 10^4$ cells reproduces $\Phi_\star$ to $2 \cdot 10^{-10}$ at 101 values of u, and $2 \cdot 10^5$ random admissible profiles per u never beat the front-loaded one.

Where the surplus lives. The constant bounds $\kappa - c_x$ below on $[0, T]$ and $c_x + \alpha_2(0)$ below on $[T, \pi/2]$ — the two branches the bang-bang serves. On the complementary branches there is no uniform surplus and none is claimed: $\kappa - c_x(\pi/2) = \alpha_1(\pi/2)$ and $c_x(0) + \alpha_2(0) = \alpha_2(0)$, which (d) makes nonnegative and nothing makes bounded away from 0. So on $\mathcal{D}$

$$\kappa - c_x \;\ge\; \Big(\cos\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\Big)\mathbf{1}_{[0,T]}, \qquad c_x + \alpha_2(0) \;\ge\; \Big(\cos\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\Big)\mathbf{1}_{[T,\pi/2]},$$

and both are strict in the interior of their branch, since equality in $\alpha_1' \ge \rho_1$ would force $\alpha_2 \equiv 0$, which (d) excludes.

Exactly what is used. (a) at 0: $H(0) = 1$, for $A(0) = 0$ and for $\alpha_2(0) = H'(\pi/2^+)$. (a) at $\pi/2$: $H(\pi/2) = 1$, for $A(\pi/2^-) = \kappa$, $A_2(\pi/2^\pm) = 0$, $\alpha_1(0) = -\frac{1}{2}$ and $\alpha_2(\pi/2) = -\frac{1}{2}$. (b): $\rho_i \ge 0$ for the easy branches and for the sign step; $\rho_i \le 1$ at the exchange step and nowhere else; the single atom at $\pi/2$, so that α_1, α_2 are absolutely continuous on $(0, \pi/2)$ and A jumps only at $\pi/2$; $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$, for $\alpha_1(0) = \alpha_2(\pi/2) = -\frac{1}{2}$ and for the two moments. (d): $\alpha_2 > 0$ on $[0, T]$ twice, and $\alpha_1 \ge 0$ on $[T, \pi/2]$ twice. Hypothesis (c) is used nowhere in this remark; it enters only through Remark 69, which needs $c_y \le 1$.

Two byproducts hold on all of $\mathcal{D}$: $\kappa \ge 1 - \sqrt{3}/2$, so the sign condition on $H'(\pi/2^-)$ assumed in Remark 50 is forced rather than assumed; and the ceiling facet has length exceeding $1 - \sqrt{3}/2$, so the atom is quantitatively nondegenerate. The first is the same front-loaded extremal run against a different objective: $\kappa = A(\pi/2^-) - A(0) = \int_0^{\pi/2} \rho_1 \sin$, and minimising $\int \rho \sin$ at fixed $\int \rho \cos = \frac{1}{2}$ has price $\gamma(s) = \tan s$, again increasing, again $\rho_\star = \mathbf{1}_{[0,\pi/6]}$, with value $\int_0^{\pi/6} \sin = 1 - \frac{\sqrt{3}}{2}$; the second is then $m = \kappa + \alpha_2(0)$ with $\alpha_2(0) > 0$ from (d). A third, $\kappa - x(t) = \sec t \int_t^{\pi/2} \rho_1(s) \sin(s - t)\, ds \ge 0$, needs neither the moments nor (d) and supersedes the monotonicity argument of Remark 40.

Scope, stated precisely. This proves $c(t) \in C_2$, one of the two endpoint conditions of the convexity reduction of Remark 44; the other, $X(\theta) - \mu_\theta \in C_2$, is the width condition of Remark 52 and remains open. And the argument does not cover its own binding direction $\omega = \pi/2$, where the bound degenerates to $\Theta \ge c_y$: that case is imported from Remark 62. So it replaces two of the three comparisons of Remark 49, not all three.

Rule 0: PROVED, subject to (d) in its repaired form and to Remark 62. VERIFIED: the reduction to the c_x interval (`refl_slack`, `refl_lower_half`, `refl_upper_half`); the easy branches and the sign step (`claim_easy`, `claim_hard`, `density_ge_moment`); the substitution into (29) (`phi_from_arm_lower`); the exchange kernel and its pointwise product signs, which are the extremal (`sin_max_mono`, `exKernel_nonpos`, `exKernel_nonneg`, `exchange_product_left`, `exchange_product_rig`
the value of the extremal in both cases (`extremal_value_far`, `extremal_value_near`); and the constant (30) (`Psi_hasDerivAt`, `Psi_at_half`, `Psi_at_pi_six`, `Psi_min_near`, `cos_half_lower`, `far_ge_surplus`, `bangbang_surplus`, `bangbang_surplus_pos`, `ceiling_facet_floor_pos`). Not formalised, and marked as such: the passage from the pointwise kernel sign to $\int \rho\, w_u \ge \int_0^{\pi/6} w_u$, which is linearity of the integral, the moment constraint and $\int$(nonnegative) ≥ 0; and the assembly of (28) itself, whose ingredients are verified separately as `bracket_deriv`, `cx_from_bracket` and `moment_from_bracket`. The remark does not by itself discharge the con-

tainment hypothesis of Theorem 115, which is about the truncated sweeps and not about the corner path alone.

Remark 68 (Retraction: the width condition is necessary, not sufficient)**.** The Scope paragraph of Remark 67 asserted that the second endpoint condition $X(\theta)-\mu_\theta \in C_2$ *is* the width condition of Remark 52. That is withdrawn. Membership in C_2 is the whole family $\langle X(\theta)-\mu_\theta, \mu_\varphi\rangle \le h_2(\varphi)$ over all φ, and the width condition is that family at the single direction $\varphi = \theta+\pi$. Remark 52 and the Lean statement `width_needed_for_endpoint` both say "only if", correctly; the prose upgraded it.

The gap is not cosmetic. A triangle with vertices $(0,0)$, $(0.1,-10)$, $(5,-0.1)$ has width 10 in the vertical direction and does not contain $X(\pi/2)-\mu_{\pi/2}$. For Σ the two quantities vanish at different orders at the contact: with $\delta = \pi/2-\theta$, $D \to m\delta$ while the true containment margin $\to \frac{1}{2}\rho_1(\pi/2^-)\delta^2$. So no bound margin $\ge cD$ holds at any constant, and Remark 67's theorem, though correct as a statement about D, cannot supply the endpoint. The substitution survived earlier scrutiny because $\varphi = \theta + \pi$ is never the binding direction except at $\theta = \pi/2$, where it coincides with the floor.

The required statement is nevertheless true on $\mathcal{D}$, by a shorter argument that uses the brackets already established: both facets have the same x-shadow $[-\alpha_2(0), \kappa]$, so C_2 contains the rectangle $R = [-\alpha_2(0),\kappa]\times[0,1]$, and the two brackets confine $X(\theta)-\mu_\theta$ to R. This is written out and formalised in Remark 69. It consumes (d) at *both* $t = 0$ and $t = \pi/2$, not at $t = 0$ alone.

Rule 0: the identification of the endpoint condition with the width condition is FALSE; Remark 67's theorem about D stands; the endpoint condition itself is PROVED by the rectangle argument of Remark 69, and VERIFIED there. Theorem 115 remains conditional on containment, but the other four defects listed in the previous version of this paragraph are repaired: (55) is discharged on $\mathcal{D}$ by Proposition 136; Remark 62 is reproved without the phase decomposition that the old (d) supplied; the concavity step now cites Corollary 123, which applies by Proposition 114, in place of the superseded Remark 89; and the $V = |N|$ sentence cites Proposition 20 with all four of its hypotheses named rather than Proposition 76.

Remark 69 (The rectangle argument, written out)**.** This supplies the second endpoint condition that Remark 68 retracted and left open. Write $X(\theta) = H(\theta)\mu_\theta + H'(\theta)\nu_\theta$ for the boundary point of C_2 with outer normal μ_θ, and put $\rho(\theta) = 1 - r_{ac}(\theta)$ on $(0,\pi)$, so that $\rho = \rho_1$ on $(0,\pi/2)$ and $\rho(\cdot+\pi/2) = \rho_2$ on $(\pi/2,\pi)$; by (b), $0 \le \rho \le 1$ and ρ carries no atom on either piece.

Claim. For every $H \in \mathcal{D}$ and every $\theta \in [0,\pi]$, $X(\theta)-\mu_\theta \in C_2$.

Step 1: the two facets have the same x-shadow, so $R \subseteq C_2$. Since $H(\pi/2) = 1$, $X(\pi/2^\pm) = \big(-H'(\pi/2^\pm),\, 1\big)$, so the ceiling of C_2 is the horizontal segment $[-H'(\pi/2^+), -H'(\pi/2^-)]\times\{1\}$, of length the atom mass $m = H'(\pi/2^+) - H'(\pi/2^-)$. Its right end is $\kappa = -H'(\pi/2^-)$ by definition, and its left end is $-\alpha_2(0)$, because the gauge $H(0) = 1$ makes $\alpha_2(0) = H(0) - 1 + H'(\pi/2^+) = H'(\pi/2^+)$. So the ceiling facet is $[-\alpha_2(0),\kappa]\times\{1\}$. Because C_2 is ρ-invariant its image under $(x,y)\mapsto(x,1-y)$ is again a facet of C_2, namely $[-\alpha_2(0),\kappa]\times\{0\}$: the *same* x-shadow, which is the step the argument turns on. Both segments lie in C_2, and the convex hull of two parallel segments with a common shadow at heights 0 and 1 is the full rectangle, so by convexity

$$R := [-\alpha_2(0),\kappa]\times[0,1] \;\subseteq\; C_2.$$

Step 2: the endpoint in coordinates. $X(\theta)-\mu_\theta = (H-1)\mu_\theta + H'\nu_\theta = \big(A(\theta), A_2(\theta)\big)$ with

$$A = (H-1)\cos\theta - H'\sin\theta, \qquad A_2 = (H-1)\sin\theta + H'\cos\theta,$$

which are the bracket behind Remark 67's first identity and the companion bracket of Remark 62, read at the angle θ rather than at the time t: at $\theta = t$ the pair $\kappa - A(t) = \int_t^{\pi/2} \rho_1 \sin$ and $c_x = A(t) - \alpha_1(t) \sin t$ is that identity. Hence on $(0, \pi/2)$ and on $(\pi/2, \pi)$

$$A' = \rho \sin\theta, \qquad A_2' = -\rho \cos\theta,$$

and the endpoint values are forced by (a) and (b):

$$\begin{array}{llll} A(0) = 0, & A(\pi/2^-) = \kappa, & A(\pi/2^+) = -\alpha_2(0), & A(\pi) = 1 - H(\pi), \\ A_2(0) = \frac{1}{2}, & A_2(\pi/2^\pm) = 0, & & A_2(\pi) = \frac{1}{2}, \end{array}$$

using $H(0) = H(\pi/2) = 1$, $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$. The only discontinuity is the drop of A by m at $\pi/2$, which is Step 1's facet again.

Step 3: $\theta \in [0, \pi/2]$. There $\sin\theta \geq 0$ and $\cos\theta \geq 0$, so $A' \geq 0$ and $A_2' \leq 0$: A is nondecreasing, from 0 to κ, and A_2 nonincreasing, from $\frac{1}{2}$ to 0. So $X(\theta) - \mu_\theta \in [0, \kappa] \times [0, \frac{1}{2}]$, and this lies in R as soon as $-\alpha_2(0) \leq 0$, that is $\alpha_2(0) \geq 0$ — hypothesis (d) at $t = 0$.

Step 4: $\theta \in [\pi/2, \pi]$. There $\sin\theta \geq 0$ and $\cos\theta \leq 0$, so $A' \geq 0$ and $A_2' \geq 0$: both are nondecreasing, A from $-\alpha_2(0)$ to $1 - H(\pi)$ and A_2 from 0 to $\frac{1}{2}$. So the point lies in $[-\alpha_2(0), 1 - H(\pi)] \times [0, \frac{1}{2}]$, and this is inside R exactly when $1 - H(\pi) \leq \kappa$. Since $\alpha_1(\pi/2) = H(\pi) - 1 - H'(\pi/2^-) = H(\pi) - 1 + \kappa$, that inequality *is* $\alpha_1(\pi/2) \geq 0$, which is hypothesis (d) at $t = \pi/2$, available because $T < \pi/2$. This proves the claim.

Exactly what is used. (a) at 0 and at $\pi/2$; (b) for $\rho \geq 0$, for $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$, and for the absence of atoms away from $\pi/2$; (d) at the two endpoints $t = 0$ and $t = \pi/2$. Hypothesis (c) is not used, and neither is the upper bound $\rho \leq 1$. Two byproducts: $\kappa = A(\pi/2^-) \geq A(0) = 0$, and then $m = \kappa + \alpha_2(0) > 0$, so on all of $\mathcal{D}$ the ceiling is a genuine facet rather than one by assumption.

What it replaces. With $R \subseteq C_2$ in hand, $c(t) \in C_2$ follows from $-\alpha_2(0) \leq c_x \leq \kappa$ together with $0 \leq c_y \leq \frac{1}{2}$, that is from Remark 67's two identities, Remark 62 and (c). So the rectangle replaces the support-function reduction of Remark 67 — including its degenerate direction $\omega = \pi/2$, which no longer has to be imported — and the comparisons of Remarks 45 through 50. It does *not* replace those two identities, which are what bound c_x.

Consequence, with its inputs named. The two truncated sweeps have four endpoints (Remark 44): $c(t)$ and $X(t + \pi/2) - \mu_{t+\pi/2}$ for face 2, and $c(t) - \alpha_1^+ \nu_t$ — which is $X(t) - \mu_t$ where $\alpha_1 \geq 0$ and $c(t)$ where $\alpha_1 < 0$ — and $c(t) - \sigma\nu_t = (x(t), 0)$ for face 1, the window being a genuine interval by Proposition 136. The claim above covers the two of the form $X - \mu$; $c(t)$ is the previous paragraph; and for the floor endpoint, $F - 1 = A \cos t + A_2 \sin t$ gives $x = (F - 1) \sec t = A + A_2 \tan t \geq 0$ while $\kappa - x = \sec t \int_t^{\pi/2} \rho_1(s) \sin(s - t)\, ds \geq 0$, both from (b) alone, so $x(t) \in [0, \kappa] \subseteq [-\alpha_2(0), \kappa]$, again by $\alpha_2(0) \geq 0$, and the floor endpoint lies in the floor facet. Since C_2 is convex, a sweep segment lies in C_2 as soon as its two endpoints do. Containment on $\mathcal{D}$ therefore reduces to Remark 67, and to nothing else.

Rule 0: the claim is PROVED on $\mathcal{D}$ and VERIFIED (`endpoint_coords`, `facet_left_from_arm`, `rectangle_of_facets`, `bracket_monotoneOn`, `moment_bracket_antitoneOn`, `moment_bracket_monotoneOn`, `endpoint_range_lower`, `endpoint_range_upper`, `endpoint_mem_rectangle_lower`, `endpoint_mem_rectangle` `alpha1_half_pi_bounds_x`, `endpoint_in_cap`). The last paragraph is a REDUCTION and is recorded as one. The reason previously given for keeping containment as a standing hypothesis of Theorem 115 — that the bang-bang step of Remark 67 was a sketch and was not written out — no longer holds: that step is now written out there, with the extremal re-derived, the constant corrected from $\pi/6 - \frac{1}{2}$ to $\cos\frac{1}{2} - \frac{\sqrt{3}}{2}$, and its scalar ingredients verified. With it, the four endpoint memberships listed above are all in place. Theorem 115 nevertheless retains containment as a standing hypothesis in this version: what would remove it is this paragraph's own bookkeeping over the four endpoints, not Remark 67, and that bookkeeping has not been audited here against Remark 51, which is where the hypothesis was introduced.

Remark 70 (The size of the gap, measured)**.** Equality in Proposition 41 holds exactly when nothing is re-swept, which is what the three hypotheses of Proposition 20 assert, and which is why $V(\Sigma) = |N(\Sigma)|$. Perturbing H by a smooth bump η supported where $H + H'' \geq \frac{1}{2}$ (so convexity survives) and measuring $|N|$ with a polygon oracle against V from the same H, the excess $V - |N|$, corrected for the oracle's known $O(1/n)$ bias, is

ε	-0.20	-0.10	-0.05	0.05	0.10	0.20
$V - \lvert N \rvert$	$5.5 \cdot 10^{-3}$	$1.4 \cdot 10^{-4}$	$4.2 \cdot 10^{-6}$	$9.3 \cdot 10^{-5}$	$8.4 \cdot 10^{-4}$	$5.0 \cdot 10^{-3}$

strictly positive on *both* sides and growing faster than ε^2. So Σ is an isolated zero of $V - |N|$: the functional Q touches the true bound at Σ and falls strictly below it in every direction. Therefore Q cannot certify Σ, and the fact that Q is tight, critical and concave there (Remark 39) does not yield an optimality statement.

This locates the difficulty exactly. The repair is the one the one-corner argument performs: replace the exact flux V by the flux over a sub-family on which the sweep is *provably* injective, so that the integral is the area of a genuine subset of N and the inequality runs the right way, while the restriction is vacuous at Σ. Producing such a sub-family is the content of that argument's Chapters 3–6 and is not done here.

One reformulation that may help. A point p lies on the face-2 line at parameter t iff $\phi(t) := \langle p, \nu_t \rangle - (G(t) - 1)$ vanishes, and with $s(p,t) := F(t) - 1 - \langle p, \mu_t \rangle$ one computes $\phi' = s - \alpha_2$ and $s' = -\phi - \alpha_1$, i.e. in complex form

$$z := s + i\phi, \qquad z' = iz - (\alpha_1 + i\alpha_2).$$

So the whole face-2 sweep is governed by one linear ODE whose homogeneous part is an exact rotation, $z(t) = \zeta(t) + e^{it} w_0$ with w_0 an affine coordinate for p; the sweep is injective iff no solution meets $\{\phi = 0,\ 0 \leq s \leq \alpha_2\}$ twice, and at every such meeting $\phi' = s - \alpha_2 \leq 0$, so the crossing is always downward. A second meeting therefore forces an upward crossing in between, at $s \geq \alpha_2$. Since the total rotation over $[0, \pi/2]$ is only $\pi/2$, the homogeneous part alone cannot supply that extra crossing; whether the forcing ζ can is the open point.

Proposition 71 (Monotonicity of the outer arm)**.** *The face-1 segment's far endpoint has x-coordinate*

$$x(t) = c_x(t) + \sigma(t) \sin t = \frac{F(t) - 1}{\cos t}, \tag{31}$$

the x-intercept of the face-1 line, and $\cos^2 t\, x'(t) = F'(t) \cos t + (F(t) - 1) \sin t$*. For* Σ*:*

- *on* $[0, \beta)$*, where* $A(t) \equiv P$ *so* $F(t) = \cos t + \frac{1}{2} \sin t$*,*

  $$x = 1 + \tfrac{1}{2} \tan t - \sec t, \qquad \cos^2 t\, x' = \tfrac{1}{2} - \sin t > 0,$$

  the last because $\sin \beta = 1/(4a_1) = 0.2856\ldots < \frac{1}{2}$ *by Theorem 14, i.e.* $\beta < \pi/6$*;*

- *on* $(\pi/2-\beta,\pi/2]$, *where* $F(t)-1 = A\cos t + \frac{1}{2}\sin t - \frac{1}{2}$ *with* $A = 1-\frac{2}{3}a_1$,

  $$x = A - \tfrac{1}{2}\tan\Big(\tfrac{\pi}{4}-\tfrac{t}{2}\Big), \qquad \cos^2 t\, x' = \tfrac{1}{2}(1-\sin t) > 0 \quad (t<\pi/2),$$

  and in particular $x(\pi/2) = A = 1-\frac{2}{3}a_1 = 0.4164750917\ldots$, *which is exactly the right endpoint of the corridor floor facet of Remark 40;*

- *on* $[\beta,\pi/2-\beta]$, *substituting* $u = t/2-\pi/8$ *and using* $-f_2/f_1 = \sqrt{2}-1 = \tan(\pi/8)$,

  $$\cos^2 t\, x' = W(u) := \tfrac{3K}{4}\sin u + \tfrac{K}{4}\cos 3u + \tfrac{1}{2} - \sin\Big(2u+\tfrac{\pi}{4}\Big), \qquad K = f_1\sqrt{4-2\sqrt{2}},$$

  on $|u| \le \pi/8-\beta/2$.

Hence x *is strictly increasing, given* $W>0$.

Proof. The face-1 line is $\{\langle p,\mu_t\rangle = F(t)-1\}$, so its x-intercept is $(F-1)/\cos t$; that it agrees with $c_x+\sigma\sin t$ is $c_x + c_y\tan t = (F-1)/\cos t$, immediate from $c = (F-1)\mu_t + (G-1)\nu_t$. The three closed forms follow by substituting $F(t)-1 = \mathrm{Re}(e^{-it}z(t))$ with the complex forms of SOL1, SOL6, SOL5. For the middle phase the κ_6 terms cancel and the triple-angle identities $\sin(3t/2) = s(3c^2-s^2)$, $\cos(3t/2) = c(c^2-3s^2)$ with $s=\sin\frac{t}{2}$, $c = \cos\frac{t}{2}$ reduce the result to the stated three-term form. □

Proposition 72 ($W>0$, by a finite certificate)**.** *$W>0$ on $|u| \le u_0 := \pi/8-\beta/2 = 0.2478721712\ldots$, so x is strictly increasing on all of $[0,\pi/2]$ and $S_1 = \sigma$.*

Proof. $K = 1.3020516916172893452\ldots$, and $3u_0 = 0.7436\ldots < \pi$, so $\cos 3u$ is unimodal on $[-u_0,u_0]$ with maximum at 0 and its minimum over any subinterval is attained at an endpoint. Also $\sin u$ is increasing, and $2u+\pi/4$ ranges over $[0.2896\ldots, 1.2812\ldots] \subset [0,\pi/2]$, so $\sin(2u+\pi/4)$ is increasing. Therefore on $[p,q] \subseteq [-u_0,u_0]$,

$$W \;\ge\; L(p,q) := \tfrac{3K}{4}\sin p + \tfrac{K}{4}\min\big(\cos 3p, \cos 3q\big) + \tfrac{1}{2} - \sin\Big(2q+\tfrac{\pi}{4}\Big).$$

Dividing $[-u_0,u_0]$ into 32 equal pieces gives $L>0$ on every piece, with $\min L = 6.13\cdot 10^{-3}$; the computation is at 40 working digits, so the margin exceeds the arithmetic error by 37 orders of magnitude. (16 pieces do not suffice: $\min L = -8.6\cdot 10^{-3}$.) □

Remark 73. W is in fact strictly decreasing, with $W(u_0) = \frac{1}{2}(1-\cos\beta) = 0.020828595599846555\ldots$ at the right junction, matching the last phase's $\frac{1}{2}(1-\sin t)$ there to $1.7\cdot 10^{-41}$, as C^1 gluing requires. The three closed forms of Proposition 71 were also checked against numerical differentiation of the path to 10^{-10}.

Remark 74 (The identity at 61 digits)**.** With all constants from their closed forms ($u^3+3u=2$, $\beta = \arctan(u/2)$, $a_1 = \sqrt{4+u^2}/(4u)$, f_1 from Proposition 6) and 60 working digits,

$$V = 0.184193197088768988251655564606\ldots, \qquad |C_2| = 2.013341612602978828172120476813\ldots,$$

$$Q(\Sigma) - A_R^* = -1.56\cdot 10^{-61}.$$

The two sides share no computation, so this rules out the possibility that the double-precision agreement of (23) was a rounding coincidence. It remains floating point, not an interval enclosure.

11 A convex subdomain on which the bound is valid

Proposition 41 says $V \geq |N|$ with equality exactly when nothing is re-swept, i.e. when the combined sweep is injective. We now show that injectivity is governed by inequalities that are *linear* in H, so the set where the bound is valid is convex.

Each sweep lies on a line: face 2 at parameter t on $\ell_2(t) = \{\langle p, \nu_t\rangle = G(t) - 1\}$, face 1 on $\ell_1(t) = \{\langle p, \mu_t\rangle = F(t) - 1\}$. Two distinct lines meet in one point, so a double cover forces that point into both segments. Solving $\ell_2(t) \cap \ell_2(t')$ for the coordinate s on $\ell_2(t)$, and using $\langle \mu_t, \nu_{t'}\rangle = \sin(t - t')$,

$$s_{22}(t, t') = \frac{\langle c(t), \nu_{t'}\rangle - (G(t') - 1)}{\sin(t - t')}, \qquad s_{11}(t, t') = \frac{\langle c(t), \mu_{t'}\rangle - (F(t') - 1)}{\sin(t' - t)}. \tag{32}$$

The numerator of s_{22} vanishes at $t' = t$ with derivative $-\alpha_2(t)$, so $s_{22}(t, t') \to \alpha_2(t)$: nearby face-2 lines meet at the *far* end of the face-2 segment $[0, \alpha_2]$. Likewise $s_{11}(t, t') \to \alpha_1(t)$, the *near* end of the face-1 segment $[\alpha_1^+, \sigma]$.

Definition 75. Let $\mathcal{K}'$ be the set of admissible H satisfying, for all $0 \leq t' < t \leq \pi/2$,

$$\text{(C22)} \quad \langle c(t), \nu_{t'}\rangle - (G(t') - 1) \ \geq\ \alpha_2(t) \sin(t - t'), \tag{33}$$

$$\text{(C11)} \quad \langle c(t), \mu_{t'}\rangle - (F(t') - 1) \ \leq\ \alpha_1(t)^+ \sin(t' - t) \ \ (\text{read with } \sin(t' - t) < 0), \tag{34}$$

together with the analogous condition for $\ell_2(t) \cap \ell_1(t')$.

Proposition 76. *$\mathcal{K}'$ is convex, and on $\mathcal{K}'$ the combined sweep is injective, hence $V = |N|$ and $Q = |C_2| - 2|N| \geq |T|$ for every ambidextrous sofa T with cap data H.*

Proof. By Lemma 18, $c(t)$ is affine in H, hence so are $\langle c(t), \nu_{t'}\rangle$, $G(t') - 1$, $F(t') - 1$, $\alpha_1(t)$ and $\alpha_2(t)$; $\sin(t - t')$ does not depend on H. So (33) and (34) are linear inequalities in H and cut out a convex set. Given (33), for any pair $t' < t$ the intersection point of $\ell_2(t)$ and $\ell_2(t')$ has $s_{22}(t, t') \geq \alpha_2(t)$, so it is not interior to the segment at t, and the pair contributes no double cover; (34) does the same for face 1 with the inequality reversed because the relevant endpoint is the near one. With the cross condition, no point is swept twice.

For $V = |N|$: every point of N has a first capture time $\tau(p)$, at which $\partial Q_{\tau(p)}$ is advancing at p (otherwise p would have been captured earlier), so by Lemma 19 it lies on face 2 with $s \leq \alpha_2$ or on face 1 with $s \geq \alpha_1$. Hence N is covered by the two sweeps, and the sweeps lie in N because an advancing face point belongs to $Q_{t+\varepsilon}$. Injectivity turns each $\int |\det|$ into the area of its image, and the two images are disjoint, so the sum is $|N|$. $\square$

Remark 77 ($\Sigma \in \mathcal{K}'$, and $\mathcal{K}'$ has nonempty interior)**.** Measured on a grid of 401 parameters, Σ satisfies all three conditions: of the 158 802 ordered pairs with $|t - t'|$ above the diagonal cutoff, *zero* give a point interior to both face-2 segments and *zero* interior to both face-1 segments, and of 58 081 cross pairs, *zero* land in both. The margins in (33), (34) vanish linearly as $t' \to t$, which is forced: that is the envelope limit. Away from the diagonal they are strictly positive.

$\mathcal{K}'$ is not a single point. For $H = H_\Sigma + \varepsilon \sin(k\theta)$, which respects the gauge for even k, all conditions still hold with zero violations at $k = 2$ for $|\varepsilon| \leq 3 \cdot 10^{-2}$ and at $k = 4$ for $|\varepsilon| \leq 10^{-2}$, in both signs, and the two margins trade off smoothly. So Σ lies in $\mathcal{K}'$ with low-frequency room around it.

The mechanism of failure is identified: for the compactly supported bump perturbations of Remark 70, where $V - |N|$ was found positive, the face-2 condition continues to hold and it is the *face-1* condition that fails, by $-4.7 \cdot 10^{-2}$ at $\varepsilon = 0.05$ and $-2.3 \cdot 10^{-1}$ at $\varepsilon = 0.20$. Those perturbations have large higher derivatives, which is what creates face-1 envelope self-intersections.

Theorem 78 (Conditional)**.** *Assume:*

(i) $Q(\Sigma) = A_R^*$*,* $\delta Q(\Sigma) = 0$ *in every admissible direction, and* $\delta^2 Q \preceq 0$ *on the cell* $\mathcal{C}$ *of support functions sharing* Σ*'s sign pattern;*

(ii) the three conditions of Definition 75 hold on $\mathcal{K}' \cap \mathcal{C}$*;*

(iii) competitors satisfy the niche separation $M < \frac{1}{2}$ *of Corollary 2.*

Then $|T| \leq A_R^*$ *for every ambidextrous sofa* T *whose cap data lies in* $\mathcal{K}' \cap \mathcal{C}$*.*

This statement is superseded *by Theorem 115, which reaches the same conclusion on an explicitly described convex domain with all three hypotheses proved. It is retained only because its three hypotheses isolate what the later argument has to supply, and because Section 12 closes hypothesis (ii) for it. Its own label is* HEURISTIC*; Theorem 115 is* PROVED.

Proof. $\mathcal{C}$ is convex because α_1, α_2 are affine in H, so each pointwise sign condition is a half-space; $\mathcal{K}'$ is convex by Proposition 76; hence so is the intersection. By (i), Q is a concave function on that convex set with a critical point at Σ, so $Q(H) \leq Q(\Sigma) = A_R^*$ throughout. By (ii) and Proposition 76, $Q(H) \geq |T|$ there, using (iii) to get the factor 2 in $|T| \leq |C_2| - 2|N|$. □

12 One curvature condition implies injectivity

The conditions of Definition 75 were verified on a grid. We now prove them, for all pairs at once, from a single linear condition on the cap.

Definition 79. Say H satisfies (RC) if the absolutely continuous part of the surface measure obeys

$$H(\theta) + H''(\theta) \;\leq\; 1 \qquad \text{for a.e. } \theta \in [0, \pi],$$

the only atoms being the corridor ceiling and floor facets at $\theta = \pm\pi/2$. Equivalently: the cap is nowhere flatter than a circle whose radius is the corridor width.

Since $H + H'' \leq 1$ is linear in H, (RC) cuts out a convex set.

Remark 80 ((RC) is classical)**.** The bound $h + h'' \leq R$ is not new. By the dual of Blaschke's rolling theorem, a convex body K slides freely inside a ball of radius R exactly when $h_K + h_K'' \leq R$, and that in turn says K is a Minkowski summand of RB; see Schneider [4, §3.2]. Our (RC) is that condition at $R = 1$, the corridor width, *relaxed* to permit atoms at $\theta = \pm\pi/2$. Since Minkowski addition adds surface area measures, and a body whose surface measure is two equal atoms at $\pm\pi/2$ is a horizontal segment,

$$\text{(RC)} \quad \Longleftrightarrow \quad C_2 = \big[\text{a horizontal segment}\big] \oplus D, \qquad D \text{ a Minkowski summand of the unit disc.}$$

For Σ the segment has length $1.1670498\ldots$, the corridor facet, and D has curvature radius at most $0.8388253\ldots$. What we claim is not the condition but its use: that it forces the niche sweep to be injective (Theorems 81, 85).

Theorem 81. *If H satisfies (RC), then (33) and (34) hold for every pair $0 \le t' < t \le \pi/2$. Hence each of the two sweeps is injective.*

Proof. Fix t and set $\tau = t - t' \in (0, t]$. For (33) put $n(t') = \langle c(t), \nu_{t'} \rangle - (G(t') - 1)$ and

$$\Phi(\tau) = n(t - \tau) - \alpha_2(t) \sin \tau,$$

so that (33) reads $\Phi \ge 0$. Now $n(t) = 0$ because $\langle c(t), \nu_t \rangle = G(t) - 1$, and $n'(t') = -\langle c(t), \mu_{t'} \rangle - G'(t')$ equals $-(F(t) - 1) - G'(t) = -\alpha_2(t)$ at $t' = t$ by (18); hence $\Phi(0) = \Phi'(0) = 0$. Differentiating again and using $\nu'' = -\nu$,

$$n''(t') = -\langle c(t), \nu_{t'} \rangle - G''(t') = -\big(n(t') + G(t') - 1\big) - G''(t'),$$

so with $n(t - \tau) = \Phi(\tau) + \alpha_2(t) \sin \tau$ the α_2 terms cancel and

$$\Phi''(\tau) + \Phi(\tau) = 1 - (G + G'')(t - \tau) =: R(\tau). \tag{35}$$

With $\Phi(0) = \Phi'(0) = 0$, (35) gives $\Phi(\tau) = \int_0^\tau \sin(\tau - u) R(u)\, du$. Since $\tau \le t \le \pi/2 < \pi$ we have $\sin(\tau - u) \ge 0$ for $u \in [0, \tau]$, and $R \ge 0$ by (RC) because $G(s) = H(s + \pi/2)$. Therefore $\Phi \ge 0$.

The ceiling atom of H sits at $\theta = \pi/2$, i.e. at $t - u + \pi/2 = \pi/2$, i.e. $u = t$. For $\tau < t$ that is outside $[0, \tau]$, and for $\tau = t$ it is the endpoint, where the kernel $\sin(\tau - u)$ vanishes. So the atom contributes nothing.

For (34) put $m(t') = \langle c(t), \mu_{t'} \rangle - (F(t') - 1)$ and $\Psi(\tau) = m(t - \tau) + \alpha_1(t) \sin \tau$. Then $m(t) = 0$, $m'(t) = \alpha_1(t)$ by (17), so $\Psi(0) = \Psi'(0) = 0$, and $\mu'' = -\mu$ gives in the same way

$$\Psi''(\tau) + \Psi(\tau) = 1 - (F + F'')(t - \tau),$$

whence $\Psi(\tau) = \int_0^\tau \sin(\tau - u)\big[1 - (H + H'')(t - u)\big] du \ge 0$. Multiplying by $-\sin \tau < 0$ turns $\Psi \ge 0$ into $s_{11} \le \alpha_1$, which is (34). Here the atom needs $t - u = \pi/2$, i.e. $u = t - \pi/2 \le 0$, reachable only at $t = \pi/2$, where $\sigma(\pi/2) = \alpha_1(\pi/2)$ makes the face-1 segment empty and the condition vacuous. $\square$

Remark 82 (Verification)**.** For Σ the absolutely continuous part of $H + H''$ attains its maximum at $\theta = \beta$ and at $\theta = \pi - \beta$, where it equals

$$\tfrac{3}{4} f_1 \sqrt{4 - 2\sqrt{2}}\, \cos\!\big(\tfrac{\beta}{2} + \tfrac{\pi}{8}\big) \;=\; 0.8388253\ldots$$

in closed form, the value used in Proposition 137, so (RC) holds for Σ with margin $0.1611747\ldots$; second differences approach it from below and converge too slowly to confirm more than four digits, so the closed form is what is quoted; the atom at $\theta = \pi/2$ has mass $1.1670498\ldots$, the ceiling facet length. The reduction (35) was checked against direct evaluation of $n(t - \tau) - \alpha_2(t) \sin \tau$ at several (t, τ), agreeing to $2.4 \cdot 10^{-5}$, the finite-difference floor for H''.

Theorem 81 also explains the failure mode of Remark 77. Across twelve perturbations, (RC) holds exactly when (34) holds: the compactly supported bumps give $\max(H + H'')_{\mathrm{ac}} = 1.08, 1.45, 2.18, 2.67, 4.58, 8.41$ and all violate (34); $H_\Sigma + \varepsilon \sin(2\theta)$ at $|\varepsilon| = 0.03$ gives 0.888 and holds; $H_\Sigma + \varepsilon \sin(4\theta)$ gives 0.976 at $|\varepsilon| = 0.01$ and holds, 1.267 at $|\varepsilon| = 0.03$ and fails. The agreement is exact on both sides of the transition.

Remark 83 (What Theorem 81 does and does not give)**.** It replaces hypothesis (ii) of Theorem 78, for the two self-intersection conditions, by (RC): a single explicit linear inequality with a geometric reading. The cross condition, that $\ell_2(t) \cap \ell_1(t')$ avoid both segments, is not covered; it is measured to hold at Σ (0 of 58 081 pairs) and remains a hypothesis. So the hypothesis class of Theorem 78 may be replaced by $\{(\mathrm{RC})\} \cap \{\mathrm{cross}\} \cap \mathcal{C}$, an intersection of two convex sets with one measured condition.

Remark 84 (The first variation, pointwise)**.** Hypothesis (i) of Theorem 78 contains $\delta Q(\Sigma) = 0$. Integrating every η' in δQ by parts and collecting the coefficient of $\eta(\theta)$ gives the Euler–Lagrange equations of Q:

$$H + H'' = \alpha_2^+ + (\sigma - \alpha_1)\tan\theta - (\sigma - \alpha_1)' + (\alpha_1^-)' \qquad (0 < \theta < \pi/2),$$

$$H + H'' = -(\alpha_2^+)'(s) + \alpha_1^-(s), \qquad s = \theta - \pi/2 \qquad (\pi/2 < \theta < \pi).$$

Evaluated on Σ at eighteen values of θ spanning all six regions, the residual is at most $9.4 \cdot 10^{-6}$, which is the finite-difference floor. So $\delta Q(\Sigma) = 0$ has an explicit pointwise form; proving it amounts to substituting the closed forms of SOL1, SOL5, SOL6 phase by phase, and is not carried out here.

13 The cross condition, the first variation, and a Gårding bound

Theorem 85. *(RC) implies the cross condition: for all t, t' the point $\ell_2(t) \cap \ell_1(t')$ is not interior to both sweep segments. Consequently, under (RC) the combined sweep is injective and $V = |N|$, so $Q = |C_2| - 2|N| \geq |T|$.*

Proof. Work in the frame at t and write $p = c(t) + a\mu_t + b\nu_t$. The equation $\langle p, \nu_t\rangle = G(t) - 1$ forces $b = 0$, so p moves along $\ell_2(t)$ with $s_2 = -a$; the equation $\langle p, \mu_{t'}\rangle = F(t') - 1$ then gives, for $t' = t - \tau$,

$$s_2 = \frac{m(t-\tau)}{\cos\tau}, \qquad s_1 = -\Phi(\tau) - \big(\alpha_2(t) - s_2\big)\sin\tau,$$

using $\langle \mu_t, \nu_{t-\tau}\rangle = \sin\tau$ and $n(t-\tau) = \Phi(\tau) + \alpha_2(t)\sin\tau$. If $\tau > 0$ and p is interior to the face-2 segment, then $0 < s_2 < \alpha_2(t)$, and $\Phi \geq 0$ by Theorem 81 with $\sin\tau \geq 0$ because $\tau \leq \pi/2$; hence $s_1 < 0 \leq \alpha_1(t')^+$ and p lies below the face-1 segment.

For $t' > t$ work in the frame at t': the same computation with the roles of the two families exchanged gives, for $t = t' - \tau'$ with $\tau' > 0$,

$$s_2 = -\Psi(\tau') - \big(s_1 - \alpha_1(t')\big)\sin\tau',$$

so if p is interior to the face-1 segment then $s_1 > \alpha_1(t')^+ \geq \alpha_1(t')$ and $s_2 < 0$, outside $[0, \alpha_2(t)]$. At $t' = t$ the two lines are perpendicular and meet at $c(t)$, where $s_1 = s_2 = 0$, interior to neither. Both displayed identities were checked against a direct linear solve, agreeing to 10^{-16}. $\square$

So hypothesis (ii) of Theorem 78 may be replaced throughout by (RC).

Theorem 86. $\delta Q(\Sigma) = 0$. *Two things are needed and both hold: the Euler–Lagrange equations of Remark 84 hold identically on each of the six regions, as trigonometric identities in a_1, f_1, f_2 with no relation among the constants used; and the boundary term of δQ vanishes, because $H'(\pi) = -\frac{1}{2}$.*

Proof. On each phase, $F(t)-1=\mathrm{Re}(e^{-it}z(t))$ and $G(t)-1=\mathrm{Im}(e^{-it}z(t))$ evaluate to

$$\begin{array}{lll} \mathrm{SOL1} & F-1=\cos t+\frac{1}{2}\sin t-1, & G-1=(2a_1-1)\sin t+\frac{1}{2}\cos t-\frac{1}{2}, \\ \mathrm{SOL6} & F-1=f_1\cos\frac{t}{2}+f_2\sin\frac{t}{2}-1+\kappa\cos t+\frac{1}{2}\sin t, & G-1=-f_2\cos\frac{t}{2}+f_1\sin\frac{t}{2}-1+\frac{1}{2}\cos t-\kappa\sin t, \\ \mathrm{SOL5} & F-1=(1-\frac{2}{3}a_1)\cos t+\frac{1}{2}\sin t-\frac{1}{2}, & G-1=(\frac{8}{3}a_1-1)\sin t+\frac{1}{2}\cos t-1, \end{array}$$

with $\kappa=1-\frac{4}{3}a_1$. Substituting these into $\alpha_1=G-1-F'$, $\alpha_2=F-1+G'$, $\sigma=(F-1)\tan t+G-1$ and into the two Euler–Lagrange equations, with the sign pattern of each region fixed ($\alpha_2>0$ except on the last phase, $\alpha_1<0$ only on the first), every residual reduces identically to 0.

For the boundary term, integrating the η' of $\delta|C_2|=2\int_0^\pi(H\eta-H'\eta')d\theta-\eta(0)-\eta(\pi)$ by parts and using $\eta(0)=0$ leaves $-(2H'(\pi)+1)\eta(\pi)$ beyond the pointwise equations. The variation δV contributes nothing there: its only boundary term at $\theta=\pi$ carries the factor $\alpha_2^+(\pi/2)$, and $\alpha_2(\pi/2)<0$, so the clamp kills it; every other boundary sits at $\theta=0$ or $\pi/2$, where η vanishes. Finally the boundary point of C_2 with outer normal $\mu_\pi=(-1,0)$ is $H(\pi)\mu_\pi+H'(\pi)\nu_\pi=(-H(\pi),-H'(\pi))$, so its height is $-H'(\pi)$; since C_2 is ρ-symmetric about $y=\frac{1}{2}$ its leftmost point is ρ-fixed when unique, forcing $H'(\pi)=-\frac{1}{2}$ and the boundary term to vanish. The same mechanism at $\theta=0$ gives $H'(0)=\frac{1}{2}$: both extreme points of the cap sit at height $\frac{1}{2}$. □

Remark 87. The computation is a computer algebra verification, six residuals, each returning 0 with a_1, f_1, f_2 free. It was controlled against five deliberately perturbed variants of the middle-phase equation (sign flip on the tan term, each of the two terms dropped, α_1 and α_2 exchanged, $H+H''$ replaced by $H-H''$), all of which return nonzero; and the three pairs of phase formulas were checked against the reference path to $2.2\cdot 10^{-16}$. The content of Theorem 86 is that Romik's ODE families are exactly the critical-point equations of Q.

Remark 88 (Towards $\delta^2 Q\preceq 0$)**.** The remaining part of hypothesis (i) is the sign of the second variation. The step that makes it accessible is an exact algebraic cancellation. On $[0,\beta)$ write $p=\eta'(t)$, $q=\eta(t)$, $r=\eta(t+\pi/2)$, $T=\tan t$; the three terms of (24) that carry $\eta'(t)$ combine as

$$-p^2-(qT+p)^2+(r-p)^2=-(p+qT+r)^2+2r^2+2qTr, \tag{36}$$

so the added term is absorbed exactly and the remainder carries no derivative of η. What is left is a lower-order estimate. With $\eta(0)=\eta(\pi/2)=0$ the relevant Poincaré constants are 4 on $[0,\pi/2]$, 1 on $[\pi/2,\pi]$ (Dirichlet–Neumann, $\eta(\pi)$ free) and $(\pi/2\beta)^2=29.41$ on $[0,\beta]$. Retaining a fraction δ of $-\int_0^\beta\eta'^2$ before applying (36) and splitting $(qT+r)^2$ with a parameter κ, the coefficient of $\int_0^\beta\eta^2$ is

$$\left(\tfrac{1+\kappa}{1-\delta}-1\right)\tan^2\beta+1-\delta\left(\tfrac{\pi}{2\beta}\right)^2=-1.83\ldots<0\qquad(\delta=\tfrac{1}{10},\ \kappa=1),$$

and the r^2 remainder, of coefficient $3.22\ldots$, is absorbed by the E_2 term of (24), which supplies $-\frac{1}{2}\int_{\pi/2}^{\pi-\beta}\eta'^2$: since $\eta(\pi/2)=0$ gives $\int_{\pi/2}^{\pi/2+\beta}\eta^2\le(\beta^2/2)\int\eta'^2$, the requirement is $3.22\cdot 0.0420=0.135\le\frac{1}{2}$. So every ingredient of a Gårding inequality $\delta^2 Q\le -c\|\eta\|_{L^2}^2$ is in place; Theorem 89 assembles it and Theorem 118 sharpens the constant to $\frac{2}{3}$.

The sharp value is $\frac{1}{2}c=0.7309566\ldots$ by Theorem 119. A P1 finite-element computation approaches it from above, as Rayleigh–Ritz must: $-0.735818, -0.733924, -0.733086, -0.732285, -0.731311, -0.7312$ at $m=32,\ldots,2048$.

14 The area identity

The concavity of Q is established in Section 20; what follows here is the earlier and weaker route to the constant, retained for its elementary ingredients, and the closed-form evaluation of $Q(\Sigma)$.

Remark 89 (An earlier route to the constant)**.** Superseded by Theorem 119, which gives the exact constant, and by Theorem 122, which gives concavity on the whole ordered anchored family. It is retained because its ingredients are elementary and its failure locates the loss. On the cell $\mathcal{C}$ of Σ's sign pattern, $\delta^2 Q \le 0$, with equality only at $\eta = 0$. Explicitly, writing $J_1 = \int_0^\beta \eta^2$, $J_2 = \int_\beta^{\pi/2} \eta^2$, $J_3 = \int_{\pi/2}^{\pi} \eta^2$,

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -1.6013\,J_1 - 0.3710\,J_2 + 0\cdot J_3 - 0.005\int_{\pi/2}^{\pi-\beta} \eta'^2. \tag{37}$$

Proof of the statement in Remark 89. On $[0,\beta)$ retain a fraction δ of $-\int_0^\beta \eta'^2$ and apply (36) with the remaining $-(1-\delta)\eta'^2$; completing the square in $p = \eta'(t)$ and discarding the resulting negative square,

$$-(1-\delta)p^2 - (qT+p)^2 + (r-p)^2 \;\le\; \frac{(qT+r)^2}{1-\delta} - q^2T^2 + r^2 \;\le\; A\,q^2T^2 + B\,r^2,$$

with $A = \frac{1+\kappa}{1-\delta} - 1$, $B = \frac{1+1/\kappa}{1-\delta} + 1$ from $(qT+r)^2 \le (1+\kappa)q^2T^2 + (1+1/\kappa)r^2$, and $T = \tan t \le \tan\beta$ on $[0,\beta)$. The three Poincaré constants for $\eta(0) = \eta(\pi/2) = 0$ are $P_1 = (\pi/2\beta)^2 = 29.4135$ on $[0,\beta]$, $P_2 = (\pi/2(\pi/2-\beta))^2 = 1.5035$ on $[\beta,\pi/2]$ and $P_3 = 1$ on $[\pi/2,\pi]$, each with a Dirichlet endpoint and a free one. Finally $-\big(\eta(t) + \eta'(t+\pi/2)\big)^2 \le -\lambda\,\eta'(t+\pi/2)^2 + \frac{\lambda}{1-\lambda}\eta(t)^2$ turns the E_2 term into $-\lambda\int_{\pi/2}^{\pi-\beta} \eta'^2 + \frac{\lambda}{1-\lambda}(J_1+J_2)$, and $\eta(\pi/2) = 0$ gives $\int_{\pi/2}^{\pi/2+\beta} \eta^2 \le (\beta^2/2)\int_{\pi/2}^{\pi-\beta} \eta'^2$, so the $B\,r^2$ remainder is absorbed as soon as $\lambda \ge B\beta^2/2$. Collecting,

$$\text{coefficient of } J_1 = A\tan^2\beta - \delta P_1 + 1 + \tfrac{\lambda}{1-\lambda}, \qquad \text{of } J_2 = 1 - P_2 + \tfrac{\lambda}{1-\lambda},$$
$$\text{of } J_3 = 1 - P_3 = 0, \qquad \text{of } \int_{\pi/2}^{\pi-\beta} \eta'^2 = B\beta^2/2 - \lambda.$$

At $\delta = \frac{1}{10}$, $\kappa = 2$, $\lambda = 0.117$ these are -1.6013, -0.3710, 0 and -0.005, giving (37). For strictness, equality forces $\eta \equiv 0$ on $[0,\pi/2]$ and η the first Dirichlet–Neumann mode on $[\pi/2,\pi]$; but with $\eta \equiv 0$ on $[0,\pi/2]$ the E_2 term contributes $-\int_0^{\pi/2-\beta} \eta'(t+\pi/2)^2 < 0$. □

The bound $\frac{1}{2}\delta^2 Q \le -0.3710\|\eta\|^2_{L^2(0,\pi/2)}$ that (37) yields is about half of the sharp $-0.7309566\ldots$ for the full interval, the shortfall being the J_3 coefficient that (37) leaves at zero. Theorems 118 and 119 recover it.

Proposition 90. *$\sigma - \alpha_1 = \cos t\,x'(t)$ on each phase, with $x = (F-1)/\cos t$ as in (31). Hence the middle term of (20) is $\frac{1}{2}\int \cos^2 t\,x'(t)^2\,dt$, an energy of the intercept path.*

Proof. $\sigma - \alpha_1 = (F-1)\tan t + F'$ by (17) and (19), and $\cos^2 t\,x' = F'\cos t + (F-1)\sin t$ by (31); dividing by $\cos t$ gives the claim. Verified symbolically on SOL1, SOL5 and SOL6, residual 0 in each case, which also recovers the first and third closed forms of Proposition 71. □

Remark 91 (The area identity)**.** Since $u = 2\tan\beta$, Romik's area is

$$A_R^* = u^2 + 1 + \arctan(u/2) = 1 + 4\tan^2\beta + \beta.$$

Using Proposition 90 to make every integrand elementary, the six blocks of $|C_2|$ and the three blocks of V evaluate at 50 working digits to $|C_2| = 2.0133416126029788281721\ldots$, $V = 0.1841931970887689882517\ldots$, and

$$Q(\Sigma) - \big(1 + 4\tan^2\beta + \beta\big) = -2.7 \cdot 10^{-51}.$$

A symbolic evaluation of the same integrals was attempted and did not terminate; the identity therefore remains HEURISTIC, now against a closed-form target and with a phase-by-phase decomposition.

15 Σ satisfies (RC), and $Q(\Sigma) = A_R^*$

Proposition 92. *Σ satisfies (RC). The surface-measure density is, block by block,*

$$H+H'' = \begin{cases} 0 & \theta \in [0,\beta) \text{ and } \theta \in (\pi-\beta,\pi], \\ \frac{1}{2} & \theta \in (\pi/2-\beta,\pi/2) \text{ and } \theta \in (\pi/2,\pi/2+\beta), \\ \frac{3K}{4}\cos\!\big(\frac{\theta}{2}+\frac{\pi}{8}\big) & \theta \in [\beta,\pi/2-\beta], \\ \frac{3K}{4}\sin\!\big(\frac{\theta-\pi/2}{2}+\frac{\pi}{8}\big) & \theta \in [\pi/2+\beta,\pi-\beta], \end{cases} \qquad K = f_1\sqrt{4-2\sqrt{2}},$$

plus the ceiling atom at $\theta = \pi/2$. Hence

$$\max(H+H'')_{\mathrm{ac}} = \tfrac{3K}{4}\cos\!\big(\tfrac{\beta}{2}+\tfrac{\pi}{8}\big) = 0.8388253494\ldots < 1,$$

and already the amplitude satisfies $\frac{3}{4}K = 0.9765387687\ldots < 1$, so (RC) holds as soon as

$$f_1^2 \;\le\; \frac{16}{9\,(4-2\sqrt{2})} = 1.5174\ldots, \qquad \textit{against } f_1^2 = 1.4470\ldots \tag{38}$$

Proof. Each block of H is read off the three closed forms of Theorem 86. On the first phase $H = \cos\theta + \frac{1}{2}\sin\theta$, so $H + H'' = 0$; the shifted last phase is the same. On the last phase $H = (1-\frac{2}{3}a_1)\cos\theta + \frac{1}{2}\sin\theta + \frac{1}{2}$ and on the shifted first phase $H = (2a_1-1)\sin\theta + \frac{1}{2}\cos\theta + \frac{1}{2}$, each giving $H + H'' = \frac{1}{2}$. On the middle phase only the half-angle terms survive differentiation twice, contributing a factor $1 - \frac{1}{4}$, so $H + H'' = \frac{3}{4}(f_1\cos\frac{\theta}{2} + f_2\sin\frac{\theta}{2})$ and, on the shifted middle phase, $\frac{3}{4}(f_1\sin\frac{\theta}{2} - f_2\cos\frac{\theta}{2})$. Both have amplitude $\frac{3}{4}\sqrt{f_1^2+f_2^2} = \frac{3}{4}f_1\sqrt{4-2\sqrt{2}}$ by $f_2 = (1-\sqrt{2})f_1$, and phase $\pi/8$ because $-f_2/f_1 = \sqrt{2}-1 = \tan(\pi/8)$. On $[\beta,\pi/2-\beta]$ the argument $\frac{\theta}{2}+\frac{\pi}{8}$ runs over $[0.5375, 1.0333] \subset [0,\pi/2]$, where cosine decreases, so the maximum is at $\theta = \beta$. $\square$

Criterion (38) is strictly stronger than the separation criterion $f_1^2 < (2+\sqrt{2})/2 = 1.7071\ldots$ of Theorem 8: within the family in which f_1 is the free parameter, (RC) implies $M < \frac{1}{2}$.

Theorem 93 (conditional on containment)**.** *Assume the truncated sweeps for Σ lie in C_2. Then $Q(\Sigma) = A_R^*$.*

Proof. Σ satisfies (RC) by Proposition 92, so the combined sweep is injective by Theorems 81 and 85; the two sweeps are disjoint and cover N by Remarks 42 and 43; (55) holds by Proposition 136; and containment is the standing hypothesis. With the four, $V(\Sigma) = |N(\Sigma)|$ by Proposition 20. The niches are disjoint by Theorem 8, and C_2 is ρ-invariant, so $|\Sigma| = |C_2| - |N| - |\rho N| = |C_2| - 2|N|$. Hence $Q(\Sigma) = |C_2| - 2V(\Sigma) = |C_2| - 2|N| = |\Sigma| = A_R^*$. □

Remark 94 (The gap this theorem inherits, stated)**.** The proof above previously read "Σ satisfies (RC), so $V(\Sigma) = |N(\Sigma)|$ by Theorems 81 and 85". That is the inference Remark 51 found unsupported: those two theorems give injectivity of the combined sweep and nothing else, whereas $V = |N|$ is Proposition 20 and needs four hypotheses. Three of the four are available for Σ — disjointness and covering from Remarks 42 and 43, (55) from Proposition 136 — and the fourth, containment of the truncated sweeps in C_2, is not proved here. So $Q(\Sigma) = A_R^*$ carries exactly the hypothesis Theorem 115 carries and is not independent evidence for it. Rule 0: Theorem 93 is CONDITIONAL ON CONTAINMENT, not PROVED. The 50-digit numerical agreement below does not use Proposition 20 for Σ and is unaffected; it remains the independent evidence for the value.

No integral is evaluated. Independently, with $A_R^* = 1 + 4\tan^2\beta + \beta$ and the integrands made elementary by Proposition 90, the two sides agree to $2.7 \cdot 10^{-51}$ at 50 working digits, which is a consistency check on the whole chain rather than a proof of it.

Remark 95 (Regression against the one-corner problem)**.** The construction does *not* recover Gerver's sofa, and this should be stated plainly. Applying the same machinery to Gerver's rotation path:

- Gerver satisfies the one-corner injectivity condition everywhere: $\alpha_1 > 0$ and $\alpha_2 > 0$ on all of $[0, \pi/2]$, as expected.
- (RC) *fails* for Gerver's cap: $\max(H + H'')_{\mathrm{ac}} \geq 1.399$ near $\theta = \pi$.
- Gerver's face-2 sweep is genuinely not injective in this decomposition: 33 002 of 158 802 tested pairs meet interior to both segments, with $\min(s_{22} - \alpha_2) = -0.0944$. The face-1 sweep is injective.
- Consistently, $V = 0.6426146$ against $|N| = |C| - |S| = 0.6406$ to 0.6412 at 481 and 721 constraints, i.e. $V > |N|$ as Proposition 41 requires.

So (RC) and the one-corner injectivity condition are independent: Σ satisfies the first and not the second, Gerver the second and not the first. By Remark 80 the difference has a classical reading: Σ's cap is a corridor facet plus a body that slides freely inside the unit disc, whereas part of Gerver's cap boundary is *flatter than the corridor is wide*, its curvature radius reaching at least 1.399. The decomposition $N = W_2 \sqcup W_1^{\mathrm{out}}$ is therefore not the one the one-corner argument uses, and the fact that it fails on the solved case is a standing caution about the framework, recorded rather than explained.

Remark 96 (The separation hypothesis for competitors)**.** Hypothesis (iii) remains open. Over 250 random perturbations $H_\Sigma + \sum_{k=1}^{3} c_k \sin(2k\theta)$ with $|c_k| \leq a/k^2$ for $a = 0.02, 0.05, 0.10$, every H satisfying (RC) had $M \leq 0.3976$, against the required $M < \frac{1}{2}$; the (RC)-satisfying fraction fell from $250/250$ to $34/250$ as a grew. No counterexample and no implication.

16 Two negative findings

Remark 97 (The mirror decomposition does not rescue Gerver)**.** The decomposition of Proposition 20 has a mirror image under $t \mapsto \pi/2 - t$, $\mu \leftrightarrow \nu$: sweep the inner part of face 1 and the outer part of face 2, the latter reaching the floor at distance $\sigma_2 = c_y/\sin t$. For Gerver both give the same value, $0.6426144\ldots$, because Gerver's rotation path is itself symmetric under $t \mapsto \pi/2 - t$; both exceed $|N| \approx 0.641$. So the failure of Remark 95 is not an artefact of orientation.

The structural reason is visible in Proposition 92. Σ's cap has $H + H'' = 0$ on $[0, \beta)$ and $(\pi - \beta, \pi]$, i.e. *vertices*, because Σ's outer phases have constant contact arcs pinned at $(1, \frac{1}{2})$. Gerver's cap has no such vertices: there $H + H''$ reaches at least 1.399 near $\theta = \pi$. The degeneracy that makes Σ hard for local analysis is exactly what makes (RC) hold for it.

Remark 98 (A false bound, retracted)**.** The following was attempted for hypothesis (iii) and is **false**. Solving $H''+H = W$ with $H(0) = 1$ gives $H(\theta) = \cos\theta + H'(0)\sin\theta + \int_0^\theta \sin(\theta-s)W(s)\,ds$, and substituting into $c_y(t) = (H(t) - 1)\sin t + (H(t + \pi/2) - 1)\cos t$ yields

$$c_y(t) = H'(0) - \sin t - \cos t + \sin t \int_0^t \sin(t-s)W\,ds + \cos t \int_0^{t+\pi/2} \cos(t-s)W\,ds.$$

Both kernels are non-negative on their ranges, so "$W \le 1$" would give $c_y \le H'(0)$, and for a ρ-symmetric cap $H'(0) = \frac{1}{2} + \frac{1}{2}$(vertical right edge), apparently yielding $M \le \frac{1}{2}$.

The error is that W is *not* ≤ 1 as a measure: it carries the ceiling atom of mass $1.1670\ldots$ at $s = \pi/2$, and here that atom meets the kernel $\cos(t - \pi/2) = \sin t$, which does not vanish. The corrected bound is $M \le H'(0) + \frac{1}{2} \cdot 1.1670 = 1.083\ldots$, useless against the required $\frac{1}{2}$.

This does *not* affect Theorem 81: there the same atom lands at $u = t$, which is outside the integration range when $\tau < t$ and is the endpoint where the kernel $\sin(\tau - u)$ vanishes when $\tau = t$; and the face-1 atom needs $u = t - \pi/2 \le 0$. The placement of the atom relative to the kernel is what distinguishes the two arguments, and it was checked in both.

Hypothesis (iii) therefore remains open.

17 A finite-angle relaxation, and what it can give

Everything above is local. We record here the one construction in this note that bears on the *global* problem, together with an honest account of how far it has been carried.

Proposition 99 (Finite-angle relaxation)**.** *Let S be an ambidextrous moving sofa. For any finite $F_1, F_2 \subset [0, \pi/2]$,*

$$|S| \;\le\; \sup\Bigg|\bigcap_{t\in F_1} L(t, c_t) \;\cap\; \bigcap_{s\in F_2} R(s, d_s)\Bigg|,$$

the supremum being over all placements $c_t, d_s \in \mathbb{R}^2$, where $L(t, c)$ is the left-turning hallway rotated by t with inner corner c, and $R(s, d)$ the right-turning one, that is, the mirror image rotated by s and placed at d.

Proof. S lies in the hallway at every instant of each of its two motions, so it lies in the intersection over any finite subset of instants, for the placements those instants realise. Dropping the constraints linking the placements to one another enlarges the right-hand side. $\square$

The bound this yields depends on F_1, F_2 and requires the supremum, which is a global maximisation over $2(|F_1|+|F_2|)-2$ real parameters. Two facts are recorded.

Proposition 100 (A limitation at three angles)**.** *For $F_1 = F_2 = \{0, \pi/4, \pi/2\}$ the supremum in Proposition 99 is at least $2\sqrt{2}-1 = 1.8284271\ldots$, so no bound below that value is obtainable at those angles.*

Proof. An explicit placement attaining it is recorded in the accompanying computation; exhibiting one placement suffices. □

Three objects already force this value, the surrounding structure is classical, and the constant has an exact geometric meaning.

A moving sofa lies in the incoming straight corridor before it reaches the corner, so $S \subseteq \mathcal{S}$ for a strip $\mathcal{S}$ of width 1; an ambidextrous one meets a $\pi/4$ hallway of each handedness. Hence

$$|S| \;\le\; \sup_{c,d} \big|\mathcal{S} \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\big|. \tag{39}$$

With only $L_{\pi/4}$ this is Hammersley's bound: each arm of that hallway is a strip of width 1 at $\pi/4$ to $\mathcal{S}$, meeting it in a parallelogram of area $1/\sin(\pi/4) = \sqrt{2}$, so $|S| \le 2\sqrt{2}$. The second handedness changes the geometry, and the following is what the constant $2\sqrt{2}-1$ measures.

Lemma 101 (Cap lemma)**.** *Let T be a unit square whose diagonals are horizontal and vertical, and $\mathcal{S}$ a horizontal strip of width 1. Then $|T \cap \mathcal{S}| \le \sqrt{2} - \frac{1}{2}$, with equality when the strip is centred on T.*

Proof. T has vertical extent $\sqrt{2} > 1$, and its horizontal cross-section at height y from the centre has length $\sqrt{2} - 2|y|$ for $|y| \le \sqrt{2}/2$. The cap above height h has area $\int_h^{\sqrt{2}/2} (\sqrt{2} - 2y)\, dy = \frac{1}{2} - \sqrt{2}\, h + h^2$, decreasing in h on $[0, \sqrt{2}/2]$, so a strip of width 1 removes least when centred, at $h = \frac{1}{2}$:

$$|T \cap \mathcal{S}| \;\le\; 1 - 2\big(\tfrac{1}{2} - \tfrac{\sqrt{2}}{2} + \tfrac{1}{4}\big) = \sqrt{2} - \tfrac{1}{2}.$$

□

All four arms of the two $\pi/4$ hallways are strips of width 1 with normal $(1,1)/\sqrt{2}$ or $(-1,1)/\sqrt{2}$. Two arms from different hallways with perpendicular normals meet in exactly such a square, so each perpendicular pair contributes at most $\sqrt{2} - \frac{1}{2}$ by Lemma 101, and

$$2\big(\sqrt{2} - \tfrac{1}{2}\big) = 2\sqrt{2} - 1.$$

This is not merely numerology: the maximising configuration found for (39) splits into exactly two connected components, each of area $0.914213562 = \sqrt{2} - \frac{1}{2}$, one for each perpendicular pair, with the two parallel pairs empty. The corners sit at heights 0.995169 and 0.004831, symmetric about the midline.

The conjecture below reduces to a one-dimensional problem. Let $\mu = (1,1)/\sqrt{2}$, $\nu = (-1,1)/\sqrt{2}$ and use the orthonormal coordinates $u = \langle p, \mu\rangle$, $v = \langle p, \nu\rangle$; write $a = \langle c, \mu\rangle$, $a' = \langle c, \nu\rangle$, $b = \langle d, \mu\rangle$, $b' = \langle d, \nu\rangle$.

Lemma 102 (Rotated coordinates)**.** *In these coordinates both hallways are axis-aligned and*

$$L_{\pi/4}(c) \cap R_{\pi/4}(d) = \big([b-1, a+1] \times [b'-1, a'+1]\big) \setminus \big(\{u < a,\ v < a'\} \cup \{u > b,\ v > b'\}\big),$$

a rectangle less two opposite quadrants, while $\mathcal{S} = \{0 \le u + v \le \sqrt{2}\}$*. Consequently, with* $s = u + v$ *and* $t = u - v$*,*

$$\left|\mathcal{S} \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\right| = \tfrac{1}{2} \int_0^{\sqrt{2}} \ell(s)\, ds, \tag{40}$$

where $\ell(s)$ *is the length of* $\big[\max(p, r) - 2,\ \min(q, w) + 2\big] \setminus \big((p, q) \cup (r, w)\big)$ *with* $p = s - 2a'$*,* $q = 2a - s$*,* $r = 2b - s$*,* $w = s - 2b'$*. In particular* ℓ *is piecewise linear in* s*, and the two removed intervals are nonempty exactly for* $s < a + a'$ *and* $s > b + b'$ *respectively.*

Proof. $L_{\pi/4}(c) = \{u \le a + 1, v \le a' + 1\} \setminus \{u < a, v < a'\}$ and $R_{\pi/4}(d) = \{u \ge b - 1, v \ge b' - 1\} \setminus \{u > b, v > b'\}$ directly from the definitions, since reflection carries μ, ν to $-\nu, -\mu$. The cross-section of the rectangle at level s is $[\max(p, r) - 2, \min(q, w) + 2]$ in t, the two quadrants cut out (p, q) and (r, w), and $du\, dv = \frac{1}{2}\, ds\, dt$. □

Formula (40) agrees with direct polygon computation to $2.5 \cdot 10^{-11}$ on test placements. The crude bound $\ell \le 4$ recovers Hammersley: it gives $\frac{1}{2} \cdot 4\sqrt{2} = 2\sqrt{2}$. The conjectured value is attained, and can be computed exactly.

Proposition 103 (The extremal configuration)**.** *At* $a = a' = b = b' = \sqrt{2}/4$*, that is with both corners at the same point at mid-corridor height,* $\ell(s) = 4 - 4\left|s - \frac{\sqrt{2}}{2}\right|$ *and* (40) *equals exactly* $2\sqrt{2} - 1$*.*

Proof. Put $\delta = s - \sqrt{2}/2$. Then $p = w = \delta$ and $q = r = -\delta$, so the outer interval is $[|\delta| - 2,\ 2 - |\delta|]$, of length $4 - 2|\delta|$. Exactly one of $(p, q) = (\delta, -\delta)$ and $(r, w) = (-\delta, \delta)$ is nonempty, according to the sign of δ, and it has length $2|\delta|$ and lies inside the outer interval because $|\delta| \le \sqrt{2}/2 < 1$. Hence $\ell = 4 - 4|\delta|$ and $\frac{1}{2} \int_0^{\sqrt{2}} \big(4 - 4|\delta|\big) ds = \frac{1}{2}\big(4\sqrt{2} - 4 \cdot \frac{1}{2}\big) = 2\sqrt{2} - 1$. □

A search of 600 multistarts over $[-3, 3]^4$ in exact piecewise-linear arithmetic returns 1.828427124746, agreeing with $2\sqrt{2} - 1$ to $2.7 \cdot 10^{-15}$, and no configuration exceeding it was found.

The supremum in (39) can be evaluated in closed form on most of the parameter space, and this gives the first upper bound specific to the ambidextrous problem. The key is that the set left after the two quadrants are removed is confined to two short end pieces, whose lengths are controlled by two gauge-invariant combinations of c and d.

Lemma 104 (Four pieces)**.** *With the notation of Lemma 102, set*

$$m_1 = \langle c, \nu\rangle + \langle d, \mu\rangle = a' + b, \qquad m_2 = \langle c, \mu\rangle + \langle d, \nu\rangle = a + b',$$

$$\delta_\mu = \langle d - c, \mu\rangle = b - a, \qquad \delta_\nu = \langle d - c, \nu\rangle = b' - a'.$$

All four are unchanged by the translation $(a, a', b, b') \mapsto (a + \varepsilon, a' - \varepsilon, b + \varepsilon, b' - \varepsilon)$*, and for every* s

$$\ell(s) \ \le\ \big(2 - 2|s - m_1|\big)^+ + \big(2 - 2|s - m_2|\big)^+ + \min\big(2(s - m_1)^+, 2\delta_\mu^+\big) + \min\big(2(s - m_2)^+, 2\delta_\nu^+\big). \tag{41}$$

Proof. Write $I = [\max(p, r) - 2, \min(q, w) + 2]$. Since $I \setminus (p, q) = \{x \in I : x \le p\} \cup \{x \in I : x \ge q\}$ and likewise for (r, w), intersecting the two gives

$$I \setminus \big((p, q) \cup (r, w)\big) \ =\ \big(A \cup B \cup C \cup D\big) \cap I,$$

$$A=(-\infty,\min(p,r)],\quad B=[q,r],\quad C=[w,p],\quad D=[\max(q,w),\infty).$$

Now $|A\cap I|\le \min(p,r)-\max(p,r)+2=(2-|p-r|)^+$ and $p-r=2s-2(a'+b)$, giving the first term; $|D\cap I|\le \min(q,w)+2-\max(q,w)=(2-|q-w|)^+$ and $q-w=2(a+b')-2s$, giving the second. For the cross piece $C=[w,p]$: it is empty unless $p>w$, i.e. $\delta_\nu>0$, and $C\setminus A\subseteq[\min(p,r),p]$, so $|C\setminus A|\le \min\big((p-r)^+,(p-w)^+\big)=\min\big(2(s-m_1)^+,2\delta_\nu^+\big)$. Symmetrically B is empty unless $\delta_\mu>0$ and $|B\setminus D|\le \min\big((w-q)^+,(r-q)^+\big)=\min\big(2(s-m_2)^+,2\delta_\mu^+\big)$. Adding the four bounds can only over-count overlaps. $\square$

Proposition 105 (The profile integral)**.** *Let $g(x)=\int_0^{\sqrt{2}}\big(2-2|s-x|\big)^+ds$. Then $g(\sqrt{2}-x)=g(x)$, g vanishes off $(-1,\sqrt{2}+1)$, on $[\sqrt{2}-1,1]$ it equals $2\sqrt{2}-x^2-(\sqrt{2}-x)^2$, and*

$$\max_{x\in\mathbb{R}} g(x) \;=\; g\big(\tfrac{1}{\sqrt{2}}\big) \;=\; 2\sqrt{2}-1.$$

Proof. The substitution $s\mapsto\sqrt{2}-s$ gives the symmetry. Differentiating under the integral, $g'(x)=2\big|[0,\sqrt{2}]\cap(x-1,x)\big|-2\big|[0,\sqrt{2}]\cap(x,x+1)\big|$, which is ≤ 0 for $x\ge\sqrt{2}/2$ because the second window then contains at least as much of $[0,\sqrt{2}]$ as the first; with the symmetry, g increases up to $\sqrt{2}/2$ and decreases after. For $\sqrt{2}-1\le x\le 1$ the integrand is nonnegative throughout $[0,\sqrt{2}]$, so $g(x)=2\sqrt{2}-2\int_0^{\sqrt{2}}|s-x|\,ds=2\sqrt{2}-x^2-(\sqrt{2}-x)^2$, and at $x=\sqrt{2}/2$ this is $2\sqrt{2}-\frac12-\frac12$. $\square$

Combining (41) with $\ell\ge 0$ and integrating over the band gives a bound in closed form, with no computation of any kind.

Theorem 106 (The sharp constant, off the cross region)**.** *For all c,d,*

$$\big|\mathcal{C}_2\cap L_{\pi/4}(c)\cap R_{\pi/4}(d)\big| \;\le\; \tfrac12\big[g(m_1)+g(m_2)\big]+\sqrt{2}\big[\min(1,\delta_\mu^+)+\min(1,\delta_\nu^+)\big]. \tag{42}$$

In particular, if $\langle d-c,\mu\rangle\le 0$ and $\langle d-c,\nu\rangle\le 0$ then

$$\big|\mathcal{C}_2\cap L_{\pi/4}(c)\cap R_{\pi/4}(d)\big| \;\le\; 2\sqrt{2}-1 \;=\; 1.8284271\ldots,$$

with equality exactly when $m_1=m_2=1/\sqrt{2}$ and $\delta_\mu=\delta_\nu=0$. Consequently every ambidextrous moving sofa whose two corner positions at angle $\pi/4$ satisfy those two inequalities has area at most $2\sqrt{2}-1$.

Proof. Integrate (41) over $s\in[0,\sqrt{2}]$ and halve, as in (40). The first two terms give $\frac12 g(m_1)+\frac12 g(m_2)$ by definition of g; the third is at most $\frac12\int_0^{\sqrt{2}}2\min(1,\delta_\mu^+)\,ds=\sqrt{2}\min(1,\delta_\mu^+)$ and likewise the fourth. If $\delta_\mu\le 0$ and $\delta_\nu\le 0$ the cross terms are absent and Proposition 105 bounds the rest by $\frac12\cdot 2(2\sqrt{2}-1)$. Equality forces both g's to be maximal, hence $m_1=m_2=1/\sqrt{2}$; the configuration $a=a'=b=b'=\sqrt{2}/4$ of Proposition 103 realises it, and (42) is therefore sharp. $\square$

The bound (42) was tested against the objective computed independently from the definition at $4\cdot 10^4$ random parameter triples over scales from 0.1 to 160: the minimum slack is $-1.0\cdot 10^{-14}$, i.e. zero to rounding, so the bound is both correct and attained.

When $\delta_\mu>0$ or $\delta_\nu>0$ the cross terms in (42) are genuinely present and the estimate exceeds $2\sqrt{2}-1$. The next three lemmas remove the sign hypothesis. Throughout, the reflection across the diagonal $u=v$ preserves the band and exchanges (a,a') with (a',a) and (b,b') with (b',b), hence $\delta_\mu\leftrightarrow\delta_\nu$ and $m_1\leftrightarrow m_2$; so assume without loss of generality $t:=\delta_\nu=\max(\delta_\mu,\delta_\nu)>0$.

Lemma 107 (Wide offsets)**.** *If* $t \ge 1$ *then* $\left|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\right| \le \sqrt{2}\,(2-t)^+ \le \sqrt{2}$.

Proof. $w - p = (s - 2b') - (s - 2a') = -2t$ identically in s, so $\ell(s) \le |I| \le 4 + w - p = 4 - 2t$ for every s, and the band has length $\sqrt{2}$: $\frac{1}{2}\int_0^{\sqrt{2}} \ell \le \frac{1}{2}\sqrt{2}\,(4-2t)^+$. □

Lemma 108 (Both offsets positive)**.** *If* $0 < \delta_\mu \le \delta_\nu = t \le 1$ *then* $\ell(s) \le (2 - 2|s - m_1|)^+ + (2 - 2|s - m_2|)^+$ *for every* s*, so the area is at most* $\frac{1}{2}[g(m_1) + g(m_2)] \le 2\sqrt{2} - 1$.

Proof. Write, as in Lemma 104, the kept set as $(A \cup B \cup C \cup D) \cap I$ with $A = (-\infty, \min(p,r)]$, $B = [q,r]$, $C = [w,p]$, $D = [\max(q,w), \infty)$, and note $m_1 \le m_2$ since $\delta_\mu \le \delta_\nu$. For $s \le m_1$ we have $p \le r$ and $w \le q$, so $C \subseteq A$ and $B \subseteq D$, and $\ell \le |A \cap I| + |D \cap I| \le (2 - 2(m_1 - s))^+ + (2 - 2(m_2 - s))^+$ exactly as in the proof of Theorem 106; symmetrically for $s \ge m_2$. On the middle window $m_1 < s < m_2$: $A = (-\infty, r]$, $D = [q, \infty)$, and since $\delta_\mu > 0$ gives $q \le r$,

$$A \cap D = [q,r] = B, \qquad A \cap C \cap D = [q,r] \cap [w,p] = [q,r] = B,$$

the latter because $w \le q$ and $r \le p$ on the window. In the inclusion–exclusion for $|A \cup C \cup D|$ the terms $-|A \cap D|$ and $+|A \cap C \cap D|$ therefore cancel, leaving

$$\ell = |A \cap I| + |D \cap I| + \big[\,|C \cap I| - |A \cap C \cap I| - |C \cap D \cap I|\,\big].$$

Since $t \le 1$, $C = [w,p] \subseteq [p-2, w+2] = I$, so $|C \cap I| = 2t$; and $A \cap C = [w,r]$, $C \cap D = [q,p]$ are subsets of C, of lengths $2(n_2 - s)$ and $2(s - n_1)$ with $n_1 = a + a'$, $n_2 = b + b'$, both nonnegative on the window. The bracket equals $2t - 2(n_2 - n_1) = 2t - 2(\delta_\mu + \delta_\nu) = -2\delta_\mu < 0$, and the first two terms are the two tents. □

Lemma 109 (Mixed offsets, pointwise)**.** *If* $\delta_\mu \le 0 < \delta_\nu = t \le 1$*, write* $u = -\delta_\mu$*, so* $m_2 - m_1 = t + u$*. Then for every* s

$$\ell(s) \;\le\; (2 - 2|s - m_1|)^+ + (2 - 2|s - m_2|)^+ + \min\!\big(2(s - m_1)^+,\ 2(m_2 - s)^+,\ 2\min(t,u)\big).$$

Proof. Outside the middle window the absorption of the previous proof applies verbatim (here $B = \emptyset$). On the window, $A \cap D = [q,r] = \emptyset$ since $\delta_\mu < 0$, and

$$\ell = |A \cap I| + |D \cap I| + c(s), \qquad c(s) = |C \cap I| - |A \cap C \cap I| - |C \cap D \cap I| = 2t - 2(n_2 - s)^+ - 2(s - n_1)^+,$$

by the same computations ($C \subseteq I$ from $t \le 1$; $A \cap C = [w,r]$ when $s \le n_2$ and empty otherwise, likewise $C \cap D$). Now $n_2 = m_1 + t$ and $n_1 = m_2 - t$ give $c \le 2t - 2(n_2 - s) = 2(s - m_1)$ and $c \le 2(m_2 - s)$; and $(n_2 - s)^+ + (s - n_1)^+ \ge (n_2 - n_1)^+ = (t - u)^+$ gives $c \le 2t - 2(t-u)^+ = 2\min(t,u)$. The first two terms are at most the two tents, as before. □

Integrating Lemma 109 over the band reduces everything to one two-variable inequality about the profile g.

Lemma 110 (The profile inequality)**.** *For* $m_1 \le m_2$ *set* $D = m_2 - m_1$, $\tau^* = \min(1, D/2)$ *and*

$$E(m_1, m_2) = \int_0^{\sqrt{2}} \min\!\big(2(s - m_1)^+,\ 2(m_2 - s)^+,\ 2\tau^*\big)\, ds.$$

Then $g(m_1) + g(m_2) + E(m_1, m_2) \le 2\big(2\sqrt{2} - 1\big)$*, with equality exactly at* $m_1 = m_2 = 1/\sqrt{2}$.

Before the four regions, the statement simplifies. Write $C(x) = \int_0^{\sqrt{2}} \min(|s-x|,1)\,ds$. The identity $\max(0,y) = y + \max(0,-y)$ gives $(2-2|s-x|)^+ = 2 - 2\min(|s-x|,1)$ pointwise, hence $g(x) = 2\sqrt{2} - 2C(x)$; and $E = 2\int_0^{\sqrt{2}} F$ with $F(s) = \min\big((s-m_1)^+, (m_2-s)^+, \tau^*\big)$. Substituting, the inequality $g(m_1) + g(m_2) + E \le 2(2\sqrt{2}-1)$ is *equivalent* to

$$\int_0^{\sqrt{2}} \Big[\min(|s-m_1|,1) + \min(|s-m_2|,1) - F(s)\Big] ds \;\ge\; 1. \tag{43}$$

Neither g nor E appears. Since $\int_0^{\sqrt{2}} \min(|s-x|,1)ds \ge \frac{1}{2}$ for every x, which is the pairing argument of the formalisation section, the two positive terms already contribute 1, and (43) says exactly that $\int F$ does not exceed the excess of those two terms over $\frac{1}{2}$ each. On the central square that excess is $x_1^2 + x_2^2$ with $x_i = m_i - \sqrt{2}/2$, and $\int F \le D^2/4 = (x_2-x_1)^2/4$, so region (i) reduces to $(x_2-x_1)^2 \le 4(x_1^2+x_2^2)$, which is $(x_1+x_2)^2 \ge 0$ together with $(x_2-x_1)^2 \le 2(x_1^2+x_2^2)$.

Remark 111 (A tempting simplification that is false)**.** $F \le \min\big(\min(|s-m_1|,1), \min(|s-m_2|,1), \tau^*\big) =: M$ pointwise, so replacing F by M in (43) would give a symmetric statement with no one-sided parts. It is false: on a 241×241 grid of $[-1.6, 3.2]^2$ the left side with M has minimum 0.8751, at $(m_1, m_2) = (0.520, 0.880)$. The loss is at $m_1 = m_2$, where $F \equiv 0$ but $M = \min(|s-m_1|,1)$ does not vanish. The one-sided form is not a convenience.

Proof. Write $C(x) = \int_0^{\sqrt{2}} \min(|s-x|,1)\,ds$ and $e(x) = C(x) - \frac{1}{2}$, which is nonnegative by the pairing argument. By (43) the claim is $\int_0^{\sqrt{2}} F \le e(m_1) + e(m_2)$. Set $D = m_2 - m_1 \ge 0$ and $u_i = m_i - \sqrt{2}/2$.

(L1) $\int F \le D^2/4$. F is supported on $[m_1, m_2]$ and dominated by $\min(s-m_1, m_2-s)$, a triangle of base D and height $D/2$, of area $D^2/4$; when $\tau^* = 1 < D/2$ the profile is instead a trapezoid of area $D-1$, and $D-1 \le D^2/4$ because $(D-2)^2 \ge 0$. Clipping the domain to $[0,\sqrt{2}]$ only decreases the integral.

(L2) $\int F \le \min(C(m_1), C(m_2))$. Pointwise $F \le \tau^* \le 1$ and $F \le (s-m_1)^+ \le |s-m_1|$, so $F \le \min(|s-m_1|,1)$; likewise for m_2.

(L3) $C(x) < 1$ forces $|x - \sqrt{2}/2| \le 1$, and (L4) on that range $e(x) \ge (x-\sqrt{2}/2)^2/2$. Both are finite piecewise-polynomial verifications: with no clipping the pairing identity gives $e(x) = \int_0^{\sqrt{2}/2} 2\,\mathrm{dist}\big(x, [s, s+\sqrt{2}/2]\big)ds$, equal to u^2 for $|u| \le 1 - \sqrt{2}/2$, and clipping contributes the remaining two polynomial pieces.

Assembly. If $\max(C(m_1), C(m_2)) \ge 1$ then by (L2)

$$\int F \;\le\; \min(C_1, C_2) \;=\; C_1 + C_2 - \max(C_1, C_2) \;\le\; C_1 + C_2 - 1 \;=\; e_1 + e_2.$$

Otherwise both $C(m_i) < 1$, so (L3) gives $|u_i| \le 1$ and (L4) gives $e(m_i) \ge u_i^2/2$; then

$$e_1 + e_2 \;\ge\; \tfrac{1}{2}\big(u_1^2 + u_2^2\big) \;\ge\; \tfrac{1}{4}(u_2-u_1)^2 \;=\; \tfrac{D^2}{4} \;\ge\; \int F$$

by (L1), the middle step being $(u_1+u_2)^2 \ge 0$. Equality throughout requires $u_1 = u_2 = 0$, that is $m_1 = m_2 = 1/\sqrt{2}$. ∎

The case split is on a comparison of real numbers, not on a box: no covering, and no interval arithmetic, appears in the proof. An earlier version certified this lemma by an adaptive covering of 28 512 boxes in ball arithmetic. Numerically the assembled inequality has minimum slack $6.6 \cdot 10^{-7}$ over $6 \cdot 10^4$ sampled pairs, attained at $m_1 = m_2 = \sqrt{2}/2$, which is the extremal placement of Proposition 103.

Theorem 112 (The unconditional bound)**.** *Every ambidextrous moving sofa satisfies*

$$|S| \;\leq\; 2\sqrt{2} - 1 \;=\; 1.8284271\ldots$$

Proof. By (39) it suffices to bound the two-hallway area for every placement. If $\delta_\mu \leq 0$ and $\delta_\nu \leq 0$ this is Theorem 106. Otherwise, WLOG $t = \delta_\nu = \max > 0$. If $t \geq 1$, Lemma 107 gives $\sqrt{2} < 2\sqrt{2} - 1$. If $t < 1$ and $\delta_\mu > 0$, Lemma 108 and Proposition 105 give $2\sqrt{2} - 1$. If $t < 1$ and $\delta_\mu \leq 0$, integrate Lemma 109 over the band: the tents give $\frac{1}{2}[g(m_1) + g(m_2)]$ and the flat-top term gives $\frac{1}{2}$ of an integral that is monotone in its height, so with $\min(t, u) \leq \min(1, D/2) = \tau^*$ it is at most $\frac{1}{2}E(m_1, m_2)$; Lemma 110 finishes. □

Since Romik's sofa is ambidextrous, Theorem 112 brackets the ambidextrous supremum:

$$1.6449552\ldots \;=\; A_R^* \;\leq\; A_{\mathrm{ambi}} \;\leq\; 2\sqrt{2} - 1 = 1.8284271\ldots,$$

a factor of 1.1115. The previously available bound was $2.2195\ldots$: an ambidextrous sofa is in particular a one-corner sofa, so it inherits whatever the one-corner problem gives, and since [1] that is Gerver's area itself. Before [1] the inherited value was the 2.37 of [3].

Improving on the inherited bound is a question Kallus and Romik put in print. Their concluding section observes that any one-corner upper bound applies to the ambidextrous variant, calls the gap between the bounds an appealing opportunity, and asks for their techniques to be extended to the ambidextrous setting so as to obtain better upper bounds on the ambidextrous moving sofa constant [3, §6]. Theorem 112 answers that question, though by evaluating a small relaxation in closed form rather than by extending their branch and bound. As of the most recent survey-level treatment [7, Problem 6.63], which records the lower bound $1.6449\ldots$ and states that whether it is sharp remains open, no upper bound specific to the problem had been given.

The constant $2\sqrt{2} - 1$ cannot be improved by adding angles to this pair: it is attained, by Proposition 103, so it is exactly the value of the two-hallway relaxation. The route to a smaller constant is more constraints, not a better estimate of these.

The construction is not new, and its lineage should be stated. Relaxing a sofa to the intersection of finitely many placed hallways is Hammersley's idea [6] and is the framework of Kallus and Romik [3]; the worked example following their Proposition 4 evaluates the one-angle case in closed form, finds the optimum at $\alpha = \pi/4$, and recovers exactly Hammersley's $2\sqrt{2}$, which those authors describe as Hammersley's proof rewritten in their notation. What follows is the two-hallway ambidextrous analogue of that example. The novelty claimed here is the resulting bound, not the method that produces it: we know of no closed-form evaluation of a relaxation with more than one hallway, in either handedness, and Romik has described the multi-hallway maximum as very hard to obtain by hand, which is why [3] passes to interval-arithmetic branch and bound.

Four of the six constraints of Proposition 99 also force this value.

Proposition 113 (Four constraints suffice, and they recover Hammersley)**.** *Write L_θ, R_θ for a left- and a right-turning hallway at angle θ, with free placements. Then*

$$\sup\bigl|L_0 \cap L_{\pi/4} \cap L_{\pi/2}\bigr| = 2\sqrt{2}, \qquad \sup\bigl|L_0 \cap R_0 \cap L_{\pi/4} \cap R_{\pi/4}\bigr| \;\geq\; 2\sqrt{2} - 1,$$

the first being Hammersley's classical bound for the one-corner problem.

The first equality is a regression check rather than a new result: it says the construction reproduces the known elementary bound $2\sqrt{2}$ for one corner, from three left-turning hallways.

That $2\sqrt{2}-1$ is Hammersley's constant less exactly 1 is structural, not a coincidence, and Lemma 101 says where the 1 goes. In Hammersley's configuration the strip meets the $\pi/4$-rotated corridor in two unit squares set on their diagonals, of total area $2\sqrt{2}$. Adding the oppositely-handed hallway at the same angle cuts a quadrant from each square, and Lemma 101 evaluates what survives in one of them as $\sqrt{2}-\frac{1}{2}$. Two of those give $2\sqrt{2}-1$: each square loses exactly $\frac{1}{2}$. Since dropping constraints only enlarges the supremum, the four-constraint value already bounds the six-constraint one, so a proof of

$$\left|L_0 \cap R_0 \cap L_{\pi/4} \cap R_{\pi/4}\right| \;\leq\; 2\sqrt{2}-1 \qquad \text{for all placements} \tag{44}$$

gives $A_{\text{ambi}} \leq 2\sqrt{2}-1$ outright, with no branch and bound and no appeal to the remaining angles. That is what Theorem 106 and Lemmas 107–110 establish: (44) *holds*, since the corridor pair $L_0 \cap R_0$ is the strip whose rotated coordinates give the band, and the four sign cases of the offsets exhaust the placements. Theorem 112 is its restatement for sofas. Two structurally different placements attain the value: one splits the region into two congruent pieces of area $\sqrt{2}-\frac{1}{2}$, one leaves it connected. So $2\sqrt{2}-1$ *is* the supremum at those angles, attained by Proposition 103 and bounded above by Theorem 112, with no interval arithmetic anywhere in the two-hallway argument.

The construction was validated before use: intersecting along Σ's own corner path must give an area decreasing to $|\Sigma|$, and it does, through 2.050458, 2.000690, 1.991986, 1.990120, 1.645249 at $3, 9, 33, 129, 513$ angles per side for the two-family intersection. Two errors were caught by that check, one in the side of the inner corner quadrant and one in the identification of the mirrored family, before any bound was computed.

18 Relation to prior work

The one-corner problem was settled by Baek [1], who proved Gerver's sofa optimal. Our reading of that paper, for the purpose of separating what is his from what is not:

- The *arm lengths* f, g are his (Definition 6.2.1), defined through the contact points A and C; our α_1, α_2 are these shifted by the corridor width.
- His injectivity condition (Definition 6.1.2) has three parts. Part (1) requires the surface measure σ_K to be absolutely continuous on $[0, \pi/2)$ and on $(\pi/2, \pi]$, so an atom is permitted only at $\theta = \pi/2$; part (3) is $x'_K(t) \cdot u_t < 0$ and $x'_K(t) \cdot v_t > 0$ on $(0, \pi/2)$.
- Our (RC) shares part (1) verbatim and *adds* a bound on the density, while dropping part (3) entirely. No curvature bound appears in [1]; its only mentions of curvature are the standard identification of the surface-measure density with the radius of curvature, which we also use. The bound itself is classical, however: it is the sliding-ball condition of Remark 80, so (RC) is a known condition put to a new use rather than a new condition.
- His concavity argument attaches Mamikon regions to the overestimating region so that the total is linear in the cap; ours subtracts them from the cap. He works on a space of triples (K, B, D) of convex bodies; we work with the single support function H on $[0, \pi]$. Whether the two bookkeepings are equivalent we do not determine.

- The ambidextrous problem is not treated in [1], and we have found no upper bound for it in the literature.

The constants a_1, β, f_1 and the ODE families are Romik's [5]; the cubics of Section 7 are the standard ones and we claim no priority for them. This audit was made against [1] and the Kallus–Romik line [3] only, not against the wider literature.

19 The main theorem

The concavity input is stated below for the fixed-T form of (d). The point is that (d) does not merely make the two sign sets ordered; together with (b) it makes the larger one anchored, which is what Corollary 123 asks for.

Proposition 114 ((b) and (d) put every competitor on an ordered anchored cell)**.** *Let H satisfy (b) and (d). Then $E_1 := \{\alpha_1 < 0\} \subseteq [0, T)$ and $E_2 := \{\alpha_2 > 0\}$ is an initial segment of $[0, \pi/2]$ containing $[0, T]$. In particular $E_1 \subseteq E_2$ with E_2 anchored, which is the hypothesis of Corollary 123.*

Proof. By (d), $\alpha_1 \geq 0$ on $[T, \pi/2]$, so $E_1 \subseteq [0, T)$; and $\alpha_2 > 0$ on $[0, T]$, so $[0, T] \subseteq E_2$ and hence $E_1 \subseteq E_2$. On $[T, \pi/2]$ the arm sandwich (Lemma 132) and $\alpha_1 \geq 0$ give $\alpha_2' \leq -\alpha_1 \leq 0$, so α_2 is non-increasing there; therefore $E_2 \cap [T, \pi/2]$ is an interval with left endpoint T, and E_2 is $[0, \tau)$ or $[0, \tau]$ for some $\tau \in [T, \pi/2]$. The two differ in one point, which the cell form (48) does not see.

Note that no non-degeneracy hypothesis is needed, unlike in Proposition 27: (d) supplies the strict sign $\alpha_2 > 0$ on $[0, T]$ outright, so there is no touching point to exclude. □

Theorem 115 (conditional on containment)**.** *Assume the truncated sweeps lie in C_2, which by Remarks 44 and 51 is not implied by the other hypotheses and is open for Σ. Let $\mathcal{D}$ be the set of support functions H on $[0, \pi]$ of admissible caps such that*

(a) $H(0) = H(\pi/2) = 1$ (the gauge: horizontal translation, unit corridor)*;*

(b) (RC) *σ_K is absolutely continuous on $[0, \pi/2)$ and $(\pi/2, \pi]$ with density at most 1, the only atom being at $\theta = \pi/2$; equivalently, on $(0, \pi)$ $0 \leq (H + H'')_{\mathrm{ac}} \leq 1$ with no atom, together with $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$, the last two being the vanishing of the junction atoms of the ρ-extension (Lemma 133);*

(c) $c_y(t) = (H(t) - 1)\sin t + (H(t + \pi/2) - 1)\cos t \leq \frac{1}{2}$ *for $t \in [0, \pi/2]$* (note $\leq$, not $<$: this gives $|U \cap \rho U| = 0$, which is all an area bound needs)*;*

(d) there is a fixed $T \in (0, \pi/2)$ with $\alpha_2 > 0$ on $[0, T]$ and $\alpha_1 \geq 0$ on $[T, \pi/2]$.

Then $\mathcal{D}$ is convex, $\Sigma \in \mathcal{D}$, and

$$|T| \;\leq\; A_R^*$$

for every ambidextrous moving sofa T whose cap data lies in $\mathcal{D}$.

Proof. Each of (a)–(d) is a family of affine inequalities or equalities in H, by Lemma 18 and (17)–(19), so $\mathcal{D}$ is convex. Σ satisfies (b) by Proposition 92, (c) with margin $\frac{1}{2} - M = 0.1121\ldots$ by Theorem 8, and (d) by definition of β.

Proposition 20 is applied with all four of its hypotheses named. By (b) and Theorems 81 and 85 the combined sweep is injective and the two sweeps are disjoint; the covering argument inside the proof of Proposition 76 gives that the two truncated sweeps cover N; (55), which Proposition 20 needs for its face-1 window to be the interval $[\alpha_1^+, \sigma]$, holds by Proposition 136; and containment of the truncated sweeps in C_2 is the standing hypothesis of this theorem. With the four, $V = |N|$ by Proposition 20. Proposition 76 is *not* cited for containment: its argument places an advancing face point in $Q_{t+\varepsilon}$, which gives sweep $\subseteq U$ and not sweep $\subseteq U \cap C_2 = N$, as Remark 51 records. By (c) and Lemma 1, $U \subseteq \{y \le \frac{1}{2}\}$ and $\rho U \subseteq \{y \ge \frac{1}{2}\}$, so $|U \cap \rho U| = 0$; with C_2 ρ-invariant this gives $|T| \le |C_2| - |N| - |\rho N| = |C_2| - 2|N| = Q$. (Corollary 2 states the strict form $M < \frac{1}{2}$, which is what Σ satisfies; the null-set form used here is what the hypothesis $\le$ delivers.) By (b), (d) and Proposition 114 every $H \in \mathcal{D}$ sits on an ordered anchored cell, so $\delta^2 Q \le 0$ on $\mathcal{D}$ by Corollary 123. (Earlier versions cited Remark 89 here. That is superseded, and it covers only the single cell of Σ's own sign pattern, so it cannot serve a statement about all of $\mathcal{D}$; Corollary 123 is the live result, and Proposition 114 is what checks that the fixed-T form of (d) meets its hypothesis.) By Theorem 86, $\delta Q(\Sigma) = 0$. A concave function with a critical point on a convex set attains its maximum there, so $Q(H) \le Q(\Sigma)$, and $Q(\Sigma) = A_R^*$ by Theorem 93. Combining, $|T| \le Q(H) \le A_R^*$. □

20 The second variation

The bound above needs $\delta^2 Q \le 0$ on the cells the segment meets. This section establishes that, with the best constant, and on a family of cells rather than one.

Two routes to the constant are recorded and only the second is load-bearing. The first splits $[0, \pi/2]$ into $[0, \beta]$ and $[\beta, \pi/2]$, applies a separate Poincaré inequality on each, and discards the E_2 term, giving the weak constants of Remark 89. Treating each half as a *single* weighted eigenvalue problem removes both losses; not splitting at all is better still and gives the exact value.

Proposition 116 (An elementary constant, no computer)**.** *On the cell $\mathcal{C}$, for every η with $\eta(0) = \eta(\pi/2) = 0$,*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -\tfrac{1}{12}\,\|\eta\|^2_{L^2(0,\pi)},$$

by elementary inequalities alone: no transfer matrix, no oscillation argument, and no ball arithmetic anywhere in the proof.

Proof. Run the decoupling of Theorem 118 at $\lambda = \frac{9}{20}$, $\kappa = \frac{1}{4}$, so $r = \frac{9}{11}$ and $q = 5$, and bound the two halves by hand, using only $\beta \le \frac{3}{10}$ and $\frac{\pi}{2} \le 1.5708$.

Left half. $w_L \ge 1 - \kappa = \frac{3}{4}$ and $m_L \le 2 + r$, and the Dirichlet–Dirichlet Poincaré inequality on $[0, \frac{\pi}{2}]$ gives $\int p'^2 \ge 4 \int p^2$, so

$$\int w_L p'^2 - \int m_L p^2 \;\ge\; \left[3 - 2 - \tfrac{9}{11}\right] \int p^2 \;=\; \tfrac{2}{11} \int p^2.$$

Right half. Write $c = \frac{\pi}{2} - \beta$ and $s = \lambda - q\beta^2/2$. Three elementary steps. The heavy mass sits on $[0, \beta)$, next to the Dirichlet end, so $q \int_0^\beta \eta^2 \le q \frac{\beta^2}{2} \int_0^\beta \eta'^2 \le \frac{9}{40} \int_0^c \eta'^2$ by the half-Poincaré inequality $\eta(t)^2 \le t \int_0^t \eta'^2$; this absorbs the mass into the weight surplus and leaves $s = \frac{9}{20} - \frac{9}{40} = \frac{9}{40}$. The surplus converts to mass through the cut at c: by the same half-Poincaré inequality and $\eta(t)^2 \le 2\eta(c)^2 + 2\beta \int_c^{\pi/2} \eta'^2$ for $t \ge c$,

$$\int_0^{\pi/2} \eta^2 \;\le\; \left(\tfrac{c^2}{2} + 2\beta c\right) \int_0^c \eta'^2 + 2\beta^2 \int_c^{\pi/2} \eta'^2 \;\le\; \tfrac{11}{5} \int_0^c \eta'^2 + \tfrac{9}{50} \int_c^{\pi/2} \eta'^2,$$

so $\int_0^c \eta'^2 \geq \frac{5}{11}\int \eta^2 - \frac{9}{110}\int_c^{\pi/2}\eta'^2$. With the Dirichlet–Neumann Poincaré inequality $\int \eta'^2 \geq \int \eta^2$ on $[0, \frac{\pi}{2}]$,

$$\int w_R \eta'^2 - \int m_R \eta^2 \;\geq\; \Big(1 - \tfrac{9}{40}\cdot\tfrac{9}{110}\Big)\int \eta'^2 - \Big(1 - \tfrac{9}{40}\cdot\tfrac{5}{11}\Big)\int \eta^2 \;\geq\; \tfrac{369}{4400}\int \eta^2.$$

Since $\frac{369}{4400} > \frac{1}{12}$ and $\frac{2}{11} > \frac{1}{12}$, the two halves give the claim. Every constant is rational and every inequality is Cauchy–Schwarz, a Poincaré inequality on an interval, or arithmetic. □

Remark 117 (A rational certificate for the decoupled half)**.** The gap between $\frac{1}{12}$ and the decoupled truth is closed almost entirely by a certificate that is still exact arithmetic. In Prüfer form the half-problem $-(w\eta')' - m\eta = c\eta$ becomes $\theta' = \cos^2\theta/w + (m+c)\sin^2\theta$ with $\theta(0) = 0$, and $c_1 > c$ exactly when θ stays below $\pi/2$; a piecewise-linear B with $B(0) = 0$ and $B' \geq$ the right side dominates θ, so a barrier reaching less than $\pi/2$ is a proof and every check is an inequality between rationals. The identity

$$\frac{\cos^2\theta}{w} + (m+c)\sin^2\theta \;=\; \frac{1}{w} + \Big(m + c - \frac{1}{w}\Big)\sin^2\theta$$

is what makes it work: the coefficient is positive on every piece, so an upper bound on $\sin^2\theta$ gives one on the whole, which is exactly what an upper barrier supplies. Bounding $\cos^2 \leq 1$ instead is fatal, since the slope then never decays and the barrier crosses $\pi/2$ prematurely. With sin bounded above by $x - x^3/6 + x^5/120$, every constant rounded in the safe direction, and slopes rounded upward to denominator 10^7, 1200 segments give

$$c_1 \;>\; \tfrac{37}{100}$$

for the right half. The left half is Dirichlet–Dirichlet, so the phase runs to π, and there the method has a genuine weakness: once an upper barrier passes $\pi/2$ it knows only $\sin^2 \leq 1$, because an upper bound on θ is a *lower* bound on sin beyond the maximum. At the note's $(\lambda,\kappa) = (\frac{9}{20}, \frac{1}{4})$ that made the left half, not the right, the limit. Raising κ shrinks the left weight until the left half is slack; scanning both halves, $(\lambda,\kappa) = (\frac{11}{20}, \frac{1}{5})$ gives

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\leq\; -\tfrac{9}{20}\,\|\eta\|^2_{L^2(0,\pi)}$$

on the cell, by exact rational arithmetic: 5.4 times the $\frac{1}{12}$ of Proposition 116 and 61.6% of the sharp c^*. The true decoupled values there are $c_R = 0.4748$ and $c_L = 2.4414$, so the certificate captures 95% of what is available and the right half binds. Both negative controls behave: the right half at $c = 1$ crosses $\pi/2$ and the left at $c = 3$ crosses π (`ambi_barrier.py`). A two-sided barrier, tracking a lower phase as well as an upper one, does *not* remove the limitation, and the reason is worth recording. It does recover information past $\pi/2$: with a bracket $[L, B]$ the bound $\sin^2 \leq 1$ is replaced by $\max(\sin^2 L, \sin^2 B)$, which the trace shows falling back to 0.55 and then 0.26 once the bracket clears the maximum. But beyond $\pi/2$ the upper barrier's slope must be driven by $\sin^2 L$, the value at the *lower* endpoint, because sin decreases there; the lagging endpoint therefore sets the pace and the bracket widens. At $(\frac{9}{20}, \frac{1}{4})$ the left half still fails at $c = 1$, the barrier reaching π at $t = 1.5624$ against $T = 1.5708$, and refining the mesh to 20 000 segments does not close that gap, so the loss is structural rather than a discretisation artefact. At $(\frac{7}{10}, \frac{2}{5})$ it fails at every c tested. Reaching $\frac{2}{3}$ by a barrier would need a genuinely different device, not a finer one. Both the monotone bound and the comparison principle that turns a barrier into a bound on the phase are machine-checked (`prufer_rhs_eq`, `prufer_rhs_le`, `sin_sq_le_of_le`, `coef_pos_right`, `coef_pos_left`, `prufer_comparison`, `barrier_certifies`). The comparison is not an ODE-theoretic fact requiring a Lipschitz constant or a Grönwall bound: it is the fencing lemma of the mean value theory, whose hypothesis is checked only where the two functions touch, and at a touching point the barrier's own strictness gives the conclusion.

The constant is 11% of the sharp c^*, and that is the price of dispensing with the transfer matrices: Theorem 118 reaches $\frac{2}{3}$ and Theorem 119 the sharp $\frac{73}{100}$, both through certified evaluations. What Proposition 116 buys is qualitative: strict concavity of the form on the cell, hence the local maximality statements at Σ, no longer depends on any computation.

Theorem 118 (Concavity, sharp order)**.** *On the cell $\mathcal{C}$, for every η with $\eta(0) = \eta(\pi/2) = 0$,*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -\tfrac{2}{3}\,\|\eta\|^2_{L^2(0,\pi)}. \tag{45}$$

Proof. Fix $\lambda \in (0,1)$ and $\kappa \in (0,1)$, and put $r = \lambda/(1-\lambda)$ and $q = 1 + 1/\kappa$. Two elementary inequalities dispose of the two coupling terms of (24). Since $(1+r)(1-\lambda) = 1$,

$$(1+r)a^2+2ab+(1-\lambda)b^2 = \big(\sqrt{1+r}\,a+\sqrt{1-\lambda}\,b\big)^2 \;\ge\; 0, \qquad \text{that is} \qquad -(a+b)^2 \;\le\; -\lambda\,b^2+r\,a^2,$$

applied with $a = \eta(t)$, $b = \eta'(t+\pi/2)$ on $E_2 = [0,\pi/2-\beta)$; it buys gradient weight λ on $[\pi/2, \pi-\beta]$ at the price of mass r on $[0,\pi/2-\beta)$. And $(x-y)^2 \le (1+1/\kappa)x^2 + (1+\kappa)y^2$ with $x = \eta(t+\pi/2)$, $y = \eta'(t)$ on $E_1 = [0,\beta)$ pays mass q on $[\pi/2, \pi/2+\beta]$ and gradient weight $1+\kappa$ on $[0,\beta)$. Substituting $s = t + \pi/2$ in both, the two halves decouple:

$$\tfrac{1}{2}\delta^2 Q[\eta] \;\le\; \int_0^{\pi/2}\big(m_L\eta^2 - w_L\eta'^2\big) + \int_{\pi/2}^{\pi}\big(m_R\eta^2 - w_R\eta'^2\big) \;=:\; B[\eta],$$

$$\begin{aligned} &w_L = 1-\kappa \text{ on } [0,\beta),\ 2 \text{ after}, && m_L = 2+r \text{ on } [0,\tfrac{\pi}{2}-\beta),\ 2 \text{ after},\\ &w_R = 1+\lambda \text{ on } [\tfrac{\pi}{2},\pi-\beta),\ 1 \text{ after}, && m_R = 1+q \text{ on } [\tfrac{\pi}{2},\tfrac{\pi}{2}+\beta),\ 1 \text{ after}. \end{aligned}$$

Each half is a Rayleigh quotient, so $B[\eta] \le -\min(c_L, c_R)\,\|\eta\|^2_{L^2(0,\pi)}$, where c_L and c_R are the first eigenvalues of

$$-(w_L\eta')'-m_L\eta = c\,\eta \ \text{ on } (0,\tfrac{\pi}{2}), \quad \eta(0)=\eta(\tfrac{\pi}{2})=0; \qquad -(w_R\eta')'-m_R\eta = c\,\eta \ \text{ on } (\tfrac{\pi}{2},\pi), \quad \eta(\tfrac{\pi}{2})=0,\ \eta'(\pi)=0.$$

The Dirichlet conditions at 0 and $\pi/2$ are the gauge, not a restriction, and nothing pins $\eta(\pi)$.

Both coefficients are piecewise constant with three pieces, so the solution of the initial value problem $\eta = 0$, $w\eta' = 1$ at the left endpoint is a product of three transfer matrices acting on $(\eta,\, w\eta')^\top$,

$$T(\ell, w, m; c) = \begin{pmatrix} \cos k\ell & (wk)^{-1}\sin k\ell \\ -wk\sin k\ell & \cos k\ell \end{pmatrix}, \qquad k = \sqrt{(m+c)/w}$$

(hyperbolic when $m + c < 0$). Write $\Phi_L(c)$ for the resulting $\eta(\pi/2)$ and $\Phi_R(c)$ for $(w\eta')(\pi)$. Its zeros are exactly the eigenvalues, and by Sturm oscillation $\Phi > 0$ on $(-\infty, c_1)$, so a single sign evaluation bounds c_1 from *below*, which is the direction a certificate needs and the one a Rayleigh–Ritz or finite-element computation cannot supply.

Positivity alone does not locate c, because $\Phi > 0$ again on (c_2, c_3); one also needs $c < c_2$. That comes from min–max, crudely. Since $\int w\eta'^2 \ge w_{\min}\int\eta'^2$ and $-\int m\eta^2 \ge -m_{\max}\int\eta^2$, the form dominates the constant-coefficient one, so $c_k \geq w_{\min}\lambda_k - m_{\max}$ with λ_k the eigenvalues on an interval of length $\pi/2$, namely $(2k)^2$ for Dirichlet–Dirichlet and $(2k-1)^2$ for Dirichlet–Neumann. Hence

$$c_2^L \;\geq\; \tfrac{3}{4}\cdot 16 - \tfrac{63}{13} = 7.153\ldots, \qquad c_2^R \;\geq\; 1\cdot 9 - 6 = 3,$$

both above $\frac{2}{3}$. (At $k=1$ the same bound gives -1.85 and -5, useless; the second eigenvalue is far enough away to be reached crudely while the first is not, which is exactly why the transfer matrix is needed.)

It remains to evaluate two signs. At $\kappa = \frac{1}{4}$, $\lambda = \frac{37}{50}$ and $c = \frac{2}{3}$ every weight and mass is rational ($w_L = \frac{3}{4}, 2, 2$ and $m_L = \frac{63}{13}, \frac{63}{13}, 2$; $w_R = \frac{87}{50}, \frac{87}{50}, 1$ and $m_R = 6, 1, 1$), so each $k^2 = (m+c)/w$ is exact, and the only inexact inputs are the three lengths β, $\frac{\pi}{2} - 2\beta$, β and the trigonometric functions of $k\ell$. In ball arithmetic at 300 bits, with $\beta = \arctan(u/2)$ enclosed from $u^3 + 3u = 2$ by exact rational bisection,

$$\Phi_L(\tfrac{2}{3}) \in 0.0046739486452026048\ldots{\pm}6.4{\cdot}10^{-73}, \qquad \Phi_R(\tfrac{2}{3}) \in 0.0026447861413929500\ldots{\pm}8.3{\cdot}10^{-73},$$

both certified strictly positive. Hence $c_L, c_R > \frac{2}{3}$, which is (45). □

Optimising the two parameters gives $\min(c_L, c_R) = 0.674944$ at $(\kappa, \lambda) = (0.26, 0.74)$. Since $\|\eta\|_{L^2(0,\pi/2)} \leq \|\eta\|_{L^2(0,\pi)}$, (45) also improves the $[0, \pi/2]$ constant of Theorem 89 from 0.371 to $\frac{2}{3}$.

The splitting into two scalar problems is itself avoidable, and avoiding it gives the sharp constant.

Theorem 119 (The sharp constant)**.** *The best constant in* (45) *is the first eigenvalue c^* of an explicit 2×2 Sturm–Liouville system, and*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\leq\; -\tfrac{73}{100}\,\|\eta\|^2_{L^2(0,\pi)}, \qquad c^* = 0.7309566\ldots \tag{46}$$

Proof. The two halves of $[0, \pi]$ are the same interval re-indexed. Put $p(t) = \eta(t)$ and $q(t) = \eta(t + \pi/2)$ for $t \in [0, \pi/2]$; then (24) reads $\frac{1}{2}\delta^2 Q = \int_0^{\pi/2} L$ with

$$L = 2p^2 - 2p'^2 + q^2 - q'^2 - \mathbf{1}_{E_2}(p + q')^2 + \mathbf{1}_{E_1}(q - p')^2,$$

and every boundary condition is forced: $p(0) = p(\pi/2) = 0$ is the gauge, $q(0) = \eta(\pi/2) = 0$ is the same gauge seen from the other half, and $q(\pi/2) = \eta(\pi)$ is free. Nothing has been discarded, so the first eigenvalue of this system is the best constant. Writing $M := -L$ as $ap'^2 + bq'^2 + gp^2 + dq^2 + 2e\,pq' + 2f\,qp'$,

piece	a	b	g	d	e	f
$[0, \beta)$	1	2	-1	-2	1	1
$[\beta, \frac{\pi}{2} - \beta)$	2	2	-1	-1	1	0
$[\frac{\pi}{2} - \beta, \frac{\pi}{2})$	2	1	-2	-1	0	0

On $[0, \beta)$, exactly where the obstruction sits, $e = f = 1$ and $2pq' + 2qp' = 2(pq)'$ is a total derivative: the E_1 obstruction and the E_2 resource cancel there up to the single boundary value $2p(\beta)q(\beta)$. The coupling is not pointwise, which is why every pointwise splitting loses and this one does not.

Eliminating p', q' in favour of the momenta $P = ap' + fq$ and $Q = bq' + ep$, the Euler–Lagrange system is first order in (p, P, q, Q) with piecewise constant matrix

$$A(c) = \begin{pmatrix} 0 & 1/a & -f/a & 0 \\ g - c - e^2/b & 0 & 0 & e/b \\ -e/b & 0 & 0 & 1/b \\ 0 & f/a & d - c - f^2/a & 0 \end{pmatrix},$$

so the transfer across a piece of length ℓ is $\exp\bigl(A(c)\ell\bigr)$. The left conditions $p(0) = q(0) = 0$ leave a two-dimensional solution space; propagating a basis through the three pieces and imposing $p(\pi/2) = 0$ and $Q(\pi/2) = 0$ gives a 2×2 determinant $\Phi(c)$ whose zeros are exactly the eigenvalues. A sign change brackets $c^* \in (0.7309, 0.7310)$.

For the certificate the oscillation argument is unavailable, since for a system the count is a Maslov index rather than a zero count, and it is not needed. Two facts compose: $c^* \geq \frac{2}{3}$ by Theorem 118, and Φ has no zero on $[\frac{2}{3}, \frac{73}{100}]$. The latter is checked by covering that interval with 60 balls and verifying in 400-bit ball arithmetic that no enclosure of Φ contains 0, with β enclosed from $u^3 + 3u = 2$ by exact rational bisection. Hence no eigenvalue lies at or below $\frac{73}{100}$. □

The certified $\frac{73}{100}$ is 99.9% of c^*; the decoupled $\frac{2}{3}$ is 91.2%. Both are lower bounds, and no step in either is floating point. The covering route is also self-contained: with the crude spectral bound -4 of Theorem 122(c), covering $[-4, \frac{73}{100}]$ at the fixed lengths $(\beta, \frac{\pi}{2} - 2\beta, \beta)$ with 3074 boxes certifies $c_1 > \frac{73}{100}$ directly, without the decoupled chain or the oscillation argument; the negative control at $\frac{74}{100}$ is refused with the failing box at $c \in [0.7310, 0.7313]$, which lies just above $c^* = 0.7309566\ldots$ and within $4 \cdot 10^{-5}$ of it: the enclosure of Φ there is still wide enough to contain 0, which is how a target past c^* fails. The decoupled chain remains the hand-checkable route to $\frac{2}{3}$. Nothing in the argument stops at $\frac{73}{100}$: certifying a target c needs the enclosures to separate from 0 on every subinterval of $[\frac{2}{3}, c]$, and those must shrink as c approaches the root, so the number of balls grows: $60, 60, 800, 12000$ for targets $0.73, 0.7305, 0.7309, 0.73095$. The last is 99.9991% of c^*. We quote $\frac{73}{100}$ because a reader can check it with 60 balls; the limit is arithmetic cost, not a gap.

Remark 120 (The $[\pi/2, \pi]$ eigenvalue in closed form)**.** When only the weight is raised and the mass is left at 1 – that is, $w = 1 + \lambda_J$ on $[\pi/2, \pi - \beta]$ and 1 on $[\pi - \beta, \pi]$ – the transfer product collapses to a transcendental equation for the first eigenvalue $\Lambda(\lambda_J)$ of $-(w\eta')' = \Lambda\eta$ with $\eta(\pi/2) = 0$, $\eta'(\pi) = 0$:

$$\sqrt{w_1}\,\cot\bigl(L_1\sqrt{\Lambda/w_1}\bigr) = \tan\bigl(L_2\sqrt{\Lambda}\bigr), \qquad w_1 = 1 + \lambda_J, \quad L_1 = \tfrac{\pi}{2} - \beta, \quad L_2 = \beta. \tag{47}$$

At $\lambda_J = 0$ this reads $\cot(L_1\sqrt{\Lambda}) = \cot(\pi/2 - L_2\sqrt{\Lambda})$, so $\sqrt{\Lambda}\,(L_1 + L_2) = \pi/2$; since $L_1 + L_2 = \pi/2$ *exactly*, $\Lambda(0) = 1$ exactly. That is the marginal case $P_3 = 1$ of Theorem 89, and it explains why no gain is available on $[\pi/2, \pi]$ until the weight is raised. For $\lambda_J > 0$ put $G(w_1) = \sqrt{w_1}\cot(L_1/\sqrt{w_1}) - \tan\beta$. Then $G(1) = 0$, and G increases: the prefactor increases, while $L_1/\sqrt{w_1}$ decreases and cot decreases on $(0, \pi)$. So $G > 0$ for $w_1 > 1$. The left side of (47) decreases in Λ and the right side increases, so the root moves right and $\Lambda(\lambda_J) > 1$ for every $\lambda_J > 0$:

λ_J	0	0.05	0.10	0.16483	0.25	0.35
Λ	1	1.049467	1.098883	1.162878	1.246819	1.345183

Theorem 118 runs this mechanism on both halves at once, raising the mass as well as the weight.

Concavity on every ordered anchored cell

The concavity results so far concern single cells: Σ's own (Theorems 89, 118, 119) and the eleven measured entries of Proposition 145 below. This subsection proves concavity for the whole ordered anchored family at once. In the variables $p(t) = \eta(t)$, $q(t) = \eta(t + \pi/2)$ on $[0, \pi/2]$, with $p(0) = p(\pi/2) = q(0) = 0$ and $q(\pi/2)$ free, the cell form is

$$B_{E_1,E_2}[\eta] = \int_0^{\pi/2} \left[2p^2 - 2p'^2 + q^2 - q'^2\right] - \int_{E_2} (p + q')^2 + \int_{E_1} (q - p')^2, \tag{48}$$

where $E_1 = \{\alpha_1 < 0\}$ and $E_2 = \{\alpha_2 > 0\}$.

Lemma 121 (Monotonicity in the sign sets). *If $E_1 \subseteq E_2$ then $B_{E_1,E_2}[\eta] = B_{E_2,E_2}[\eta] - \int_{E_2 \setminus E_1} (q - p')^2 \le B_{E_2,E_2}[\eta]$.*

Proof. The two forms differ only in the set carrying the $(q - p')^2$ term. $\square$

So every cell with $E_1 \subseteq E_2$ and E_2 *anchored* is dominated by a member of the one-parameter diagonal family $D_\tau := B_{[0,\tau),[0,\tau)}$, and it suffices to treat that family.

Theorem 122 (Diagonal concavity). *$D_\tau[\eta] \le 0$ for every $\tau \in [0, \pi/2]$ and every admissible η.*

Proof. On $[0, \tau)$ the integrand of (48) combines, by the algebra of Theorem 119, into $(p^2 - p'^2) + 2(q^2 - q'^2) - 2(pq)'$, so with $p(0)q(0) = 0$,

$$D_\tau = \int_0^\tau \left[(p^2 - p'^2) + 2(q^2 - q'^2)\right] - 2p(\tau)q(\tau) + \int_\tau^{\pi/2} \left[2(p^2 - p'^2) + (q^2 - q'^2)\right]. \tag{49}$$

Three overlapping ranges of τ are treated by three arguments.

Range (a): $\tau \le 1/\sqrt{3}$. Write (49) as $B_{\text{bare}} + \int_0^\tau [(p'^2 - p^2) + (q^2 - q'^2)] - 2p(\tau)q(\tau)$ with $B_{\text{bare}} := \int_0^{\pi/2} [2(p^2 - p'^2) + (q^2 - q'^2)]$. The Poincaré constants for the boundary conditions are 4 (Dirichlet–Dirichlet, for p) and 1 (Dirichlet–Neumann, for q), so $B_{\text{bare}} \le -\frac{3}{2} \int_0^{\pi/2} p'^2$. The added terms are bounded by $\int_0^\tau p'^2 \le \int_0^{\pi/2} p'^2$, by $\int_0^\tau q^2 \le \tau^2 \int_0^\tau q'^2$ (from $q(t)^2 \le t \int_0^t q'^2$), and, using Young's inequality at weight $s = 1/(2\tau)$ together with $p(\tau)^2 \le \tau \int_0^\tau p'^2$ and $q(\tau)^2 \le \tau \int_0^\tau q'^2$,

$$2|p(\tau)q(\tau)| \;\le\; \frac{p(\tau)^2}{2\tau} + 2\tau\, q(\tau)^2 \;\le\; \tfrac{1}{2} \int_0^{\pi/2} p'^2 + 2\tau^2 \int_0^\tau q'^2.$$

Collecting, $D_\tau \le \left(-\frac{3}{2} + 1 + \frac{1}{2}\right) \int_0^{\pi/2} p'^2 + (3\tau^2 - 1) \int_0^\tau q'^2 \le 0$ for $\tau^2 \le \frac{1}{3}$.

Range (b): $\sigma := \pi/2 - \tau \le \frac{1}{18}$. The obstruction here is that the bound must degenerate: $D_{\pi/2}$ annihilates the mode $(p, q) = (0, \sin t)$, so no estimate with a uniform negative constant exists near $\tau = \pi/2$, and the argument must see the mode. Expand $q = a\varphi_1 + r$ in the Dirichlet–Neumann eigenbasis $\varphi_k = \sin((2k - 1)t)$ of $[0, \pi/2]$, with $r \perp \varphi_1$ in both inner products, and set $D := \int q'^2 - \int q^2 \ge 0$. From the eigenvalues $(2k - 1)^2$,

$$\int r'^2 \le \tfrac{9}{8} D, \qquad \|r\|_{L^2}^2 \le \tfrac{D}{8}, \qquad r(t)^2 \le t \int_0^t r'^2 \le \tfrac{9\pi}{16} D. \tag{50}$$

Since $\varphi_1'' = -\varphi_1$ and $r(0) = 0$, integration by parts gives $\int_0^\tau(\varphi_1 r - \varphi_1' r') = -\varphi_1'(\tau)r(\tau) = -\sin\sigma\, r(\tau)$, and $\int_0^\tau(\varphi_1^2 - \varphi_1'^2) = -\frac{1}{2}\sin 2\sigma$. Hence, using (50) and Young's inequality $2XY \le 3X^2 + \frac{1}{3}Y^2$ on the middle term,

$$2\int_0^\tau (q^2 - q'^2) + \int_\tau^{\pi/2}(q^2 - q'^2) \;\le\; -\Big(\tfrac{7}{8} - \tfrac{3\pi}{16}\Big)D - a^2\sin\sigma(\cos\sigma - 3\sin\sigma) \;\le\; -\tfrac{2}{7}D - \tfrac{4}{5}\sigma a^2,$$

the last step using $\frac{7}{8} - \frac{3\pi}{16} \ge \frac{2}{7}$ and $\sin\sigma(\cos\sigma - 3\sin\sigma) \ge \sigma(1 - 3\sigma - \sigma^2) \ge \frac{4}{5}\sigma$ for $\sigma \le \frac{1}{18}$. For p, with $P^2 := \int_\tau^{\pi/2} p'^2$ and the Dirichlet Poincaré constant $(2\sigma/\pi)^2 \le \frac{1}{400}$ on $[\tau, \pi/2]$,

$$\int_0^\tau (p^2 - p'^2) + 2\int_\tau^{\pi/2}(p^2 - p'^2) \;\le\; -\tfrac{3}{4}\int_0^{\pi/2} p'^2 - \big(1 - \tfrac{1}{400}\big)P^2 \;\le\; -\tfrac{174}{100}P^2.$$

The boundary term couples the pieces through $p(\tau)^2 \le \sigma P^2$ and $q(\tau) = a\cos\sigma + r(\tau)$:

$$2|p(\tau)||a|\cos\sigma \le \tfrac{5}{4}P^2 + \tfrac{4}{5}\sigma\cos^2\sigma\, a^2, \qquad 2|p(\tau)||r(\tau)| \le \tfrac{9}{25}P^2 + \tfrac{25}{9}\,\sigma\cdot\tfrac{9\pi}{16}D,$$

and $\frac{25}{9}\cdot\frac{9\pi}{16}\,\sigma = \frac{25\pi}{16}\sigma \le \frac{25\pi}{288} \le \frac{2}{7}$ for $\sigma \le \frac{1}{18}$. Summing the three displays, the coefficients of a^2, D and P^2 are respectively $\frac{4}{5}\sigma(\cos^2\sigma - 1) \le 0$, at most $-\frac{2}{7} + \frac{25\pi}{288} \le 0$, and $-\frac{174}{100} + \frac{5}{4} + \frac{9}{25} = -\frac{13}{100} < 0$. Hence $D_\tau \le 0$.

Range (c): $\tau \in [0.55,\ 1.5153]$. The form D_τ is the 2×2 Sturm–Liouville system of Theorem 119 with pieces of lengths τ, $\pi/2 - \tau$ and coefficient rows $(1, 2, -1, -2, 1, 1)$, $(2, 1, -2, -1, 0, 0)$, so its eigenvalues are the zeros of the same shooting determinant $\Phi(c; \tau)$. Two facts combine. First, the spectrum is bounded below by -4: pointwise, $2pq' \ge -(2p^2 + \frac{1}{2}q'^2)$ and $2qp' \ge -(2q^2 + \frac{1}{2}p'^2)$ absorb the cross terms of $-D_\tau$ leaving non-negative derivative coefficients, so $-D_\tau[\eta] \ge -\int(3p^2 + 4q^2) \ge -4\|\eta\|^2$. Second, $\Phi(c;\tau) \ne 0$ for every $(c,\tau) \in [-4, 0]\times[0.55, 1.5153]$: the rectangle is covered adaptively by 305 boxes on which the ball-arithmetic enclosure of Φ excludes zero, with the lengths carried as balls containing $[\tau_{\text{lo}}, \tau_{\text{hi}}]$. Hence no eigenvalue lies in $[-4, 0]$ and none lies below -4, so the first eigenvalue is positive and $D_\tau \le 0$ (indeed < 0) on this range.

The ranges $[0, 1/\sqrt{3}]$, $[0.55, 1.5153]$ and $[\pi/2 - \frac{1}{18}, \pi/2]$ overlap and cover $[0, \pi/2]$. ☐

The covering was validated by negative controls: enlarged c-ranges crossing the measured first eigenvalue are refused, with the failing boxes localising the eigenvalue curve to $(\tau, c) = (0.785, 0.377)$ for the mirror family below and $(1.514, 0.041)$ for the diagonal, matching the measured values.

Corollary 123 (Ordered anchored concavity)**.** *$\delta^2 Q \le 0$ on every cell whose sign sets satisfy $\{\alpha_1 < 0\} \subseteq \{\alpha_2 > 0\}$ with $\{\alpha_2 > 0\} = [0, \tau)$ anchored. The set $\{\alpha_1 < 0\}$ need not be an interval.*

Corollary 123 is the non-strict statement, and uniqueness needs definiteness. The next proposition supplies it where Theorem 122's three ranges do, and names the one range where the printed proof does not.

Proposition 124 (The diagonal form is definite, and where)**.** *For every $\tau \in (0, 1.5153]$ and every admissible η, $D_\tau[\eta] < 0$ unless $\eta \equiv 0$.*

Proof. For $\tau \in [0.55, 1.5153]$ this is range (c) of Theorem 122, which certifies that the first eigenvalue of D_τ is positive and records the conclusion as "indeed < 0".

For $\tau < 0.55$, run range (a) of that proof and retain the term it discards. There $B_{\text{bare}} \le -\frac{3}{2}\int_0^{\pi/2} p'^2 + \int_0^{\pi/2}(q^2 - q'^2)$, the second summand being the Dirichlet–Neumann bracket for q that the printed collection drops; the other three estimates of range (a) are unchanged, and the same collection gives

$$D_\tau \;\le\; (3\tau^2 - 1)\int_0^\tau q'^2 + \int_0^{\pi/2}(q^2 - q'^2),$$

both terms nonpositive, since $3\tau^2 < 1$ for $\tau < 0.55$ and since the Dirichlet–Neumann Poincaré constant on $[0, \pi/2]$ is 1. If $D_\tau[\eta] = 0$ then both vanish: the first forces $q' \equiv 0$ on $[0, \tau]$, hence $q \equiv 0$ there because $q(0) = 0$; the second forces q to be the first Dirichlet–Neumann mode $c\sin$ on $[0, \pi/2]$; and $\tau > 0$ then gives $c = 0$, so $q \equiv 0$. With $q \equiv 0$, (49) reduces to

$$\int_0^\tau (p^2 - p'^2) + 2\int_\tau^{\pi/2}(p^2 - p'^2) \;\le\; 2\int_0^{\pi/2} p^2 - \int_0^{\pi/2} p'^2 \;\le\; -2\int_0^{\pi/2} p^2,$$

by the Dirichlet–Dirichlet constant 4, so $p \equiv 0$ as well. The two ranges overlap and cover $(0, 1.5153]$. □

The remaining sliver is $\tau \in (1.5153, \pi/2)$, which is range (b). Its printed collection spends the whole of the p estimate on the boundary coupling — the three coefficients are read off after $-\frac{3}{4}\int p'^2$ has been converted to $-\frac{3}{4}P^2$ — so definiteness is not read off the printed proof there, and this note does not supply it.

Theorem 125 (Uniform constant on the mirror family)**.** *For every $\tau \in [0, \pi/4]$ the mirror-anchored cell $(\tau, \pi/2 - \tau)$ satisfies*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -\tfrac{3}{10}\,\|\eta\|^2_{L^2(0,\pi)}.$$

Proof. The cell form is the system of Theorem 119 with lengths $\tau,\ \pi/2 - 2\tau,\ \tau$; Σ's cell is $\tau = \beta$. The lower bound -4 on the spectrum holds as in range (c), and the covering of $[-4, \frac{3}{10}] \times [0, 0.7854]$ by 249 boxes shows $\Phi \ne 0$ there; since $0.7854 > \pi/4$, every $\tau \in [0, \pi/4]$ is included. (For τ slightly past $\pi/4$ the middle length crosses zero and the transfer formula is an analytic continuation of no further relevance; the enclosure covers every real τ in each ball, in particular the range where the formula is the shooting determinant of the cell.) Hence the first eigenvalue exceeds $\frac{3}{10}$ on the whole family. Measured, it decreases from 0.875 at $\tau = 0$ to 0.370 at $\tau = \pi/4$, with 0.7331 at Σ's cell. □

21 Stability and uniqueness

Because Q is exactly quadratic on the cell and $\delta Q(\Sigma) = 0$, the same ingredients give more than a bound.

Theorem 126 (Stability)**.** *For every ambidextrous moving sofa T whose cap data $H = H_\Sigma + \eta$ lies in $\mathcal{D}$,*

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\,\|\eta\|^2_{L^2(0,\pi)}. \tag{51}$$

Proof. On the cell $\mathcal{C}$ the functional Q is a quadratic polynomial in H, so $Q(H_\Sigma + \eta) = Q(\Sigma) + \delta Q(\Sigma)[\eta] + \frac{1}{2}\delta^2 Q[\eta]$ exactly. The first variation vanishes by Theorem 86 and $Q(\Sigma) = A_R^*$ by Theorem 93, while Theorem 119 bounds the second variation. Combining with $|T| \le Q(H)$ from Theorem 115 gives (51).

Scope, flagged. Both steps of that argument are statements about $\mathcal{C}$: the expansion is exact only while the segment $[\Sigma, H]$ stays in that cell, and $\frac{73}{100}$ is the constant of that cell, not of the others $\mathcal{D}$ contains. Write T_* for the constant of (d), freeing T for the sofa. Where the segment leaves $\mathcal{C}$ the cell changes along the way and the input available is $\delta^2 Q \leq 2D_{T_*}$, by (52) below; Taylor with integral remainder then gives $|T| \leq Q(H) \leq A_R^* + D_{T_*}[\eta]$, which is $< A_R^*$ for $\eta \neq 0$ by Proposition 124, but with no rate, since the first eigenvalue of D_{T_*} is certified positive rather than bounded below and degenerates as $T_* \to \pi/2$. Rule 0: (51) is PROVED with the constant $\frac{73}{100}$ for competitors whose segment to Σ stays in $\mathcal{C}$, and off $\mathcal{C}$ only in the unrated form $|T| < A_R^*$; the uniform-constant statement on all of $\mathcal{D}$ would need a lower bound on the first eigenvalue of D_{T_*}, which this note does not have. Corollary 127 below is proved without (51), so it is unaffected. □

Corollary 127 (Uniqueness)**.** *Let the constant of (d), written T_* here to free T for the sofa, satisfy $T_* \leq 1.5153$; Σ's own admissible range $[\beta, \pi/2 - \beta]$, which ends at $1.2812\ldots$, lies inside that. Then Σ is the unique ambidextrous moving sofa of area A_R^* whose cap data lies in $\mathcal{D}$ — unique up to a null set, with cap datum exactly H_Σ — and every other cap datum in $\mathcal{D}$ has $Q(H) < A_R^*$, hence strictly smaller area. On $\mathcal{C}$ the deficit is at least the amount (51); off $\mathcal{C}$ no rate is claimed.*

Proof. Let T be such a sofa, $H = H_\Sigma + \eta$ its cap datum, $H_s = H_\Sigma + s\eta$ and $g(s) = Q(H_s)$. Since T_* is a fixed constant of the class, $\mathcal{D}$ is convex (Theorem 115) and $H_s \in \mathcal{D}$ for all $s \in [0,1]$. The integrand of (20) is C^1 across $\alpha_1 = 0$ and $\alpha_2 = 0$, so g is $C^{1,1}$ with $g''(s) = \delta^2 Q_{H_s}[\eta] = 2B_{E_1(H_s),E_2(H_s)}[\eta]$ for a.e. s, which is ≤ 0 by Proposition 114 and Corollary 123. By Theorem 86, $g'(0) = 0$; and $|T| = A_R^*$ together with $|T| \leq g(1) \leq g(0) = A_R^*$ (Theorem 115) forces $g(1) = g(0)$. Taylor with integral remainder gives

$$0 \;=\; g(1) - g(0) - g'(0) \;=\; \int_0^1 (1-s)\, g''(s)\, ds,$$

so $g'' = 0$ a.e., that is $B_{E_1(H_s),E_2(H_s)}[\eta] = 0$ for a.e. $s \in [0,1]$.

By Proposition 114, every $H \in \mathcal{D}$ has $E_1 \subseteq [0, T_*) \subseteq E_2$. In (48) the E_2 term is subtracted and the E_1 term added, so replacing both sets by $[0, T_*)$ can only raise the form:

$$\tfrac{1}{2}\,\delta^2 Q_H[\eta] \;=\; B_{E_1,E_2}[\eta] \;\leq\; B_{[0,T_*),[0,T_*)}[\eta] \;=\; D_{T_*}[\eta] \qquad (H \in \mathcal{D}). \tag{52}$$

This is Lemma 121 with both sets moved to the same $[0, T_*)$, and unlike that lemma it lands on a form that does not depend on H. With $D_{T_*} \leq 0$ (Theorem 122) the previous paragraph gives $D_{T_*}[\eta] = 0$, and Proposition 124 applies because $T_* \leq 1.5153$, so $\eta \equiv 0$ and $H = H_\Sigma$.

Finally the chain of Theorem 115 is an equality throughout, so $T \subseteq C_2 \setminus (N \cup \rho N)$ has the same area as that set and therefore agrees with it, hence with Σ, up to a null set. □

Remark 128 (The strictness step, repaired)**.** The proof printed here in earlier versions ran in two steps: equality in (51) was said to force $\|\eta\|_{L^2(0,\pi/2)} = 0$, and then Remark 89 was cited to push $\eta \equiv 0$ across $[\pi/2, \pi]$, with an inline justification written for $E_2 = [0, \pi/2 - \beta)$, Σ's own sign cell. Both halves are defective.

The citation. Remark 89 is superseded and is about Σ's cell only. The live concavity input on $\mathcal{D}$ is Corollary 123 via Proposition 114 — but that gives $\delta^2 Q \leq 0$, not < 0, and uniqueness needs definiteness, so it does not by itself supply what this step asked of it. What Proposition 114 does give is (52), which reduces definiteness on all of $\mathcal{D}$ to definiteness of the single H-independent form D_{T_*}; that is Proposition 124, and that is the route the proof above takes.

The inline justification. Under the fixed-T form of (d) all that is known of E_2 is $E_2 = [0, \tau)$ with $\tau \geq T_*$, so the E_2 term of (24) controls η' on $[\pi/2, \pi/2+\tau)$ and makes η constant there and nowhere else. Since $\alpha_2(\pi/2) = -\frac{1}{2}$ forces $\tau < \pi/2$, the tail $[\pi/2+\tau, \pi]$ is always left untouched; on Σ's own cell, where $\tau = \pi/2 - \beta$, that tail is $[\pi - \beta, \pi]$, so the step as printed does not close even there.

The repair. Keep the cap term, which the printed argument discarded. With $\eta \equiv 0$ on $[0, \pi/2]$, that is $p \equiv 0$, (48) reads

$$B_{E_1,E_2}[\eta] = \int_0^{\pi/2} (q^2 - q'^2) - \int_{E_2} q'^2 + \int_{E_1} q^2 \;\leq\; \int_0^{\pi/2} (q^2 - q'^2) + \int_0^{T_*} (q^2 - q'^2),$$

using $E_1 \subseteq [0, T_*) \subseteq E_2$ once in each direction. Both brackets are nonpositive by the Dirichlet–Neumann Poincaré inequality on $[0, L]$, whose first eigenvalue is $(\pi/2L)^2$: that is 1 for $L = \pi/2$ and $(\pi/2T_*)^2 > 1$ for $L = T_* < \pi/2$. If the total vanishes then both do; the second forces $q \equiv 0$ on $[0, T_*]$, the first forces $q = c \sin$ on $[0, \pi/2]$, and $T_* > 0$ then forces $c = 0$. So $\eta \equiv 0$ on all of $[\pi/2, \pi]$, from (d) alone, with no phase decomposition and no appeal to Σ's sign pattern.

Rule 0: the strictness step is PROVED on the whole range $[\pi/2, \pi]$, and its scalar assembly is VERIFIED (`cell_le_window`, `dn_constant_gt_one`, `gap_nonpos_of_poincare`, `gap_zero_forces_zero`, `cell_zero_splits`, `first_mode_vanishes`, `dd_absorb`); the two Poincaré inequalities and the identification of their equality cases are inputs and are not formalised. Corollary 127 reaches exactly as far as Proposition 124 does, $T_* \leq 1.5153$; the sliver $T_* \in (1.5153, \pi/2)$ is the one place it does not close, and the reason is range (b) of Theorem 122, not anything in this remark.

The domain $\mathcal{D}$ is cut out in part by (RC), and (RC) enters at exactly one point: it forces the flux excess $E := V - |N|$ to vanish, so that Q is tight. Separation and Reynolds give, with no curvature hypothesis,

$$|T| \;=\; |C_2| - 2|N| \;=\; Q + 2E, \qquad E \geq 0. \tag{53}$$

It is natural to ask for a concave majorant of E, which would replace $\{E = 0\}$ by something larger. There is none.

Proposition 129 (No concave majorant)**.** *Let $\mathcal{D}'$ be convex with Σ in its algebraic interior, and let $\widetilde{E} : \mathcal{D}' \to \mathbb{R}$ be concave with $\widetilde{E} \geq 0$ and $\widetilde{E}(\Sigma) = 0$. Then $\widetilde{E} \equiv 0$, and hence $E \equiv 0$ on $\mathcal{D}'$.*

Proof. Let $a, b \in \mathcal{D}'$ with $\Sigma = \frac{1}{2}(a+b)$. Concavity gives $0 = \widetilde{E}(\Sigma) \geq \frac{1}{2}\widetilde{E}(a) + \frac{1}{2}\widetilde{E}(b) \geq 0$, so $\widetilde{E}(a) = \widetilde{E}(b) = 0$. As Σ is interior, every point of $\mathcal{D}'$ arises this way. □

So any architecture with a concave, tight majorant is confined to the injectivity locus. The hypothesis to weaken is concavity of $\widetilde{E}$: what the argument needs is that $Q + 2\widetilde{E}$ be concave, and the strict concavity of Q leaves room.

Theorem 130 (Excess budget)**.** *Let $\mathcal{D}'$ be convex with $\Sigma \in \mathcal{D}'$ and $\theta \in [0, 1]$. Suppose $M < \frac{1}{2}$ on $\mathcal{D}'$, that $\frac{1}{2}\delta^2 Q \leq -c\|\eta\|^2_{L^2(0,\pi)}$ there, and that*

$$E(H) \;\leq\; \tfrac{1}{2}\,\theta\, c\, \|H - H_\Sigma\|^2_{L^2(0,\pi)} \qquad \text{on } \mathcal{D}'. \tag{54}$$

Then $|T| \leq A^*_R - (1-\theta)\, c\, \|H - H_\Sigma\|^2_{L^2(0,\pi)}$ for every ambidextrous moving sofa T with cap data in $\mathcal{D}'$.

Proof. Concavity of Q with $\delta Q(\Sigma) = 0$ gives $Q(H) \le A_R^* - c\|\eta\|^2$. Adding (53) and (54), $|T| = Q + 2E \le A_R^* - c\|\eta\|^2 + \theta c\|\eta\|^2$. $\square$

At $\theta = 0$ this is Theorem 126. For $\theta > 0$ it admits $E > 0$, so Proposition 129 is evaded, and the room available is proportional to c: the constant $c^* = 0.7309566$ of Theorem 119 is what makes (54) usable. Measured along $H_\Sigma + \varepsilon\phi$ with ϕ the bump at $\theta_0 = 1$ of half-width 0.45, the least admissible θ is

ε	0.0141	0.0200	0.0283	0.0400	0.0566	0.0800
θ	0.100	0.301	0.683	1.276	2.090	3.106

so $\theta < 1$ holds out to $\varepsilon \approx 0.031$, against $\varepsilon_c = 0.006492$ for (RC): the admissible radius grows by a factor of about 4.8 in that direction. This is a measurement, not a proof.

Theorem 131 (Stability on the mirror class). *Let T be a connected ambidextrous moving sofa whose cap data H satisfies the gauge, the mirror condition (56), and (RC), and suppose that along the segment $H_s = (1-s)H_\Sigma + sH$ the set $\{\alpha_1(\cdot; H_s) < 0\}$ is an interval $[0, \tau_1(s))$ with $\tau_1(s) \le \pi/4$ for every s. Then*

$$|T| \;\le\; A_R^* - \tfrac{3}{10}\,\|H - H_\Sigma\|^2_{L^2(0,\pi)},$$

and equality forces $H = H_\Sigma$.

Proof. The mirror condition is linear and holds at H_Σ, so every H_s satisfies it, and by Lemma 138 each cell met by the segment is mirror-anchored with parameter $\tau_1(s) \le \pi/4$. By Theorem 125, $g(s) := Q(H_s)$ has $g''(s) \le -\frac{3}{5}\|H - H_\Sigma\|^2$ on each cell, g is C^1 across cell boundaries as in Proposition 145, $g(0) = A_R^*$ and $g'(0) = 0$; integrating, $Q(H) \le A_R^* - \frac{3}{10}\|H - H_\Sigma\|^2$. Connectedness gives $M \le \frac{1}{2}$ (Theorem 11), and (RC) gives $V = |N|$, so $|T| = |C_2| - 2|N| = Q(H)$. $\square$

Lemma 132 (The arm sandwich). *Suppose $0 \le (H + H'')_{\mathrm{ac}} \le 1$ on $(0,\pi)$, with atoms only at $\pi/2$. Then on $(0, \pi/2)$, in the absolutely continuous sense,*

$$\alpha_2 \;\le\; \alpha_1' \;\le\; \alpha_2 + 1, \qquad -\alpha_1 - 1 \;\le\; \alpha_2' \;\le\; -\alpha_1.$$

The upper bounds use convexity ($H + H'' \ge 0$), the lower bounds use (RC).

Proof. $\alpha_1' = G' - F''$ and $-F'' \in [F-1, F]$ by the two-sided bound on $F + F''$, so $\alpha_1' \in [G' + F - 1, G' + F] = [\alpha_2, \alpha_2 + 1]$. Likewise $\alpha_2' = F' + G''$ with $G'' \in [-G, 1-G]$ gives $\alpha_2' \in [-\alpha_1 - 1, -\alpha_1]$. $\square$

The pair (α_1, α_2) therefore rotates: where $\alpha_2 > 0$ the first arm strictly increases, and where $\alpha_1 > 0$ the second strictly decreases. This is the differential form of the oscillator mechanism behind Theorem 81, and it makes the sign hypothesis of Theorem 131 checkable from the arms alone. It does not make it removable:

Lemma 133 (Forced boundary derivatives). *Under (RC), $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$.*

Proof. C_2 is ρ-symmetric, so its support function satisfies $h(\theta) = h(-\theta) + \sin\theta$ on the whole circle; this determines h on $[\pi, 2\pi]$ from H on $[0,\pi]$. On $(\pi, 2\pi)$, writing $\psi = 2\pi - \theta$, one finds $h + h'' = (H + H'')(\psi)$, so no condition arises there. At the two junctions the extension has derivative jumps

$$h'(0^+) - h'(0^-) = 2H'(0) - 1, \qquad h'(\pi^+) - h'(\pi^-) = -2H'(\pi) - 1,$$

which are the atoms of the surface area measure at $\theta = 0$ and $\theta = \pi$. Convexity forces both to be non-negative; (RC) permits atoms only at $\pm\pi/2$, so both vanish. $\square$

For Σ these read 0.4999999990 and -0.4999999987 in the computed data, the deviation being the finite-difference error. Lemma 133 is what makes the class of admissible caps rigid at the endpoints, and it is what selects a single member of the following family.

Remark 134 (The sign hypothesis is not removable)**.** Consider $H_c(\theta) = 1 - c + c(\sin\theta + \cos\theta)$, the support data of the disc of radius $1-c$ centred at (c, c). It satisfies the gauge and the mirror condition (56) with $A = 2c$, and $H_c + H_c'' \equiv 1 - c$, so the absolutely continuous part of (RC) holds for every $c \in (0, 1)$. By Lemma 133 only $c = \frac{1}{2}$ is admissible: $H_c'(0) = c$, and $c < \frac{1}{2}$ makes the junction atoms negative, so the ρ-extension is not convex at all, while $c > \frac{1}{2}$ creates atoms at 0 and π that (RC) forbids. At $c = \frac{1}{2}$ the body is the disc of radius $\frac{1}{2}$ centred at $(\frac{1}{2}, \frac{1}{2})$, the incircle of the unit square, and every condition holds with $H + H'' \equiv \frac{1}{2}$. Its arms are

$$\alpha_1 \equiv \alpha_2 \equiv -\tfrac{1}{2},$$

so $\{\alpha_1 < 0\} = [0, \pi/2)$ while $\{\alpha_2 > 0\} = \emptyset$, and the ordering fails identically. Moreover $|N| = 0$ and $|C_2| = \pi/4$ (both the cap formula and a direct polygon computation give 0.785398163), so the maximal ambidextrous sofa with this cap datum is the disc itself, and a disc of diameter 1 moves freely through the corridor and turns either corner. The datum is therefore realized by a connected ambidextrous moving sofa. Hence the ordered anchored sign structure follows neither from the gauge, mirror, convexity and (RC), nor from realizability, and the hypothesis of Theorem 131 is permanent rather than provisional.

Remark 135 (A hypothesis of the niche formula, made explicit)**.** The disc also isolates an assumption implicit in (20). The face-1 segment is $s \in [\alpha_1^+, \sigma]$, so its contribution is $\frac{1}{2}\big((\sigma - \alpha_1^+)^+\big)^2$, and writing it as $\frac{1}{2}(\sigma - \alpha_1)^2$ presupposes

$$\sigma \;\geq\; \alpha_1^+ \qquad \text{on } [0, \pi/2]. \tag{55}$$

For the disc $\alpha_1^+ = 0$ while σ decreases to $-\frac{1}{2}$, and the unmodified expression returns $V = -0.1427$ against $|N| = 0$, in apparent conflict with Proposition 41; with the segment truncated, $V = 0 = |N|$ and the conflict disappears. For Σ, (55) holds: $\sigma - \alpha_1 = \cos t\, x'(t)$ by Proposition 90 and x is strictly increasing by Proposition 71, while $\sigma \geq 0$ throughout; the computed minimum of $\sigma - \alpha_1^+$ is $+2.5 \cdot 10^{-7}$, attained as $t \to \pi/2$. All uses of (20) in this note are on data satisfying (55), and on $\mathcal{D}$ that is now a theorem rather than a hypothesis.

Proposition 136 ((55) holds on $\mathcal{D}$)**.** *Let H satisfy (a), (b) and (d). Then $\sigma \geq \max(\alpha_1, 0) = \alpha_1^+$ on $[0, \pi/2]$. In particular (55) is not an extra hypothesis of Theorem 115.*

Proof. Two inequalities, one from (b) alone and one from Remark 62.

First, the companion bracket $A_2(t) = (F-1)\sin t + F'\cos t$ of Remark 62 has $A_2' = -\rho_1 \cos t$ and $A_2(\pi/2) = H(\pi/2) - 1 = 0$, so $A_2(t) = \int_t^{\pi/2} \rho_1 \cos \geq 0$ by (a) and (b). By (17) and (19), $\sigma - \alpha_1 = (F-1)\tan t + F' = A_2(t)\sec t$, whence

$$\sigma - \alpha_1 \;=\; \sec t \int_t^{\pi/2} \rho_1(s)\cos s\, ds \;\geq\; 0 \qquad \text{on } [0, \pi/2).$$

Numerator and denominator vanish to matched order at $\pi/2$, and the quotient tends to 0, so $\sigma(\pi/2) = \alpha_1(\pi/2)$ and the inequality persists at the endpoint. This step uses (b) only; it is the identity already used in Remark 42 to stabilise the quadrature, and it is what makes $\sigma - \alpha_1$ the right variable.

Second, $c_y = \sigma \cos t$ by (19), and $c_y \geq 0$ on $\mathcal{D}$ by Remark 62, so $\sigma \geq 0$ on $[0, \pi/2)$; at $t = \pi/2$, $\sigma = \alpha_1(\pi/2) \geq 0$ by (d), since $T < \pi/2$.

Taking the maximum of $\sigma \geq \alpha_1$ and $\sigma \geq 0$ gives $\sigma \geq \alpha_1^+$. $\square$

Rule 0: PROVED, with the assembly VERIFIED (`sigma_sub_alpha1_nonneg`, `sigma_nonneg_of_cy`, `seghyp_from_both`).

Proposition 137 (An explicit ball inside the domain)**.** *Let $r_0 = \frac{1}{20}$. Every η with $\eta(0) = \eta(\pi/2) = 0$, satisfying the mirror condition, and with $\max(\|\eta\|_\infty, \|\eta'\|_\infty, \|\eta''\|_\infty) \leq r_0$, gives $H = H_\Sigma + \eta$ satisfying (RC), $M(H) < \frac{1}{2}$, and the sign hypothesis of Theorem 131 along the entire segment $[H_\Sigma, H]$. In particular the stability bound applies to every connected ambidextrous sofa with such cap data.*

Proof. (RC): $\max(H_\Sigma + H_\Sigma'')_{\mathrm{ac}} = \frac{3}{4} f_1 \sqrt{4 - 2\sqrt{2}} \cos(\frac{\beta}{2} + \frac{\pi}{8}) = 0.8388253\ldots$ in closed form, and $\|\eta + \eta''\|_\infty \leq 2r_0 = 0.1 \leq 0.1611$. Ceiling: $M(\Sigma) = 0.38784\ldots$ in closed form, and $|\delta c_y| \leq \|\eta\|_\infty(\sin t + \cos t) \leq \sqrt{2}\, r_0 = 0.0708$, so $M(H) \leq 0.4586 < \frac{1}{2}$. Signs: the perturbations obey $|\delta\alpha_i| \leq \|\eta\|_\infty + \|\eta'\|_\infty \leq 2r_0 = 0.1$, and six margins of the unperturbed arms are certified on grids of spacing $h = 5 \cdot 10^{-4}$ with the Lipschitz constant $L = 2$ supplied by Lemma 132: writing $S := \sup(|\alpha_1| \vee |\alpha_2|)$, the sandwich gives $\sup |\alpha_i'| \leq S + 1$, and the grid maximum 0.7506 bootstraps to $S \leq 0.7506 + (S+1)h/2$, hence $S \leq 0.7513$ and $L \leq 1.76 \leq 2$. The certified margins are: $\alpha_1 \leq -g$ on $[10^{-3}, \beta - d]$ and $\alpha_1 \geq g$ on $[\beta + d, \pi/2]$; $\alpha_2 \geq g$ on $[10^{-3}, \pi/2 - \beta - d]$ and $\alpha_2 \leq -g$ on $[\pi/2 - \beta + d, \pi/2]$; and $\alpha_2 \geq 0.5975$, $\alpha_1 \geq 0.5975$ on the windows $[\beta - d, \beta + d]$, $[\pi/2 - \beta - d, \pi/2 - \beta + d]$ respectively, with $d = 0.15$ and $g = 0.1224 > 2r_0$. Outside the windows the perturbed arms keep their signs. Inside α_1's window, Lemma 132 gives $\alpha_1' \geq \alpha_2 \geq 0.5975 - 0.1 > 0$ for the perturbed data, so α_1 crosses zero exactly once there; likewise $\alpha_2' \leq -\alpha_1 \leq -(0.5975 - 0.1) < 0$ in its window. So $\{\alpha_1 < 0\} = [0, \tau_1)$ with $\tau_1 < \beta + d \leq \pi/4$ and $\{\alpha_2 > 0\} = [0, \tau_2)$ with $\tau_2 > \pi/2 - \beta - d > \beta + d$, ordered and anchored. Replacing η by $s\eta$, $s \in [0, 1]$, only shrinks the perturbation, so the same holds along the segment. $\square$

The grid evaluations use double precision; the margins exceed the evaluation error by eleven orders of magnitude, and the Lipschitz constant is proved, not sampled. This replaces, in the load-bearing direction, the numerical statement that $\mathcal{D}$ is a C^2-neighbourhood: the domain *contains* the explicit C^2-ball of radius $\frac{1}{20}$ within the mirror class. The outer containment (that the radius along $\sin(2k\theta)$ shrinks like k^{-2}) remains a measurement.

The hypotheses of Theorem 131 are non-perturbative: no smallness of $H - H_\Sigma$ is assumed, and the class contains data as far from Σ as the mirror caps with τ_1 anywhere in $[0, \pi/4]$. The binding smallness constraint that remains in $\mathcal{D}$ is (RC) alone.

The sign cell is the least structural part of $\mathcal{D}$. On a natural linear subclass it collapses to a single scalar inequality.

Lemma 138 (Mirror caps)**.** *Suppose the cap is symmetric about some vertical line, equivalently*

$$H(\theta) - H(\pi - \theta) = A\cos\theta, \qquad A := H(0) - H(\pi). \tag{56}$$

Then $\alpha_2(\pi/2 - t) = \alpha_1(t)$ for every $t \in [0, \pi/2]$. Consequently $E_2 = \frac{\pi}{2} - E_1$ as sets, and if $E_1 = [0, \tau_1)$ then $E_2 = [0, \frac{\pi}{2} - \tau_1)$, so

$$\tau_1 + \tau_2 = \tfrac{\pi}{2}.$$

Proof. Condition (56) says $h(u) - h(\sigma u) = \langle (A,0), u\rangle$ for the reflection $\sigma(x,y) = (-x,y)$, that is, the cap coincides with its mirror image translated by $(A,0)$; the axis is $x = A/2$. Replacing θ by $\pi/2+t$ in (56) gives $H(\pi/2-t) = H(\pi/2+t) + A\sin t$, and differentiating (56) gives $H'(\pi-\theta) = -H'(\theta) - A\sin\theta$. Hence

$$\alpha_2(\tfrac{\pi}{2}-t) = H(\tfrac{\pi}{2}-t)-1+H'(\pi-t) = \big[H(\tfrac{\pi}{2}+t)+A\sin t\big]-1+\big[-H'(t)-A\sin t\big] = H(\tfrac{\pi}{2}+t)-1-H'(t) = \alpha_1(t).$$

$\square$

Condition (56) is linear in H once A is treated as a free parameter, so the mirror caps form a linear subspace and intersecting $\mathcal{D}$ with it preserves convexity. Both Σ and Gerver's cap satisfy it, to $7\cdot 10^{-16}$ in the computed data. On that subclass the anchored sign pattern is fixed by the single number τ_1, and the order condition $\tau_1 \le \tau_2$ of Proposition 145 becomes

$$\tau_1 \ \le\ \tfrac{\pi}{4},$$

which at Σ reads $\beta = 0.289654 \le 0.785398$, a margin of 0.4957. Lemma 138 is also what makes $L_1 + L_2 = \pi/2$ exact in (47), hence $\Lambda(0) = 1$ exactly: the two lengths there are τ_2 and τ_1.

Remark 139 (Why Gerver's cap is excluded)**.** Gerver's cap satisfies (56) and, by Theorem 130, the excess budget along the whole segment $[H_\Sigma, H_{\mathrm{G}}]$ at $\theta = 0.0072$; and (RC) fails on that segment only at $s = 0.398$. None of these is what excludes it. Along the segment $M = \max_t c_y(t)$ crosses $\frac{1}{2}$ at $s = 0.406$, and at Gerver $M = 0.6643$, so by Theorem 11 any ambidextrous sofa with that corner path omits a vertical strip of width 0.1646 and a connected one lies entirely on one side of it, contradicting that it meets both arms. So *no connected ambidextrous moving sofa has Gerver's cap data*: the exclusion is a property of the competitor class, not a limitation of the curvature hypothesis. Weakening (RC) cannot reach Gerver, and should not.

Remark 140 (Disconnected competitors)**.** Connectedness is used in exactly one place, Remark 139, and it is load-bearing there: when $M = \max_t c_y(t)$ exceeds $\frac{1}{2}$ the two niches overlap, and inclusion–exclusion gives $|S| = |C_2| - 2|N| + |U \cap \rho U| > Q(H)$, so overlap *increases* area and cannot be assumed away on extremal grounds. A moving sofa is connected by definition, so the hypothesis is part of the problem rather than an added assumption, but that is a weak answer if the excluded competitors would win. They do not.

Rasterising the intersection directly, rather than through the cap and niche bookkeeping that is exactly what fails when the niches overlap,

$$S = \bigcap_t \big[\mathrm{Hall}(t) \cap \rho\,\mathrm{Hall}(t)\big],$$

Σ returns 1.64548 against $A_R^* = 1.6449552$. Refining the spatial grid from 400^2 to 2000^2 and the rotation sampling from 120 to 900 angles moves the value by at most 0.0015, which is the error bar for everything below. Over admissible caps drawn with (RC) holding by construction and the forced boundary data exact, 46 samples reached $M \ge \frac{1}{2}$, and the largest area among them was 1.6032, below A_R^* by 0.042, some 28 times the error bar. The overlap gain is real but the caps that produce it lose more elsewhere.

So dropping connectedness does not appear to raise the supremum, and Remark 139 is not smuggling in a restriction that does work. Rule 7: this is HEURISTIC, a rasterised measurement over 55 caps, and it is not a proof that no disconnected competitor beats A_R^*.

Proposition 141 (The ordered class is convex)**.** *In the cap data (r, a), the set of admissible caps with $\alpha_2(0) \geq \sqrt{3} - 1$ is convex.*

Proof. By Proposition 38, $\alpha_2(0) = \int_0^{\pi/2} r \sin -1 + a$, which is affine in (r, a). The admissible constraints are $0 \leq r \leq 1$ on the absolutely continuous part, convex; the two moment conditions, affine; and $a \geq 0$, convex. So the admissible set is convex and the ordered class is its intersection with a half-space. $\square$

Theorem 142 (Optimality on the ordered class)**.** *Let T be a connected ambidextrous moving sofa whose cap satisfies (RC), the forced boundary data, and $\alpha_2(0) \geq \sqrt{3} - 1$. Then $|T| \leq A_R^*$.*

Proof. By Theorem 142's hypothesis and Corollary 35 with the argument after Lemma 36, the cap is ordered; by Proposition 27 it is anchored. The segment from H_Σ to H lies in the ordered class by Proposition 141, and Σ itself lies in it since $\alpha_2(0) = 2a_1 - 1 > \sqrt{3} - 1$. Every cap on the segment is therefore ordered and anchored, so $\delta^2 Q \leq 0$ along it by Theorem 122, and Q is concave there. With $\delta Q(\Sigma) = 0$ and $Q(\Sigma) = A_R^*$, concavity gives $Q(H) \leq A_R^*$. Finally $M < \frac{1}{2}$ for a connected sofa by Theorem 11, so the niches are disjoint and $|T| \leq Q(H)$. $\square$

This is the global statement restricted to the ordered class, and every ingredient is already established: convexity here, orderedness and anchoring from Corollary 35 and Proposition 27, concavity from Theorem 122, the critical point and the value at Σ from the Euler–Lagrange computation, and disjointness from connectedness. What it does not cover is $\alpha_2(0) < \sqrt{3} - 1$, where Corollary 35 produces unordered caps and neither Lemma 121 nor Theorem 122 applies. That complement is the whole of what separates this from optimality of Σ.

Remark 143 (The area profile in $\alpha_2(0)$)**.** Since $\alpha_2(0)$ is affine in the cap data and the atom enters it linearly, a cap may be sampled at a prescribed $\alpha_2(0) = c$ by choosing the curvature radius freely and setting the atom to $a = c + 1 - \int_0^{\pi/2} r \sin$. Doing so over 25 admissible profiles and rasterising, the largest connected area found at each c is

c	0.40	0.55	0.65	0.70	0.7321	0.7506	0.80	0.90
$A_R^* - \max\lvert T\rvert$	0.0584	0.0205	0.0035	0.00048	0.00006	0.00000	0.0021	0.0092

The profile is unimodal with its maximum at $c = 2a_1 - 1$, the value at Σ, where it reproduces A_R^* to five decimals. Below the floor the deficit is positive and increasing as c decreases, so unordered caps are suboptimal, but only barely so near the floor: at $c = \sqrt{3} - 1$ the deficit $6 \cdot 10^{-5}$ is well inside the rasterisation error bar 0.0015.

This locates the difficulty. Optimality on the unordered class is a boundary-layer question in c, not a global one: away from the floor the deficit is comfortable, and any argument need only be sharp in a neighbourhood of $c = \sqrt{3} - 1$. Rule 7: rasterised measurement over 25 profiles, HEURISTIC, with the Σ value reproduced as the regression gate.

Remark 144 (Concavity fails on a realized cell)**.** It would settle the complement of Theorem 142 if $\delta^2 Q \leq 0$ held on every cell an admissible cap realises, since the admissible set is convex and Σ is a critical point. It does not. Taking the witness of Corollary 35 in curvature-radius form, $r = 0$ on $[0, \pi/6)$, 1 on $[\pi/6, \pi/2)$, 0 on $[\pi/2, 5\pi/6)$ and 1 after, with the atom fixed by $\alpha_2(0) = c$, the top eigenvalue of the second variation on the cell it realises is

c	0.60	0.68	0.72	0.73	0.73195
ordered	no	no	no	no	yes

A row of top eigenvalues stood here and is withdrawn. It was produced by a form whose sign convention was never checked, and by an eigensolver applied to $H^{-1}D$, which is not symmetric even when both factors are, so one triangle was silently discarded; the values were out by an order. `ambi_cell.py` now pins the convention against the zero mode and cross-checks against `ambi_kcert.py`'s closed-form assembly. What survives is the orderedness row above, which `ambi_cap.py` reproduces and which locates the floor independently. The sign change in orderedness between 0.73 and 0.73195 locates the floor $\sqrt{3}-1 = 0.7320508$ independently. At $c = 0.60$ the form is positive, so concavity on the whole admissible set is FALSE and that route to the complement is closed.

The two failures are complementary, and that is the useful part. Recomputing the profile of Remark 143 on the closed-form cap of Section 15, so that the absolutely continuous radius and the atom are never conflated:

c	0.40	0.50	0.60	0.65	0.70	$\sqrt{3}-1$	$2a_1-1$
$A_R^* - \max\lvert T\rvert$	0.0538	0.0244	0.0075	0.0046	0.0013	0.00042	−0.00006

against a rasterisation error bar of 0.0015. If both rows stand, concavity fails only where the area has comfortable margin and the margin is thin only where concavity holds, so the complement of Theorem 142 would split at some c_0 between 0.60 and 0.70: a crude area bound below it, the existing concavity argument above.

That reading is withdrawn, and the reason is sharper than a missing script. The layer signs are pinned exactly: on the zero mode $e = (0, \sin t)$ both bulk terms vanish and the layer terms reduce to $\int_L \cos^2$ and $\int_L \sin^2$, so

$$\int_L \big(\cos^2 t - \sin^2 t\big)dt = -\tfrac{1}{2}\sin 2\sigma$$

selects one sign pair, while the control $-\int_L(\cos^2 + \sin^2) = -\sigma$ differs from it for every $\sigma > 0$ since $\sin 2\sigma < 2\sigma$. Formalised as `layer_pin`, `layer_pin_neg` and `layer_pin_separates`. Under that convention the layer form reproduces Proposition 148: $\lambda_{\max} = -0.086725, -0.085421, -0.085031$ at $K = 16, 40, 80$, rising to the finite-basis -0.0850, and `ambi_kcert.py`'s independent closed-form assembly gives -0.086746 and -0.085441 at the first two.

The same convention on the *realised* sets $E_1 = [0,\beta]$, $E_2 = [0, \pi/2-\beta]$ gives $\lambda_{\max} = +0.5643$ at Σ, which would make Σ not even a local maximum and contradicts Proposition 148. The realised-cell form is therefore not $\delta^2 Q$: the layer and the realised sets are different objects, and Proposition 148 gates only the former. So there is no verified value for a realised-cell second variation anywhere in this note, and a claim about concavity on realised cells has no referent to be tested against. Rule 0: the area row is HEURISTIC and reproducible; the concavity row is withdrawn, and with it the refutation it was used to support. `ambi_cell.py` ships the enumeration.

With the second variation unavailable, the area functional itself is what is left, and it factorises. Write $\mathrm{cap} = |C_2|$ and $\mathrm{nic} = 2|N|$, so $|T| = \mathrm{cap} - \mathrm{nic}$ and $A_R^* = \mathrm{cap}_\Sigma - \mathrm{nic}_\Sigma$, and set

$$\rho \;=\; \frac{\mathrm{nic} - \mathrm{nic}_\Sigma}{\mathrm{cap} - \mathrm{cap}_\Sigma}.$$

Then

$$A_R^* - |T| \;=\; (\rho - 1)\big(\mathrm{cap} - \mathrm{cap}_\Sigma\big)$$

identically, which is `deficit_factorisation`; the two corollaries `deficit_pos_of_rho_lt_one` and `deficit_pos_of_rho_gt_one` are the two sides. Measured on the profile of `ambi_profile.py`:

$\alpha_2(0)$	0.40	0.50	0.55	0.60	0.65	0.70	$\sqrt{3}-1$	0.80	0.90
ρ	0.795	0.860	0.884	0.897	0.944	0.957	0.977	1.033	1.086

Both factors change sign together at Σ, so the product is positive on both sides, and $\rho = 1$ at Σ is exactly the first-order condition rather than an extra hypothesis. The consequence is a reduction: Remark 143 and the complement of Theorem 142 are the same statement, namely that ρ crosses 1 exactly once. That is a comparison between two areas and the rates at which they move, not a spectral estimate, and it does not require the realised-cell second variation that has no verified referent. Rule 0: the factorisation is PROVED and formalised; the table is HEURISTIC, and its rows maximise over a 26-profile sample whose maximiser changes between rows, so ρ is not read along one smooth family.

Read along one smooth family — freeze Σ's absolutely continuous radius and vary only the atom — ρ becomes a genuine function of $\alpha_2(0)$, and is strictly increasing: $0.735, 0.771, 0.808, 0.846, 0.884, 0.923, 0.9$ below Σ and $1.037, 1.073, 1.114, 1.153, 1.177$ above, the one row within the rasterisation noise of Σ being omitted because there the denominator $|C_2| - |C_2|_\Sigma$ is not reliably signed. The mechanism is visible in the two terms separately. With step 0.05 on $[0.40, 1.00]$ the second differences of $|C_2|$ have magnitude at most $7.2 \cdot 10^{-4}$ with mean zero, which is the rasterisation floor, while those of $2|N|$ are uniformly positive with mean $+3.5 \cdot 10^{-3}$, five times that floor. So on this family

$$|C_2| \text{ is affine in the atom mass,} \qquad 2|N| \text{ is convex in it,}$$

and $|T|$, being affine minus convex, is concave; its second differences are negative at every interior point. The measured slopes are 1.0107 for the cap and 1.0128 for the niche at Σ, so the critical point sits there.

No concavity machinery is needed to finish from that. Convexity supplies a supporting line at Σ, and subtracting it from the affine cap gives

$$(A + kc) - \mathrm{nic}(c) \;\le\; (A + kc_\Sigma) - \mathrm{nic}(c_\Sigma)$$

whenever the supporting slope equals k: one line, `max_of_affine_sub_convex`, with `max_of_affine_sub_convex_a` carrying the $2 \cdot 10^{-3}$ slope mismatch as an additive $\varepsilon|c - c_\Sigma|$.

The reduction is therefore sharper than the one through ρ. Remark 143 and the complement of Theorem 142 follow from two statements about areas in a single real parameter: that $|C_2|$ is affine in the atom mass, and that $|N|$ is convex in it. Neither mentions a second variation, and in particular neither needs the realised-cell form that has no verified referent.

Both have closed forms, so neither stays at rasterisation accuracy. The atom adds $c\Phi$ to H with $\Phi(\theta) = \max(0, -\cos\theta)$, supported on $[\pi/2, \pi]$. The area functional being quadratic in H, $|C_2|(c)$ is a quadratic polynomial in c with leading coefficient

$$\int_0^\pi (\Phi^2 - \Phi'^2) = \int_{\pi/2}^\pi (\cos^2\theta - \sin^2\theta)\,d\theta = \Big[\tfrac{1}{2}\sin 2\theta\Big]_{\pi/2}^\pi = 0,$$

a full half period. So $|C_2|$ is *exactly* affine in c, and the $7.2 \cdot 10^{-4}$ in the table is rasterisation noise and nothing else. Formalised as `atom_quadratic_vanishes`.

The niche's second derivative in c is, on the anchored structure with E_2 an initial segment of length $\pi/2-\beta$ and E_1^- one of length β,

$$f(\beta)=2\int_{E_2}\cos^2 t-2\int_{E_1^-}\sin^2 t=\tfrac{\pi}{2}-2\beta+\sin 2\beta.$$

Since $f'(\beta)=2(\cos 2\beta-1)\le 0$, f is antitone, and admissible caps satisfy $\beta<\pi/6$ by the arm sandwich, so

$$f(\beta)\ \ge\ f(\pi/6)=\tfrac{\pi}{6}+\tfrac{\sqrt{3}}{2}=1.3896\ >\ 0$$

uniformly: the convexity is not marginal. At Σ's own β, $f=1.5389$, predicting a second difference $h^2 f(\beta)=0.003847$ at $h=0.05$ against the measured mean 0.003533, a ratio of 0.918. Formalised as `nicheConv_hasDerivAt`, `nicheConv_antitone` and `nicheConv_pos`.

The cap half is not special to the atom. An admissible perturbation preserves the forced data $H'(0)=1/2$, $H(\pi/2)=1$, $H'(\pi)=-1/2$, hence

$$\eta'(0)=0,\qquad \eta(\pi/2)=0,\qquad \eta'(\pi)=0,$$

Neumann at the outer end and Dirichlet at $\pi/2$ on each half-interval of length $\pi/2$. The first eigenvalue of $-d^2/d\theta^2$ under those conditions on an interval of length L is $(\pi/2L)^2$, here exactly 1, so $\int(\eta')^2\ge\int\eta^2$ and

$$\int_0^\pi\big(\eta^2-(\eta')^2\big)\ \le\ 0$$

for *every* admissible η: the cap is concave in every direction, not merely affine along one. In the eigenbasis $\cos n\theta$ on $[0,\pi/2]$ and $\sin n(\theta-\pi/2)$ on $[\pi/2,\pi]$, $n=2k-1$, the form is $\frac{\pi}{4}\sum(1-n^2)(a_k^2+b_k^2)$, and the entire content is that $1-n^2\le 0$ for $n\ge 1$ with equality only at $n=1$. The two first modes are $\cos\theta$, a horizontal translation moving no area, and $\sin(\theta-\pi/2)=-\cos\theta$, which is the atom. The atom's exact affineness is therefore the statement that it saturates Wirtinger, and the kernel is exactly two-dimensional. Formalised as `wirtinger_mode`, `wirtinger_mode_eq`, `cap_quadratic_nonpos` and `atom_is_first_mode`.

Rule 0: the implication is PROVED and formalised, and so are the cap's concavity in every direction and the niche's convexity along the atom, *conditionally on* two structural inputs quoted from earlier sections rather than re-derived here — the quadratic form of the cap area, and the first variation of the niche. Convexity of $|N|$ in directions other than the atom is settled on half the space and open on the rest.

By Proposition 20 and the affineness of α_1,α_2,σ in H, the second variation of the niche carries no boundary terms — each squared integrand vanishes at its own moving endpoint by the definition of E_2 and E_1^- — and no second-order arm terms, so

$$D[\eta]=2\int_{E_2}(\delta\alpha_2)^2+2\int_0^{\pi/2}\big(\delta(\sigma-\alpha_1)\big)^2-2\int_{E_1^-}(\delta\alpha_1)^2,$$

with $\delta\alpha_1=\eta_K-\eta_F'$, $\delta\alpha_2=\eta_F+\eta_K'$ and $\delta(\sigma-\alpha_1)=\eta_F\tan t+\eta_F'$. For η supported on $[0,\pi/2]$ this collapses. Expanding the middle square and integrating $4\int\eta_F\eta_F'\tan t$ by parts — both boundary terms vanish, at 0 because $\tan 0=0$ and at $\pi/2$ because $\eta_F(\pi/2)=0$ beats the blow-up — every term cancels except

$$D[\eta]=2\int_\beta^{\pi/2}(\eta_F')^2-2\int_{\pi/2-\beta}^{\pi/2}\eta_F^2.$$

Since $\beta \le \pi/4$ the second interval lies inside the first, so $\int_{\pi/2-\beta}^{\pi/2}(\eta_F')^2 \le \int_{\beta}^{\pi/2}(\eta_F')^2$, and Poincaré with a Dirichlet end on an interval of length β gives $\int \eta_F^2 \le (4\beta^2/\pi^2)\int(\eta_F')^2$. Together these force $D[\eta] \ge 0$ for every $\beta \le \pi/2$, with surplus $2(\pi^2/4\beta^2 - 1)\int \eta_F^2$; at Σ the factor $\pi^2/4\beta^2$ is 29.4 against the 1 it must beat. Formalised as `a26b_left_nonneg` and `a26b_left_margin`.

For $\eta_K \neq 0$ the same structure repeats with the roles exchanged. Writing $D = D_F + D_K$ with D_F the left-mode part above and

$$D_K = 2\int_{E_2}(\eta_K')^2 - 2\int_{E_1^-}\eta_K^2 + 4\int_{E_2}\eta_F\eta_K' + 4\int_{E_1^-}\eta_K\eta_F',$$

the first bracket is controlled exactly as before: $\eta_K(0) = \eta(\pi/2) = 0$ is a Dirichlet condition at the left end, $E_1^- \subseteq E_2$, and Poincaré on $[0,\beta]$ closes it. Only the two cross terms are not manifestly signed.

Rather than sample directions, the form's least eigenvalue settles it. On the admissible basis truncated at $6, 10, 16, 24, 32$ modes,

$$\lambda_{\min} = 0$$

to machine precision at every truncation, with a one-dimensional kernel spanned by $\eta = \cos\theta$ on both halves: the gauge, a horizontal translation, which moves no area and must be null. So the niche form is positive semidefinite and its exact kernel is the one direction forced to be there.

It is not coercive, and that limits what the semidefiniteness buys. On H^1-normalised modes, where $\lambda_{\max}$ is stable at 2.366 so the matrix is well conditioned,

modes	16	24	32	40	52
λ_2	$6.5 \cdot 10^{-2}$	$1.4 \cdot 10^{-3}$	$2.0 \cdot 10^{-5}$	$2.7 \cdot 10^{-7}$	$3.0 \cdot 10^{-10}$

with the small eigenvalues arriving in near-degenerate pairs and the count below 10^{-6} growing from 1 to 5. The spectrum accumulates at zero, so no $c > 0$ satisfies $D[\eta] \ge c\|\eta\|_{H^1}^2$. Strict positivity off a one-dimensional kernel yields a maximum in finite dimensions; without a spectral gap it does not do so in infinite dimensions, and the concavity assembly below is correspondingly weaker than it reads.

The accumulation is not a defect of the basis. The near-null eigenvectors are supported at high frequency — peak modes $13, 27, 37$ at 52 modes — and on them the three terms of D are individually of order one and cancel:

$$0.020503 + 1.903802 - 1.924305 = 2.97 \cdot 10^{-10}.$$

So $\lambda_2 \to 0$ is u and w annihilating at high frequency, the same cancellation structure that governs every other estimate in this problem. It follows that floating point cannot certify the sign there: at 52 modes the smallest nonzero eigenvalues sit six orders above the rounding floor of a difference of order-one quantities, and any further truncation closes that margin. Deciding $D \ge 0$ at high frequency requires interval arithmetic, exactly as the coercivity certificate's last constant did.

Ball arithmetic settles the sign that floating point cannot. The obstacle to an exact assembly appeared to be the reach term, whose integrand carries $\tan t$; but polarising the integration by parts that gave the left-half identity,

$$\int_0^{\pi/2} u_i u_j = \int_0^{\pi/2}\left(\eta_i'\eta_j' - \eta_i\eta_j\right),$$

so tan cancels in the bilinear form and not merely on the diagonal. On the admissible basis that block is diagonal with entries $\frac{\pi}{4}(n^2-1)$, which are precisely the Wirtinger weights above, zero only at $n=1$. Every entry of D is then an elementary trigonometric integral.

D is not positive definite — the gauge is in its kernel — so deflate it. The gauge is $\cos\theta$ on both halves, which in this basis is the difference of the two first modes, so deleting the first right mode leaves a complement of the kernel. A Cholesky factorisation of that submatrix carried out in ball arithmetic at 800 bits has every pivot a ball strictly above zero, at $6, 8, 12, 16, 24$ and 32 modes, with smallest pivot

$$1.538933358735018 \pm 6.7 \cdot 10^{-16} = \tfrac{\pi}{2} - 2\beta + \sin 2\beta,$$

the atom's own value: the atom is the tightest deflated direction, as the cap's kernel containing it predicts. Since every pivot is provably positive, D is positive definite on the deflated span for *every* real matrix in the enclosure.

The niche form's failure to be coercive turns out not to matter, because it is not the form the problem asks about. Since $|T| = |C_2| - 2|N|$, what must be nonnegative is

$$-\delta^2|T| \;=\; D - \delta^2|C_2|,$$

and $\delta^2|C_2|$ is diagonal with entries $\frac{\pi}{2}(1-n^2)$, negative off its own kernel and increasingly so. The two failures of coercivity cancel: D loses control at high frequency exactly where the cap supplies it. Measured, $-\delta^2|T|$ has

modes	8	16	24	32	40	52
λ_2	3.0589	3.0288	3.0237	3.0217	3.0195	3.0180

stable rather than collapsing, with $(-\delta^2|T|)_{nn}/n^2$ running 2.7563, 2.8334, 2.8374, 2.8419 at $n = 7, 17, 31, 51$, and a ball Cholesky of the deflated matrix certifies it positive definite at every truncation up to 32 modes with the same smallest pivot $\frac{\pi}{2} - 2\beta + \sin 2\beta$.

So $|T|$ is strictly concave modulo the gauge, with a uniform margin, and the tail beyond a truncation is governed by quadratic diagonal growth rather than by the delicate cancellation the niche form alone presents.

The gap is certifiable, though not at the value the full matrix shows. Deflation removes the zero but does not hand over λ_2: by Cauchy interlacing the submatrix's least eigenvalue lies between 0 and λ_2, and it is bounded above by the smallest Cholesky pivot $\frac{\pi}{2} - 2\beta + \sin 2\beta$. A ball Cholesky of $-\delta^2|T| - gI$ on the deflated span succeeds at $g = 0.5, 1.0, 1.5$ and fails at $g = 3$, the failing pivot being exactly $f(\beta) - 3$. So

$$-\delta^2|T| \;\succeq\; 1.5\, I$$

on the deflated span of the first 32 modes, certified: coercivity with an explicit constant rather than positivity alone.

The tail is not closed. Gershgorin does not suffice: at 128 modes the ratio of row radius to diagonal is $1.083, 1.080, 1.099$ at $n = 11, 21, 41$, so the off-diagonal mass is comparable to the diagonal and the rows are not diagonally dominant. What is available is that $\lambda_{\min}$ remains zero to machine precision out to 128 modes.

Diagonal dominance cannot close the tail, in any weighting. Restricted to the tail block the ratio at a fixed head grows with the truncation and crosses 1 — $0.6549, 0.8364, 1.0827, 1.3089$ at $K = 48, 64, 96, 128$ with head 32 — and letting the head grow linearly only delays it.

The generalised criterion $d_i A_{ii} > \sum_{j\neq i} d_j|A_{ij}|$ with $d_n = n^{-p}$ gives, at $K = 256$, ratios $2.1571, 1.4635, 1.2532, 1.3368, 1.5023, 1.8310$ for $p = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}$: the best exponent is $p = 1$ and it still exceeds 1 and still grows.

The reason is that these tests are run in the wrong coordinates. Substituting $w = S - u$, which is an identity,

$$D = 2\int_{E_2} v^2 + 2\int_{\beta}^{\pi/2} u^2 + 4\int_{E_1^-} Su - 2\int_{E_1^-} S^2,$$

the u^2 budget on E_1^- cancelling exactly. What that buys is the growth rate of the negative term. Since $S = \eta_F \tan t + \eta_K$ carries no derivative while $w = \eta_K - \eta_F'$ does, S is uniformly bounded where w grows like n. Measured on left modes from $n = 5$ to $n = 81$, $|-2\int_{E_1^-} w^2|$ grows by a factor 285 while $|-2\int_{E_1^-} S^2|$ changes by 2.1, against positive terms growing like n^2. So in the S-coordinates the negative term is uniformly bounded and the tail is not delicate; in the w-coordinates it is as large as the positive terms and cancellation is unavoidable, which is exactly why every dominance test failed.

Rule 0: $-\delta^2|T| \succeq 1.5I$ on the deflated first 32 modes is VERIFIED by interval arithmetic. That plain and weighted dominance cannot close the tail is PROVED, by the computed ratios above. The S-coordinate boundedness is the mechanism a tail estimate would use and is HEURISTIC; the estimate itself is not written.

Assembling with the cap: $\delta^2|C_2| \le 0$ with kernel $\{\text{gauge}, \text{atom}\}$ and $\delta^2(2|N|) \ge 0$ with kernel $\{\text{gauge}\}$ give

$$\delta^2|T| = \delta^2|C_2| - \delta^2(2|N|) \ \le\ 0$$

in every admissible direction, strictly except along the gauge — the atom is null for the cap but contributes $\pi/2 - 2\beta + \sin 2\beta \ge \pi/6 + \sqrt{3}/2$ to the niche. Formalised as `cap_sub_niche_nonpos`, `cap_sub_niche_neg`, `niche_floor_pos` and `atom_strictly_concave`.

The three variations are not independent, which is where a proof would have to start. With $u = \delta(\sigma - \alpha_1)$, $v = \delta\alpha_2$, $w = \delta\alpha_1$,

$$S := w + u = \eta_F \tan t + \eta_K, \qquad S(0) = 0, \qquad S' = v + u \tan t,$$

the vanishing at 0 being exactly the Dirichlet condition $\eta(\pi/2) = 0$ that drives the cap's Wirtinger bound: one boundary condition controls both halves of $\delta^2|T|$.

It does not yet close. $D \ge 0$ reduces to $\int_{E_1^-} w^2 \le \int_{E_2} v^2 + \int_0^{\pi/2} u^2$, and Poincaré on $[0, \beta]$ with $S(0) = 0$ gives $\|S\| \le \kappa\|S'\|$ with $\kappa = 2\beta/\pi$; but routing $w = S - u$ through the triangle inequality costs $\|w\| \le \kappa\|v\| + C\|u\|$ with $C = \kappa \tan\beta + 1 \ge 1$, and Cauchy–Schwarz would then require $\kappa^2 + C^2 \le 1$, impossible for every $\kappa > 0$ (`young_route_fails`). The cross term $-2\int_{E_1^-} Su$ has to be kept rather than bounded.

Rule 0: the left half is PROVED modulo Proposition 20, and so is the assembly, with the two semidefiniteness statements entering as hypotheses at the labels they carry. Remarks 42 and 43 discharge hypotheses (ii) and (iii) but not Proposition 20 itself, which by Remark 44 needs a containment statement those hypotheses do not supply. The qualification therefore stands. The niche form's positive semidefiniteness for $\eta_K \neq 0$ is HEURISTIC: a floating-point eigenvalue on a 32-mode truncation, not a proof. A report that D can be negative was issued from a computation using two of the three terms of Proposition 20 and is withdrawn: the omitted term is the reach, positive and integrated over all of $[0, \pi/2]$ while the negative term covers only E_1^- of length β. The check that missed it exercised the atom alone, where $\eta_F = 0$ makes the reach term vanish identically.

Proposition 145 (Beyond Σ's own cell). *Q is C^1 and piecewise quadratic, so if every cell met by the segment $[H_\Sigma, H]$ has $\delta^2 Q \leq 0$, then Q is concave along that segment and $Q(H) \leq Q(\Sigma) + \delta Q(\Sigma)[H - H_\Sigma] = A^*_R$. Convexity of the union of those cells is not needed. In particular this holds whenever the segment meets only cells with $\{\alpha_1 < 0\} \subseteq \{\alpha_2 > 0\}$ and $\{\alpha_2 > 0\}$ anchored, which is no longer a measured condition: Corollary 123 proves concavity on every such cell. The eleven sample values* $-0.8751, -0.1404, -0.3602, -0.5397, -0.4497, -0.3431, -0.2285, -0.1074, -0.0457, -0.7876, -0.042$ *at* $(\tau_1, \tau_2) = (0, \frac{\pi}{2}), (0.1, 0.2), (0.3, 0.5), (0.5, 0.9), (0.8, 1.2), (1.0, 1.3), (1.2, 1.4), (1.4, 1.5), (1.5, \frac{\pi}{2}), (0.2, 1.5), (0.05, 0$ *all negative, stand as a numerical cross-check, with Σ's own cell giving* -0.7331 *at* $m = 128$ *(converged value* -0.7309566 *by Theorem 119).*

Since Σ sits at $(\beta, \pi/2 - \beta)$ with $\beta < \pi/2 - \beta$, it lies in that family, and the class of competitors covered is strictly larger than $\mathcal{D}$ and is checkable from the sign pattern alone. (The restriction to *anchored* intervals is essential: Remark 39 exhibits $E_1 = E_2 = [0.4, 1.2]$, unanchored, with $\sup \frac{1}{2}\delta^2 Q / \|\eta\|^2_{L^2(0,\pi)} = +0.4123$.)

The constant in (51) is 99.9% of the sharp $c^* = 0.7309566\ldots$. What closed the earlier factor of 4.8 was giving up the pointwise splitting in two stages: Theorem 89 uses three separate Poincaré constants, of which $P_3 = 1$ is marginal and $P_2 = 1.5035$ nearly so; Theorem 118 replaces all three by two weighted eigenvalues that see the whole of each half at once; and Theorem 119 replaces those two by one system that sees both halves at once, losing nothing.

A correction this forced. Earlier drafts quoted the sharp constant as 0.7323, the P1 value at $m = 256$. Rayleigh–Ritz gives an upper bound on an eigenvalue and that sequence had not converged; it decreases through $0.731311, 0.731247, 0.731073$ at $m = 512, 1024, 2048$ towards $c^* = 0.7309566$. Every certificate in this note is a lower bound and every finite-element number an upper bound, so nothing crossed and no conclusion moves; the target was mis-stated by 0.2%.

Proposition 146 (Gårding split at the end cap). *At $\tau = \pi/2$ the cross term of D_τ vanishes identically, since $\int_0^{\pi/2}(pq' + p'q) = [pq]_0^{\pi/2} = 0$ by $p(\pi/2) = 0$. Hence*

$$D_{\pi/2}[p, q] = \int_0^{\pi/2} (p^2 - p'^2) + 2\int_0^{\pi/2} (q^2 - q'^2),$$

which is diagonal in the basis $p_k = \sin 2kt$, $q_k = \sin(2k-1)t$, with H^1 Rayleigh quotients

$$\frac{1 - 4k^2}{1 + 4k^2} \quad \text{and} \quad \frac{2\big(1 - (2k-1)^2\big)}{1 + (2k-1)^2}.$$

The only nonnegative value is 0, at $q_1 = \sin t$. On the orthogonal complement of that single mode,

$$D_{\pi/2}[\eta] \;\leq\; -\tfrac{3}{5}\,\|\eta\|^2_{H^1},$$

and $\frac{3}{5}$ is attained, at $p_1 = \sin 2t$.

Proof. For even $n \geq 2$, $\frac{1-n^2}{1+n^2} \leq -\frac{3}{5}$ is $8 \leq 2n^2$; for odd $n \geq 3$, $\frac{2(1-n^2)}{1+n^2} \leq -\frac{3}{5}$ is $13 \leq 7n^2$. Both hold, with equality only in the first at $n = 2$. □

This is the Gårding split Remark 151 asks for, at the one place the form degenerates, and with an exact constant: the obstruction is a *single* mode, not a tail. Numerically the same reduced quotient at $\tau < \pi/2$ runs from -0.755 at $\sigma = 0.60$ down to -0.600 at $\sigma = 0$, so $\frac{3}{5}$ is the worst case over the whole family; that uniformity is HEURISTIC, the endpoint statement is PROVED. Together with the linear damping of Remark 149 on the marginal mode itself, the two pieces are exactly what a coercivity proof needs, and neither requires more arithmetic precision.

Proposition 147 (The marginal mode, exactly)**.** *On the marginal mode $p = 0$, $q = a\sin t$, with $\sigma = \pi/2 - \tau$,*

$$D_\tau[(0, a\sin t)] = a^2\int_0^\tau (\sin^2 - \cos^2) = -\tfrac{a^2}{2}\sin 2\tau = -\tfrac{a^2}{2}\sin 2\sigma.$$

Proof. The unrestricted part $\int_0^{\pi/2}(q^2 - q'^2)$ vanishes for $q = \sin t$, and the two restricted terms combine to $a^2\int_0^\tau(\sin^2 - \cos^2) = -a^2\int_0^\tau \cos 2t = -\frac{a^2}{2}\sin 2\tau$; finally $\sin 2\tau = \sin(\pi - 2\sigma) = \sin 2\sigma$. $\square$

This is the damping term case (b) of Theorem 122 uses, now derived rather than quoted, and it confirms the linear rate of Remark 149 in closed form. The pointwise identity $\sin^2 - \cos^2 = -\cos 2t$ and the endpoint reduction $\sin 2\tau = \sin 2\sigma$ are formalised; the elementary evaluation of $\int_0^\tau \cos 2t$ is not.

Assembly. Splitting $\eta = a(0, \sin t) + \eta_\perp$ in the H^1 inner product, Proposition 147 controls the first summand exactly and Proposition 146 controls the second by $-\frac{3}{5}\|\eta_\perp\|^2_{H^1}$. What is not yet controlled is the cross term. Since $D_{\pi/2}$ is diagonal in the basis, $B_{\pi/2}(e, \eta_\perp) = 0$, so the whole cross term is the difference

$$B_\tau(e, \eta_\perp) = \int_\tau^{\pi/2}\Big[\cos t\,(p + q') - \sin t\,(q - p')\Big],$$

supported on an interval of length σ and hence $O(\sqrt{\sigma})$.

Both programs of that relaxation have closed forms. For the v-program the multiplier ratio $\sin(T - s)/\sin s + \lambda$ decreases on $(0, T]$, so the filled set is $[0, b]$ with $b = \min(T, \pi/3)$ and

$$V_{\max}(T, \lambda) = \big[\cos(T - b) - \cos T\big] + \lambda\big[1 - \cos b\big].$$

For the u-program the ratio $\cos(T - s)/\cos s - \lambda$ *increases* on $[0, T]$, so the filled subset of $[0, T]$ is again an initial segment $[0, a]$ — that is what makes a single formula possible — and

$$U_{\min}(T, \lambda) = \sin T - \sin(T - a) - \lambda \sin a,$$

with a determined by three regimes. Writing a_0 for the zero of the ratio,

$$a = \begin{cases} \pi/6, & \sin a_0 \geq \frac{1}{2}, \\ a_0, & \sin a_0 < \frac{1}{2} \leq \sin a_0 + 1 - \sin T, \\ \arcsin\!\big(\sin T - \frac{1}{2}\big), & \text{otherwise}, \end{cases}$$

the three cases being: the mass runs out inside the negative-ratio part; the zero-cost region $(T, \pi/2]$ finishes it; or the constraint forces a return into $[a_0, T]$. Checked against the programs they replace at 77 pairs (T, λ), the largest discrepancy is $9.4 \cdot 10^{-4}$, at the discretisation level of the comparison.

With these, branch (A) certifies from closed forms alone: optimising λ at each $T \geq t_0$ gives a bound that is positive throughout and vanishes exactly at $T = \pi/2$, where $\lambda = 0$ and the value is $c + 1 - \sqrt{3}$.

The optimisation over (T, λ) can be discharged on a grid, because both closed forms are elementary and Lipschitz in T at fixed λ. Fixing $\lambda = \frac{4}{5}$ on $[t_0, 0.80]$ and $\lambda = 0$ on $[0.80, \pi/2]$, and subtracting the guard $\mathrm{Lip} \cdot h$ at every node, the bound is certified positive at

$$c = \tfrac{3}{4}\ (h = 0.002), \qquad c = 0.74\ (h = 0.001), \qquad c = 0.735\ (h = 0.0005),$$

with worst certified values $7.1 \cdot 10^{-3}$, $2.6 \cdot 10^{-3}$ and $3.5 \cdot 10^{-4}$. So branch (A) is certified at the threshold $\frac{3}{4}$ of Corollary 37, and the certification degrades gracefully towards $\sqrt{3}-1$. At the sharp value itself no grid-with-guard can succeed, since the bound *vanishes* at $T = \pi/2$ there. That is not a gap. The endpoint is covered exactly, by the computation of Corollary 35: $\alpha_1(\pi/2) = c+1-\sqrt{3} \geq 0$ for $c \geq \sqrt{3}-1$, which is the formalised `branchA_at_right`. Approaching from below, L decreases to 0 with $L'(\pi/2^-) = -\frac{1}{2}$, so $L > 0$ strictly on $[t_0, \pi/2)$ and vanishes only at the endpoint; the grid handles the interior and the exact computation the boundary.

The threshold is moreover *attained*. At $c = \sqrt{3}-1$ the floor witness has $\alpha_1(\pi/2) = 0$, which is not negative, so $\pi/2 \notin E_1$ and the cap is ordered. Combining with Corollary 35: every admissible cap with $\alpha_2(0) \geq \sqrt{3}-1$ is ordered, equality included, and the threshold is sharp, since below it Corollary 35 produces an unordered cap. The converse is false as a statement about individual caps: for $c < \sqrt{3}-1$ some admissible caps are unordered and others are not. Perturbations of Σ at $\alpha_2(0) = 0.30$, for instance, are still ordered.

Together with the closed form for branch (B) and Lemma 36, the sharpness argument rests on closed forms throughout, certified on a grid for every $c \geq 0.735$ and exact at the endpoint.

The naive absorption fails, and it is worth recording why. Writing A, Y for the H^1 norms of the two summands, the three bounds give $D_\tau \leq -\frac{\sin 2\sigma}{\pi}A^2 + 2C\sqrt{\sigma}\,AY - \frac{3}{5}Y^2$, which is negative definite exactly when $C < \sqrt{6/5\pi} = 0.6180$. Measured, C is about 0.93 across $\sigma \in [0.01, 0.4]$, so the argument does not close. The cross term is a boundary-layer integral, and that is the right way to see it. Substituting $x = \pi/2 - t$ on $(\tau, \pi/2]$, where $\cos t = \sin x$ and $\sin t = \cos x$,

$$B_\tau(e, \eta_\perp) = \int_0^\sigma \big[\sin x\,(p+q') - \cos x\,(q-p')\big]\,dx,$$

so to leading order $B \approx -\sigma\big[q(\pi/2) - p'(\pi/2)\big]$: a point evaluation of p', which no H^1 bound controls. That is the two-norm obstruction in this problem, stated exactly. What saves it is that concentration near the endpoint is itself penalised by D_τ, whose p'^2 coefficient is at most -1. Cauchy–Schwarz on the layer alone gives

$$|B_\tau(e, \eta_\perp)| \;\leq\; \|\cos\|_{L^2(0,\sigma)}\|q - p'\|_{L^2(0,\sigma)} + \|\sin\|_{L^2(0,\sigma)}\|p + q'\|_{L^2(0,\sigma)},$$

with $\|\cos\|^2 = \frac{\sigma}{2} + \frac{\sin 2\sigma}{4}$ and $\|\sin\|^2 = \frac{\sigma}{2} - \frac{\sin 2\sigma}{4}$, every quantity localised to the layer. The Sylvester condition then reads $\sup B^2/|D_\tau| \leq \sin 2\sigma/\pi$. Measured, this holds with margins of 19 to 45 percent of the target for $\sigma \leq 0.10$, and fails for $\sigma \geq 0.4$. At Σ's own $\sigma = \beta$ it holds by 0.0012, under one percent, and the margin shrinks monotonically in the basis size, 0.00201, 0.00165, 0.00147, 0.00135, 0.00123 at $K = 25, 35, 45, 60, 80$, with no sign of the drift decaying. Since a supremum over a subspace is a lower bound for the true supremum, the computed values rise towards the truth and the margin can only shrink further. The layer bound therefore does *not* establish the assembly at Σ.

The loss is entirely in the Cauchy–Schwarz step, and it is avoidable, because $B_\tau(e, \cdot)$ is a *linear* functional. For a linear functional the extremal ratio has a closed form and needs no inequality at all:

$$\sup_{\eta_\perp} \frac{B_\tau(e, \eta_\perp)^2}{|D_\tau[\eta_\perp]|} = b^{\mathsf{T}}(-A)^{-1}b,$$

b the coefficient vector of $B_\tau(e, \cdot)$ and A the matrix of D_τ on the reduced space. Evaluated, the Sylvester condition $\sup B^2/|D| < |D_\tau[e]| = \frac{1}{2}\sin 2\sigma$ holds at every σ from 0.02 to 0.80, with margin between 51 and 61 percent of the target throughout; at $\sigma = \beta$ the value is 0.1320 against

0.2737. The margin exceeds the residual basis drift by a factor of about 700, the drift itself decaying geometrically ($5.6, 2.8, 2.4, 1.5, 0.9$ in units of 10^{-4} at $K = 45, 60, 80, 100, 120$).

Proposition 148 (Coercivity at Σ, with an explicit constant)**.** *Write $\eta = ae + \eta_\perp$ with $e = (0, \sin t)$. For $\theta \in (0,1)$ with $K/\theta < |D_\tau[e]|$, where $K = \sup B^2/|D|$,*

$$D_\tau[\eta] \;\leq\; -a^2\Big(|D_\tau[e]| - \tfrac{K}{\theta}\Big) - (1-\theta)\,|D_\tau[\eta_\perp]|.$$

With $\|ae\|_{H^1}^2 = a^2\pi/2$, Proposition 146, and the H^1-orthogonality of the splitting, this gives $D_\tau[\eta] \leq -c\|\eta\|_{H^1}^2$ with $c = \min\big\{\frac{2}{\pi}(|D_\tau[e]| - \frac{K}{\theta}),\ \frac{3}{5}(1-\theta)\big\}$.

At $\sigma = \beta$, optimising θ gives $\theta = 0.8704$ and

$$\delta^2 Q[\eta] \;\leq\; -0.0777\,\|\eta\|_{H^1}^2 \qquad \text{at } \Sigma.$$

This is an *upper* bound on the top eigenvalue, which is the direction coercivity needs, and it is consistent with the finite-basis value -0.0850 computed earlier: that one is a lower bound, so $-0.0850 \leq \lambda_{\max} \leq -0.0777$ and the two bracket the truth from opposite sides.

K admits an upper bound by a validated solve, which is the direction a certificate needs. For any z, writing $r = b - Mz$ with $M = -A$,

$$b^\mathsf{T} M^{-1} b = z^\mathsf{T} M z + 2z^\mathsf{T} r + r^\mathsf{T} M^{-1} r \;\leq\; z^\mathsf{T} M z + 2z^\mathsf{T} r + \frac{\|r\|^2}{\lambda_{\min}(M)},$$

an identity, not an estimate, in the first step. The one input needed is a lower bound on $\lambda_{\min}(M)$, and Proposition 146 supplies it: in the H^1-orthonormal basis $M \succeq \frac{3}{5}I$. In that basis the coefficients of b decay like $1/k$, so $b \in \ell^2$ and the truncation tail is summable.

Taking z supported on the first N modes and measuring the residual in a basis of $2N$:

N	30	45	60	80
upper bound	0.132739	0.132675	0.132656	0.132644

decreasing and settling near 0.1326, against the target $\frac{1}{2}\sin 2\beta = 0.2737$: a factor of two. The residual $\|r\|^2$ falls as $1.2, 0.85, 0.67, 0.52$ in units of 10^{-3}, consistent with the $O(1/N)$ decay the normalisation predicts. The residual tail is bounded analytically. For the p-modes,

$$b_k = \frac{1}{\|p_k\|_{H^1}} \int_\tau^{\pi/2} \big[\cos t\,\sin 2kt + 2k\sin t\,\cos 2kt\big]\,dt,$$

and the integrand is an exact derivative:

$$\frac{d}{dt}\big[\sin t\,\sin 2kt\big] = \cos t\,\sin 2kt + 2k\,\sin t\,\cos 2kt.$$

Integrating the second term by parts produces $-\int \cos t \sin 2kt$, which cancels the first term identically, so the numerator is a boundary term and nothing else:

$$\int_\tau^{\pi/2} \big[\cos t\,\sin 2kt + 2k\sin t\,\cos 2kt\big]\,dt = \big[\sin t\,\sin 2kt\big]_\tau^{\pi/2} = -\sin\tau\,\sin 2k\tau,$$

since $\sin k\pi = 0$. Hence $|\text{numerator}| \le \sin\tau = \cos\sigma$, not $2+2\sigma$, and with $\|p_k\|_{H^1}^2 = \pi/4 + k^2\pi \ge k^2\pi$,

$$\sum_{k>N} b_k^2 \;\le\; \frac{\cos^2\sigma}{\pi N} = \frac{0.2923}{N} \quad \text{at } \sigma = \beta,$$

a factor 7.2438 below the $(2+2\sigma)^2/\pi = 2.1177$ obtained by bounding the two summands separately. The identity is formalised as `btail_numerator` and the bound as `btail_numerator_abs_le`. This is the fourth estimate in this problem to lose an order by splitting a quantity that was small through cancellation. It dominates the true tail by a factor of 2.00 uniformly in N — the residual factor is the mean of $\sin^2 2k\tau$ — so it is both valid and nearly sharp. At $N = 120$ it contributes 0.0024, and $0.0024/\frac{3}{5}$ added to $z^\mathsf{T} M z = 0.1326$ gives

$$K \;\le\; 0.1367 \;<\; \tfrac{1}{2}\sin 2\beta = 0.2737,$$

a margin of 50 percent. The remaining ingredient is an analytic bound on $\|(Mz)_{>N}\|$, which the same integration by parts supplies since M's off-diagonal entries come only from the layer, $D_{\pi/2}$ being diagonal; measured, that term is $9\cdot 10^{-5}$ at $N = 120$, two hundred times smaller than the b-tail it accompanies.

The entries themselves are exact. Every entry of M and b is a combination of

$$\int_0^T \sin at\,\sin bt, \qquad \int_0^T \cos at\,\cos bt, \qquad \int_0^T \sin at\,\cos bt$$

at integer frequencies over $[0,\pi/2]$ and $[0,\tau]$, each elementary: for $a \ne b$ the first is $\frac{1}{2}[\frac{\sin(a-b)T}{a-b} - \frac{\sin(a+b)T}{a+b}]$ and similarly for the others, with the diagonal cases $\frac{T}{2} \mp \frac{\sin 2aT}{4a}$. Implemented in ball arithmetic these agree with quadrature to its own accuracy, 10^{-12}, while carrying radii of order 10^{-58}. So ingredient three reduces to a ball-matrix solve on certified entries, with no quadrature anywhere.

The fourth ingredient yields to the same treatment as the second, once it is written the right way. Bounding $|M_{kj}|$ termwise and applying Cauchy–Schwarz over $j \le N$ gives $\sum_{k>2N} |(Mz)_k|^2 \lesssim 9\|z\|^2/2\pi^2$, which does not decay in N at all and exceeds the measured value by four orders: the sum over j discards exactly the oscillation that makes it small. But $(Mz)_k = -D_\tau(\text{mode}_k, w)$ with $w = \sum_{j\le N} z_j\,\text{mode}_j$ a *fixed* function, so $(Mz)_k$ is one integral of a fixed function against an oscillating one, the same shape as b_k. Integrating the $2k\cos 2kt$ factor by parts as before gives $|(Mz)_k| \le C/k$, and measured, $\sup_{k>2N} k|(Mz)_k| = 0.154$, so

$$\sum_{k>M} |(Mz)_k|^2 \;\le\; \frac{0.0238}{M},$$

the same $1/M$ law as the b-tail with a constant 89 times smaller. Using $\|r\|^2 \le \big(\|b_{>N}\| + \|(Mz)_{>N}\|\big)^2$ — not $2\|b_{>N}\|^2 + 2\|(Mz)_{>N}\|^2$, which is the same term-splitting loss again and costs 13 percent here, the two tails differing by a factor of five — and both analytic tails at $N = 60$,

$$K \;\le\; 0.1910 \;<\; \tfrac{1}{2}\sin 2\beta = 0.2737,$$

a margin of 30 percent with every tail bounded rather than measured.

So the certificate has four ingredients, of which two are proved outright: $\lambda_{\min}(M) \ge \frac{3}{5}$ by Proposition 146, the b-tail bound by parts, the N-block value $z^\mathsf{T} M z$ by interval arithmetic on elementary trigonometric entries, and the (Mz)-tail by the same estimate. The computation has been carried out. Assembling M and b from the closed forms in Arb ball arithmetic at 300 bits and $K = 70$ (a system of size 139), taking z from a floating-point solve — legitimate, since z enters only as a witness and any z gives a valid bound — and evaluating every subsequent quantity as a ball:

$$\begin{array}{ll} z^{\mathsf{T}} M z = 0.131681641 & \text{radius } 2.3 \cdot 10^{-87} \\ 2z^{\mathsf{T}} r = 1.0 \cdot 10^{-16}, \quad \|r_N\|^2 = 4.2 \cdot 10^{-32} & \\ 2 \times b\text{-tail} = 0.060506, \quad 2 \times (Mz)\text{-tail} = 0.000680 & \end{array}$$

giving

$$K \;\le\; 0.233657832 \;<\; \tfrac{1}{2}\sin 2\beta = 0.273722337,$$

the comparison performed between balls, not floats. The margin is 0.0401, and the (Mz)-tail contributes only 0.0011 of the total after division by $\frac{3}{5}$: the certificate survives the one measured constant $C = 0.154$ being wrong by a factor of six.

The (Mz)-tail constant is proved as well. Writing $(Mz)_k = -\frac{1}{n_k}\left[\int \sin 2kt\, F + 2k \int \cos 2kt\, G\right]$ with $F = 2w_p - \chi w_p - \chi w_q'$ and $G = -2w_p' + \chi w_p' - \chi w_q$, and integrating the second by parts — G jumps at τ, so the boundary terms include that jump —

$$C \;\le\; \frac{\|F - G'\|_{L^1} + |G(0)| + |G(\pi/2)| + \big|[G]_\tau\big|}{\sqrt{\pi}},$$

every term computable from the witness z. Writing $\|F\|_{L^1} + \|G'\|_{L^1}$ here in place of $\|F - G'\|_{L^1}$ repeats the error just corrected in the b-tail, and by the same mechanism: on each piece, where χ is constant,

$$F - G' = (2 - \chi)\big(w_p + w_p''\big),$$

the two $\chi w_q'$ terms cancelling identically, so w_q leaves the estimate altogether. Evaluated it converges upward in K through 1.608, 1.779, 1.810, 1.862, 1.875 at $K = 20, 35, 50, 70, 90$, with increments falling to 0.013; taking $C \le 1.90$ and $K = 150$ the certificate closes with the proved constant:

$$K \;\le\; 0.259397171 \;<\; \tfrac{1}{2}\sin 2\beta = 0.273722337,$$

where $z^{\mathsf{T}} M z = 0.132114949$, $\|r_N\|^2 = 8.4 \cdot 10^{-32}$, the b-tail contributes 0.028236 and the (Mz)-tail 0.048133, margin 0.0143, the comparison again between balls. Replacing 2.1177 by the exact 0.2923 drops the b-tail contribution to 0.003898 and gives

$$K \;\le\; 0.2188337 \;<\; \tfrac{1}{2}\sin 2\beta = 0.273722337,$$

margin 0.0549, a gain of 3.83.

Rule 0: $\lambda_{\min}(M) \ge \frac{3}{5}$, the b-tail bound and the (Mz)-tail formula are all PROVED, and the arithmetic is certified. The sole remaining input at a lower label was the numerical value of C. It is now enclosed, and the enclosure is the subject of the next paragraph.

The certificate is assembled by `ambi_kcert.py` from the closed forms above, and the same integration by parts that fixed the b-tail supplies C outright. With $w = \sum_j (z_j/n_j)\,\mathrm{mode}_j$ the fixed witness function and χ the indicator of the layer, integrating the p_k' factor by parts gives

$$D_\tau(p_k, w) = \int_0^{\pi/2} (1 + \chi)\, p_k\, \big(w_p + w_p''\big) \;-\; p_k(\tau)\, J, \qquad J = w_q(\tau) - w_p'(\tau),$$

the w_q' terms cancelling identically as in the b-tail. Since $|p_k| \le 1$ and $n_k \ge k\sqrt{\pi}$,

$$C \;\le\; \frac{\big\|(1+\chi)(w_p + w_p'')\big\|_{L^1} + |J|}{\sqrt{\pi}},$$

one L^1 norm of a finite trigonometric polynomial and one point value. Both are enclosable without interval quadrature: $W = w_p + w_p'' = \sum_k c_k \sin 2kt$ has $\|W'\|_\infty \le \sum_k 2k|c_k| = 9070$ exactly, so a midpoint rule on $4 \cdot 10^6$ cells with τ on a cell boundary carries a Lipschitz slack of $5.6 \cdot 10^{-3}$ and a rounding allowance of 10^{-9}. At $K = 150$,

$$\big\|(1+\chi)W\big\|_{L^1} \le 1.921953, \qquad |J| = 0.257985, \qquad C \le 1.229898,$$

against the 2.4658 the exact b-tail admits: a factor of two in hand. The bound also dominates the measured $\sup_k k|(Mz)_k| = 0.5407$, as it must. Hence

$$K \;\le\; 0.166947978 \;<\; \tfrac{1}{2}\sin 2\beta = 0.273722337,$$

margin 0.106774, against the 0.0143 of the first assembly: seven times the room, and no input below PROVED. Rule 7 is satisfied because z is a witness — the identity $b^{\mathsf T}M^{-1}b = z^{\mathsf T}Mz + 2z^{\mathsf T}r + r^{\mathsf T}M^{-1}r$ holds for every z, so a floating-point solve costs nothing but sharpness.

Truncation is otherwise bounded from two directions. The finite-basis values K_N increase with N, so each is a lower bound for K_∞; their increments decay geometrically, $5.6, 2.8, 2.4, 1.5, 0.9$ in units of 10^{-4} at $N = 45, 60, 80, 100, 120$, giving $K_\infty \approx 0.1322$. Independently, the diagonal contribution of the modes beyond the truncation is $\sum b_k^2/|A_{kk}|$ with $b_k = O(1)$ and $|A_{kk}| \sim k^2$, hence $O(1/N)$; evaluated it is about 0.03, a crude overestimate since it ignores the off-diagonal coupling that reduces the effective contribution. Both bounds leave K well below $\frac{1}{2}\sin 2\beta = 0.2737$, the looser one still by a factor of three.

Rule 0: $|D_\tau[e]| = \frac{1}{2}\sin 2\sigma$ and the tail constant $\frac{3}{5}$ are PROVED, as is the absorption step; $K = 0.1320$ is HEURISTIC, being the value of a finite-basis computation, though of a quantity involving no inequality and with a margin three orders above the increment. The constant 0.0777 therefore carries HEURISTIC, and certifying K is the single remaining step. This replaces the measured table of Proposition 152, whose entries bound the constants from the wrong side and establish nothing.

It is the argument that is lossy, not the conclusion: the direction maximising the cross term has D-quotient about -1.85, three times the global worst case $-\frac{3}{5}$, so bounding the two terms by their separate worst cases counts a direction that realises neither. The sharp condition is

$$\sup_{\eta_\perp} \frac{B_\tau(e,\eta_\perp)^2}{|D_\tau[\eta_\perp]|} \;\le\; \frac{\sin 2\sigma}{\pi},$$

and it is an *analytic* bound on that joint quantity, not a sharper constant in either factor separately, that would finish the assembly. Consistent with this, the measured top eigenvalue on the full space is about half the marginal mode's own quotient — ratios 0.503, 0.523, 0.541, 0.670 at $\sigma = 0.10, 0.05, 0.025, 0.010$ — so the true maximiser mixes the two pieces. Closing the coupling is what case (b) does by spectral splitting, and it remains the one step between these propositions and a coercivity theorem with explicit constants.

Remark 149 (The end-cap degeneracy is linear)**.** The diagonal family D_τ degenerates as $\tau \to \pi/2$: at $\tau = \pi/2$ the mode $(0, \sin t)$ has D-value 0, which is what forces the separate treatment in case (b) of Theorem 122. The rate matters, because it decides whether a perturbative argument can close the gap. Computing the top eigenvalue of D_τ against the L^2 norm in the basis $p = \sum a_k \sin 2kt$, $q = \sum b_k \sin(2k-1)t$ — which imposes $p(0) = p(\pi/2) = q(0) = 0$ with $q(\pi/2)$ free exactly — gives, with $\sigma = \pi/2 - \tau$,

K	20	30	45	60
p	0.834	0.896	0.942	0.957

for the exponent in $|\lambda_{\max}| \sim \sigma^p$ fitted on $\sigma \le 0.10$. The sequence rises towards 1 as the basis is enlarged, matching the damping $-\frac{1}{2}a^2 \sin 2\sigma \sim \sigma$ that case (b) uses. So the degeneracy is *linear*, and the existing argument is calibrated to the right rate; there is no need for a framework tuned to a faster decay.

Every eigenvalue computed was negative, at every τ and every K, which is an independent check on Theorem 122 in a discretisation sharing no code with the proof. Rule 7: the exponent is HEURISTIC; the negativity is a regression test, not a proof.

Remark 150 (Two ways to fake a coercivity constant)**.** The 2×2 reduction of the previous paragraph invites two errors, and both were made here before being caught. They are recorded because each looks like a verification.

First, the Sylvester test is not independent of what it tests. For $\left(\begin{smallmatrix} -M & c \\ c & -c_T \end{smallmatrix}\right)$ the top eigenvalue is $-\frac{M+c_T}{2} + \sqrt{(\frac{M-c_T}{2})^2 + c^2}$ and the determinant is $Mc_T - c^2$, so *top* < 0*, det* > 0 *and* $c^2 < Mc_T$ *are one condition, not three.* Measuring the top eigenvalue, inverting the eigenvalue formula for c^2, and then checking $c^2 < Mc_T$ therefore returns the sign of its own input: it reports success precisely when the measured top was already negative, and cannot corroborate anything. The honest route is to bound c *independently*, which is what the previous paragraph does, and which fails.

Second, computing the top eigenvalue in a finite basis gives a bound in the useless direction. The supremum of a Rayleigh quotient over a subspace is at most the supremum over the whole space, so a finite-basis value is a *lower* bound for $\lambda_{\max}$, while coercivity is an upper bound on it. The symptom is visible in the convergence: at $\sigma = \beta$ the computed value runs -0.0859, -0.0854, -0.0852, -0.0850 for $K = 25, 40, 60, 80$, rising towards its limit from below. Quoting -0.0850 as a coercivity constant would repeat, in a new place, exactly the error Remark 151 identifies in Proposition 152.

What survives is the observation that made the attempt worth making: Σ's own cell has $E_1 = [0, \beta)$ and $E_2 = [0, \pi/2 - \beta)$, so by Lemma 121 it is dominated by the diagonal member at $\sigma = \beta = 0.2897$, not at $\sigma \to 0$. The degeneracy of Proposition 147 is therefore *not* active at Σ: an explicit positive constant exists there, and the obstruction to producing it is the cross-term bound of the previous paragraph, not the end-cap degeneracy.

Remark 151 (Why more precision is the wrong repair)**.** Proposition 152 is the note's main remaining measured claim, and it is natural to propose fixing it with certified arithmetic. That would not work, and the reason is worth stating because the error is easy to make.

Rayleigh–Ritz values on a finite-dimensional subspace are *upper* bounds for the largest admissible constant, since restricting the competition can only raise the supremum of the quotient. Certifying such a computation in interval arithmetic certifies an upper bound to higher precision. But coercivity is a *lower* bound on the same quantity, so the certified inequality points the wrong way, and no amount of precision reverses it. Equivalently: negative-definiteness of a finite-dimensional Hessian does not imply negative-definiteness of the form it discretises.

What does work is a Gårding split — a finite-dimensional part, where certified arithmetic is exactly right, plus an analytic tail bound on its orthogonal complement. Only the second gives a lower bound, and only the pair gives a proof. The obstruction is therefore not precision but the missing tail estimate, and the crude route to it is genuinely lossy: on the diagonal family the p'^2 and q'^2 coefficients are $-2+\chi$ and $-1-\chi$, both at most -1, but the Poincaré constants on the two spaces are 4 and 1, and the q side then leaves a positive residue for every splitting parameter. The negativity of D_τ comes from cancellation between the arms, not from the symbol.

The following is retained only for the record; it is superseded by Proposition 148, and Remark 151 explains why its entries bound the constants from the side that establishes nothing.

Proposition 152 (H^1 stability, measured, superseded)**.** *The second variation controls the derivative as well. Measured against assembled mass and stiffness matrices, the largest c_0 with $\frac{1}{2}\delta^2 Q[\eta] \le -c_0\|\eta\|^2_{L^2(0,\pi)} - c_1\|\eta'\|^2_{L^2(0,\pi)}$ is*

c_1	0	0.1	0.3	0.5
$m = 64$	0.7339	0.6132	0.3595	0.0796
$m = 128$	0.7331	0.6123	0.3583	0.0779
$m = 256$	0.7323	0.6114	0.3571	0.0760
$m = 512$	0.7313	0.6103	0.3556	0.0739

These are Rayleigh–Ritz values, hence upper bounds on the true constants, and every column is still decreasing at $m = 512$: the $c_1 = 0$ column is the one whose limit is known independently, and it is descending towards the certified $c^* = 0.7309566\ldots$, not towards its own $m = 256$ entry. So the table bounds the admissible constants from above and does not *establish any of the inequalities. Only the $c_1 = 0$ case is proved, by Theorem 119. Rule 7: the $c_1 > 0$ entries are* HEURISTIC*, and the frontier $c_0 \approx 0.731 - 1.31\,c_1$ read off them, vanishing near $c_1 = 0.56$, is a measurement rather than a theorem.*

Remark 153 (Why the earlier statement was wrong)**.** An earlier version of this proposition printed the $m = 256$ row alone, asserted that the entries were converged, and concluded $\frac{1}{2}\delta^2 Q \le -0.357\|\eta\|^2 - 0.3\|\eta'\|^2$. The assertion was false in exactly the way the paragraph above this proposition already describes for 0.7323: a Rayleigh–Ritz sequence that has not converged, read as if it had. The same mesh value appears in both places, which is how the inconsistency survived. Reproducing a printed number at the mesh it was computed on confirms its provenance, not its correctness.

Two simplifications are worth recording. First, the σ term has no singularity: since $-2\int \eta\eta' \tan t = -\int (\eta^2)' \tan t = \int \eta^2 \sec^2 t$ when $\eta(0) = \eta(\pi/2) = 0$, and $\sec^2 t - \tan^2 t = 1$,

$$-\int_0^{\pi/2} \big(\eta \tan t + \eta'\big)^2 dt = \int_0^{\pi/2} \big(\eta^2 - \eta'^2\big) dt,$$

checked to 10^{-12} on five test functions, so (24) may be written with bounded coefficients throughout. Second, the sharp L^2 constant is 0.7309566, not the 0.7285 of an earlier draft; that figure came from a mass matrix that gave the endpoint hat at $\theta = \pi$ the interior weight $2h/3$ instead of $h/3$.

Remark 154 (The size of $\mathcal{D}$, and what Theorem 115 is not)**.** It is not a proof that Σ is optimal, and the reason is quantitative.

Measuring the radius of $\mathcal{D}$ along $H_\Sigma+\varepsilon\sin(2k\theta)$ gives $\varepsilon_k = 0.0864,\ 0.0117,\ 0.0047,\ 0.0036,\ 0.0021$ at $k = 2, 4, 6, 8, 10$, so $\varepsilon_k k^2$ stays between 0.17 and 0.35: the radius scales like k^{-2}. $\mathcal{D}$ therefore contains a ball in the C^2 norm and in no weaker one, which is exactly what a bound on H'' should give. It is a genuine ball rather than a sliver, of 60 random eight-mode perturbations with coefficients of size $0.02/k^2$, 51 land but every member is a small C^2-perturbation of Σ. **Theorem 115 is thus a C^2-local statement**, with explicit convex hypotheses in place of a neighbourhood of unspecified size.

In every one of those probes the binding constraint was (b); condition (c) never became active. That is a statement about the probe and not about the geometry: (c) is *independent* of (b). Adding $\delta\sin(\theta-\pi/2)$ on $[\pi/2, \pi]$ and 0 on $[0, \pi/2]$ leaves $H + H''$ identically unchanged, so (b) and the gauge hold exactly, while the ceiling facet grows by δ and c_y grows by $\delta \sin t \cos t$; at $\delta = 0.2243\ldots$ the maximum M crosses $\frac{1}{2}$. So (c) cannot be derived and must remain a hypothesis.

$\mathcal{D}$ does not reach Gerver's cap, and by Remark 155 injectivity genuinely fails once the density exceeds 1. The failure is sharp in the condition but gentle in magnitude: comparing the flux $\frac{1}{2}\int(\alpha_2^+)^2$ with the area actually swept, the excess is $5.4\cdot 10^{-7}$ at density 1.047 and $7.4\cdot 10^{-5}$ at 1.267, and Gerver's own excess is about 0.25%. Widening $\mathcal{D}$ therefore does not need a weaker constant; it needs an upper bound on that excess which is *concave* in H, so that subtracting it preserves the concavity of Q. Truncating the sweep at $\inf_{t'} s_{22}(t, t')$ fails because that infimum is concave in H and $\frac{1}{2}\lambda^2$ then loses the convexity the Mamikon step supplies; the first-capture decomposition is exact but its parameter domain stops being of the form $0 \le s \le \alpha_2$, so its area is no longer quadratic in H. So the honest reading is that the one-corner architecture transfers to the ambidextrous problem on an explicit convex C^2-neighbourhood of Σ, and that enlarging that neighbourhood is where the remaining difficulty sits.

Remark 155 ((RC) is sharp)**.** The constant 1 in the forcing $R = 1-(G+G'')$ of Theorem 81 is the corridor width: it enters through $c(t) = (F-1)\mu_t + (G-1)\nu_t$, so $\langle c(t), \nu_{t'}\rangle = n + G - 1$ and $n'' = -n + 1 - (G + G'')$. The argument therefore cannot tolerate a density above 1. Nor can the conclusion: perturbing Σ so that the maximum density passes through 1, the face-2 sweep acquires self-intersections exactly there. At densities $0.838, 0.921, 0.976$ there are none; at $1.004, 1.047, 1.119$ there are $160, 1510, 3086$ among $\binom{301}{2}$-many tested pairs, and a second mode gives 0 at 0.976 against 200 and 526 at 1.045 and 1.114. So (b) is sharp as stated.

22 Formalisation

Lemma 1, Corollary 2 and the collapse of (3) are machine-checked in Lean 4 without Mathlib as `niche_below_apex`, `niche_disjoint` and `union_area_of_disjoint`. The identity underlying Σ's middle-phase equation is also formalised: writing D for the formal derivative in the half-angle, $D(Dv) = -(v+1)$ for every arc of constant term -1, which is $4v'' = -(v+(1,1))$ for the bracket of (4) (`Trig.sol6_bracket`, proved by `rfl` and depending on no axioms), as is the frame-speed computation behind Theorem 14 (`Trig.sol1_speed_mu`) and the covering criterion of Theorem 11 (`strip_covers_iff`).

Two steps of the unconditional bound are formalised, chosen as the ones where a slip would be fatal and where no analysis is needed to state them. The four-piece decomposition of Lemma 104 is a fact about a linear order, so $\mathbb{Z}$ is a faithful model: `four_piece` states that a point avoids both removed open intervals exactly when it lies in one of the four pieces, `four_piece_no_cross` specialises to the case where both cross pieces are empty, and `cross_absorbed` is the containment that makes Lemma 108 need no new estimate. The maximum of the profile integral reduces to $2x^2 + 2(\sqrt{2} - x)^2 = 2 + (2x - \sqrt{2})^2$, an identity in $\mathbb{Z}[\sqrt{2}]$, which is built explicitly rather than posited: a hypothesis $S^2 = 2$ over $\mathbb{Z}$ has no witness and would make the statement vacuous, so `Z2` carries the ring structure and `sqrt2_sq` exhibits the model, after which `profile_identity` is the identity and `equality_case` locates the maximiser at $2x = \sqrt{2}$.

The profile integral itself is now verified, not merely its middle-range formula. The step is `integral_tent_le`: for every x, $\int_0^{\sqrt{2}} (2 - 2|s - x|)^+ ds \le 2\sqrt{2} - 1$, with `integral_tent_at_mid` giving equality at $x = \sqrt{2}/2$, so the pair says $2\sqrt{2} - 1$ is the supremum. The Lean proof is shorter than the case analysis of Proposition 105 because it avoids it: the identity $\max(0, y) = y + \max(0, -y)$ turns the truncation into a correction term,

$$(2 - 2|s - x|)^+ = 2 - 2|s - x| + 2(|s - x| - 1)^+,$$

valid with no hypothesis on s, and since $|y| - (|y| - 1)^+ = \min(|y|, 1)$ the bound becomes

$$\int_0^{\sqrt{2}} \min\bigl(|s - x|, 1\bigr) ds \;\ge\; \tfrac{1}{2} \qquad \text{for every } x \in \mathbb{R},$$

which is `half_le_integral_cap`. That has a one-line proof: split the band at its midpoint and pair s with $s + \sqrt{2}/2$. The triangle inequality gives $|s - x| + |s + \sqrt{2}/2 - x| \ge \sqrt{2}/2$, and $\min(a, 1) + \min(b, 1) \ge \min(a + b, 1)$ with $\sqrt{2}/2 < 1$ turns that into $\min(|s - x|, 1) + \min(|s + \sqrt{2}/2 - x|, 1) \ge \sqrt{2}/2$; integrating over $[0, \sqrt{2}/2]$ gives $(\sqrt{2}/2)^2 = \frac{1}{2}$. Equality throughout at $x = \sqrt{2}/2$. The -1 in $2\sqrt{2} - 1$ is exactly twice this $\frac{1}{2}$, which is the same -1 that separates the constant from Hammersley's $2\sqrt{2}$.

`lake build` is clean with no `sorry` across the development, and `#print axioms` reports nothing beyond `propext`, `Quot.sound` and `Classical.choice`. The rectangle argument of Remark 69 is verified in that sense end to end: the two bracket monotonicities are derived from `bracket_deriv` and `moment_bracket_deriv` through Mathlib's mean-value machinery rather than assumed (`bracket_monotoneOn`, `moment_bracket_antitoneOn`, `moment_bracket_monotoneOn`), and `rectangle_of_facets` carries the convexity step that turns the two facets into the rectangle. The near case of the bathtub inequality of Remark 62 is likewise verified from its derivative (`Lam_hasDerivAt`, `Lam_nonneg_near`); what is *not* formalised there is the bathtub optimisation itself, which is stated and proved only in the note. The second variation is partially formalised. Both decoupling inequalities of Theorem 118 are verified outright, each as an exact square over the field (`decouple_E2`, `decouple_E1`), and the assembly of Proposition 116 from its analytic inputs to the constant $\frac{1}{12}$ is verified as pure arithmetic (`secondvar_assembly`); three of the five inputs are theorems: the absorption inequality, the subset monotonicity and the tail cut are verified for continuously differentiable data (`habs_verified`, `hsub_verified`, `hcut_verified`), all from one engine, $(f(b) - f(a))^2 \le (b - a) \int_a^b f'^2$ (`sq_sub_le`), proved by monotonicity of $t \mapsto (t - a) \int_a^t f'^2 - (f(t) - f(a))^2$, whose derivative is $\int_a^t (f'(s) - f'(t))^2 \, ds \ge 0$; no Cauchy–Schwarz machinery is needed. The two Wirtinger inequalities are also verified, and by a route worth recording: the classical ground-state substitution divides by sin, which vanishes at the Dirichlet end, and the resulting integration by parts has a removable singularity that is expensive to formalise. Shifting the Riccati supersolution avoids it entirely. The function $v = k \cot(kt + c)$ with $c > 0$ is regular on all of $[0, T]$ when $kT + c < \pi$ and satisfies $v' + v^2 + k^2 = 0$ exactly, so the pointwise square $(f' \sin - kf \cos)^2 \ge 0$ and the ordinary fundamental theorem of calculus give

$$k^2 \int_0^T f^2 \;\le\; \int_0^T f'^2 - v(T) f(T)^2 + v(0) f(0)^2,$$

with no limits taken anywhere (`riccati_block`). At the Dirichlet end $f(0) = 0$ absorbs the $v(0)$ term however large $\cot c$ is, and the loss $k \tan(c)\, f(T)^2$ at the other end is controlled by the verified engine $f(T)^2 \le T \int f'^2$. The result is $\int f^2 \le (1 + \pi c) \int f'^2$ for the Dirichlet–Neumann problem and $4 \int f^2 \le (1 + 2\pi c) \int f'^2$ for the Dirichlet–Dirichlet one (`wirtinger_DN`, `wirtinger_DD`, the latter glued from two shifted blocks by reflection), for every $c > 0$; the elementary chain has slack $1/1886$, and $c = 1/8000$ fits inside it (`secondvar_assembly'`). The capstone `elementary_constant_verified` carries *no analytic hypothesis*: for continuously differentiable data with the boundary conditions and $\beta \le \frac{3}{10}$, the decoupled forms dominate $\frac{1}{12}$ of the mass, machine-checked end to end. The ball-arithmetic certificate of Theorem 119 for the sharp $\frac{73}{100}$ remains outside Lean, and its label remains PROVED, not VERIFIED. The geometric reduction is verified, including the containment. The placed region is defined (`placedRegion`) and shown measurable; its fibres lie in the four-piece union (`fibre_subset_four_piece`); and, in the sign-restricted case that Σ satisfies, the slicing identity is *proved*: Fubini restricts the fibre integral to the band because the region is empty outside it (`placedRegion_volume_eq`), each fibre is bounded by the two tents (`fibre_volume_le`), and the verified profile bound evaluates the result (`placedRegion_volume_le`). The containment is supplied by `Containment.lean`: `AmbiSofa` records a sofa together with the two hallway placements it occupies at angle $\pi/4$, which is what "lies in the hallway at every instant" means, so `sofa_subset_pair` is definitional; and `toFibre_mem_placedRegion` carries the intersection into `placedRegion` by affine arithmetic, which is the content of Lemma 102. The hallways are *defined* there by their half-plane descriptions in the rotated frame rather than derived from a definition of rotation, so that identification is recorded rather than re-derived; it is the one modelling step of the chain, and it is not an estimate. Lemma 110 is verified end to end: `lemmaE_verified` carries no hypotheses, and `profile_bound_verified` is the lemma in its original g-form, $g(m_1)+g(m_2)+2\int F \le 2(2\sqrt{2}-1)$ for every $m_1 \le m_2$. The analytic inputs are discharged as follows: (L1) by `L1_triangle` (extend the domain, split at the midpoint, two ramp integrals); (L2) by pointwise domination; (L3) and (L4) by `L3_full` and `L4_full`, which rest on the splitting $C = A - K$ of `Excess.lean`, the four ramp integrals `ramp_up_in` through `ramp_down_le`, the zone analysis `Kloss_le`, the closed forms `Cfun_right` and `Cfun_far` beyond the band, and the reflection `Cfun_symm`, which delivers the left side for free. No covering, no interval arithmetic and no numerics of any kind appear in the verified chain. Trigonometric quantities are carried as formal symbols with the sign hypotheses $\sin t \ge 0$, $\cos t \ge 0$, which is all Lemma 1 uses.

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